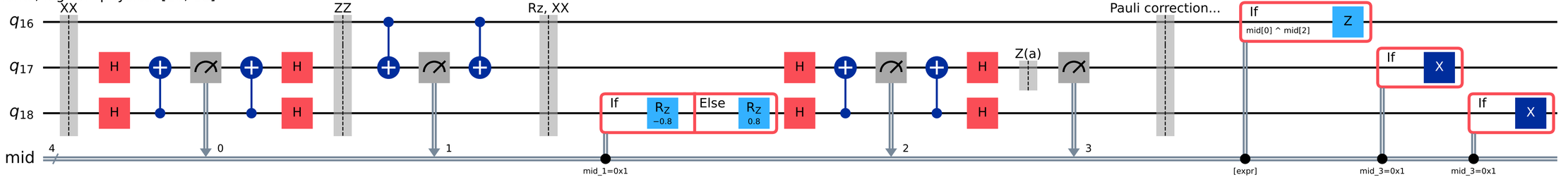
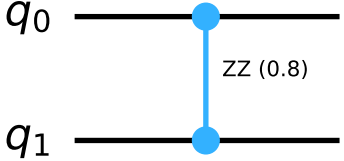
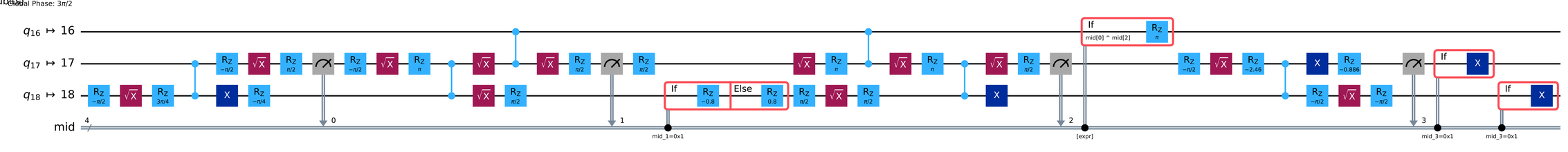
[Z2] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, 4 mid-circuit measurements, logical→physical [16, 18]



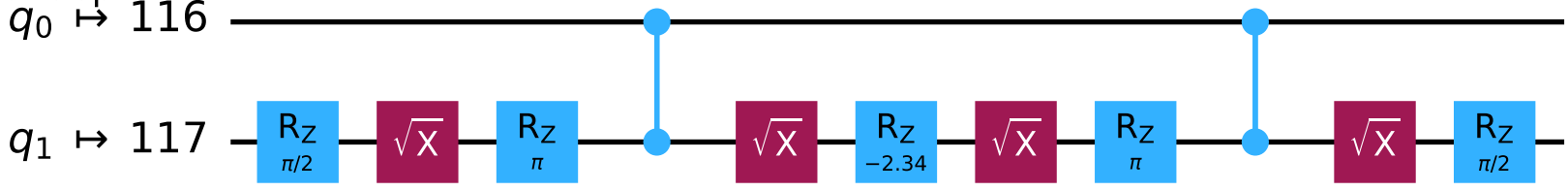
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1

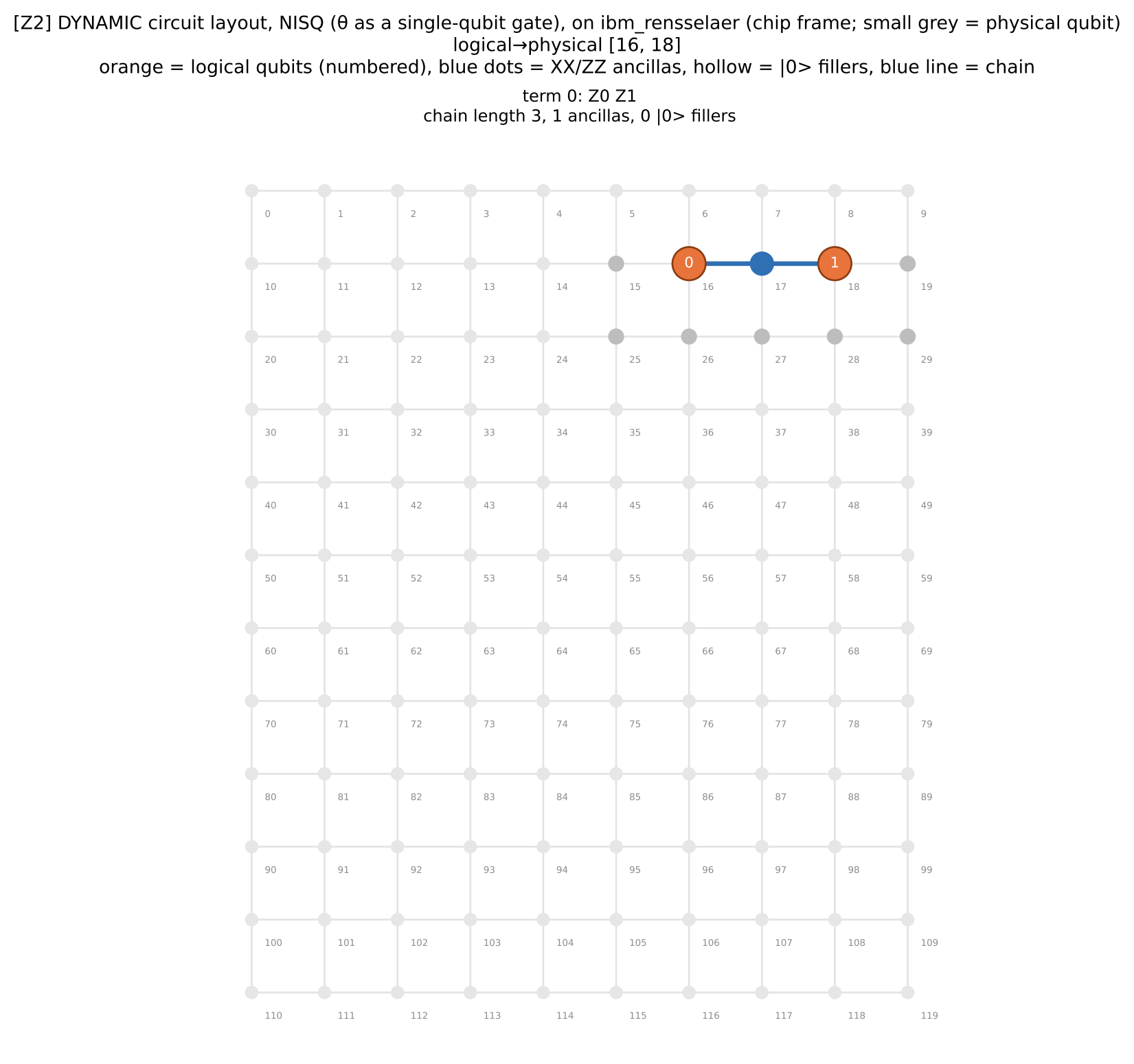


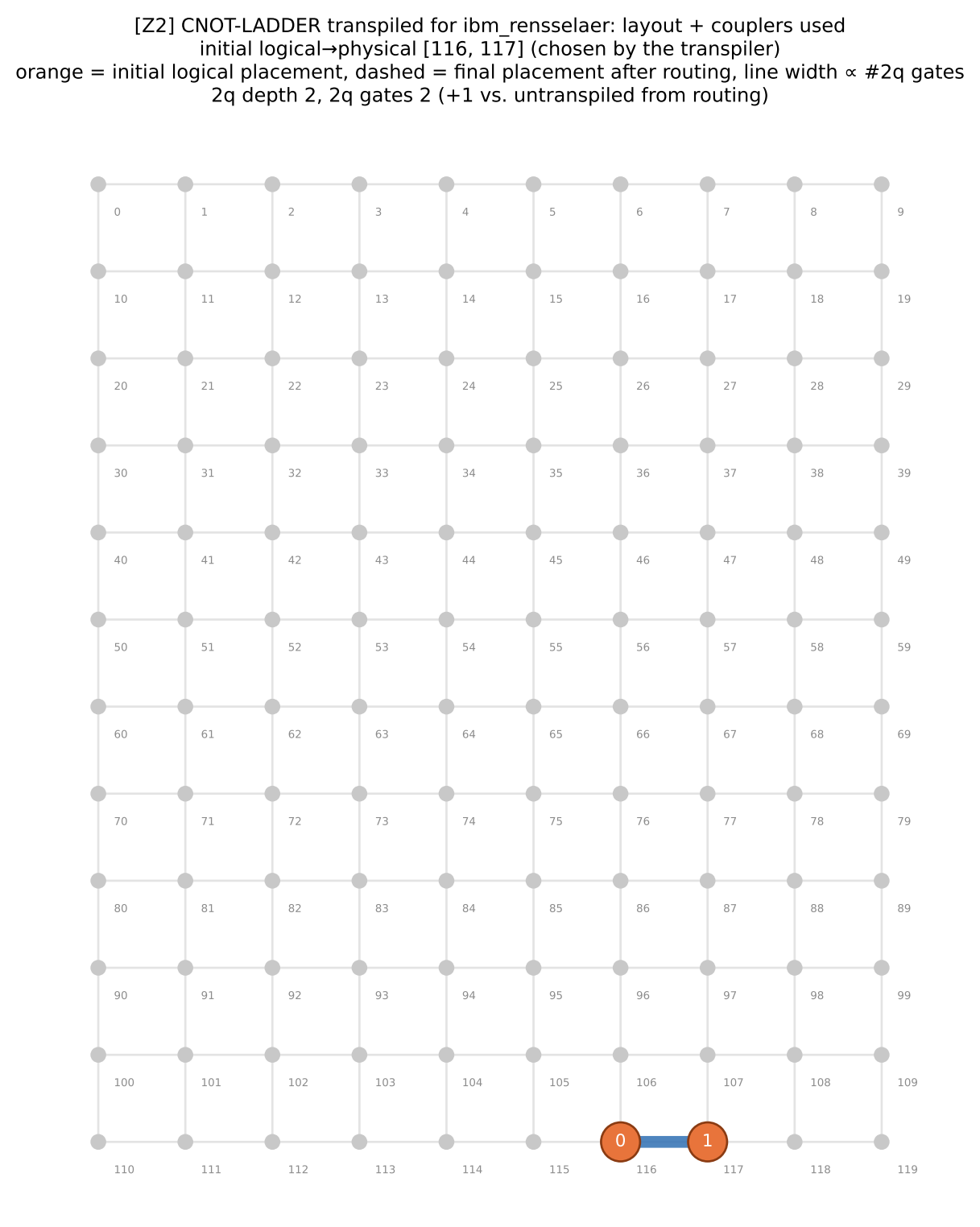
[Z2] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:20 sx:12 cz:6 measure:4 if_else:4 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits).
2q depth 6, total depth 36



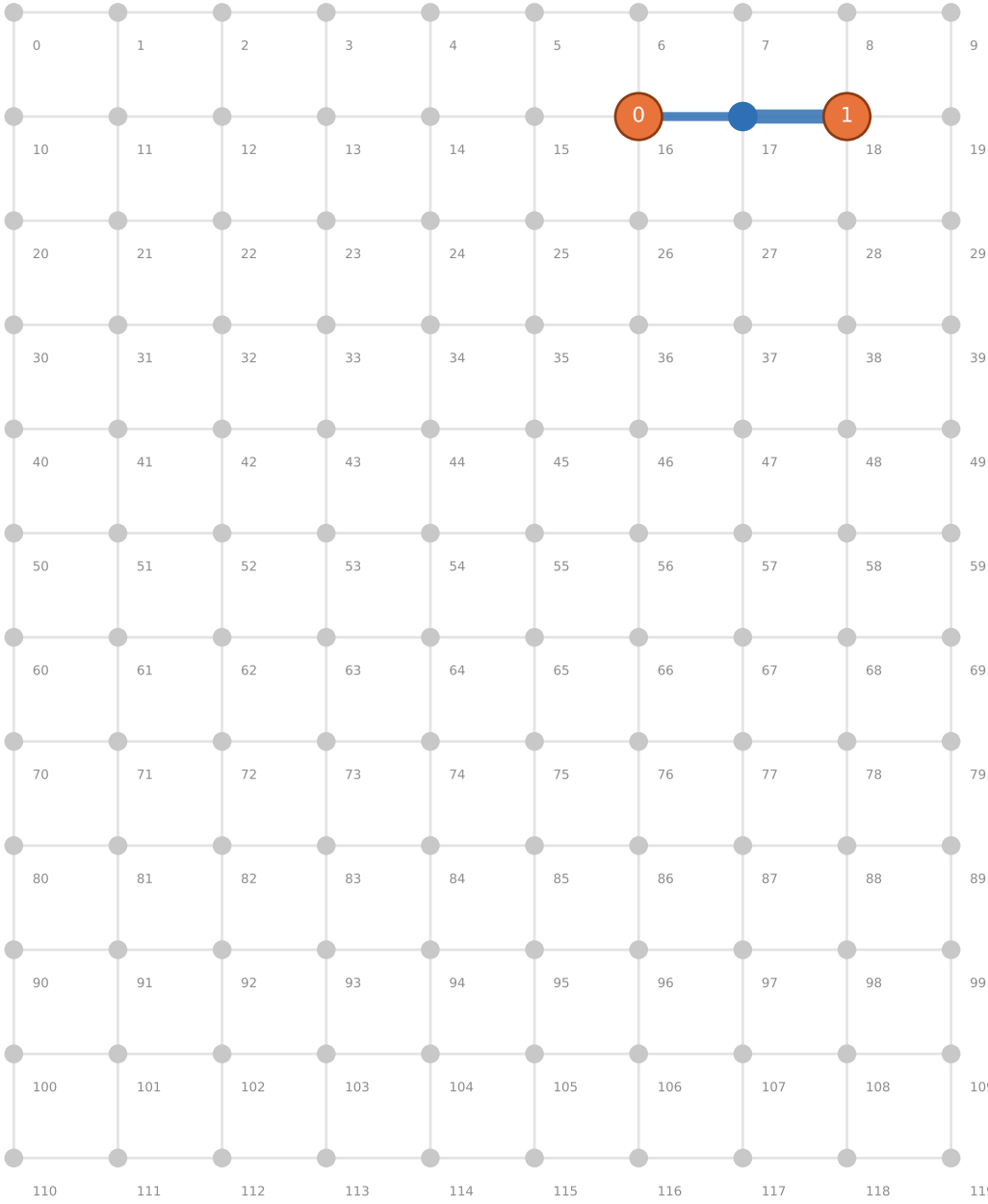
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11



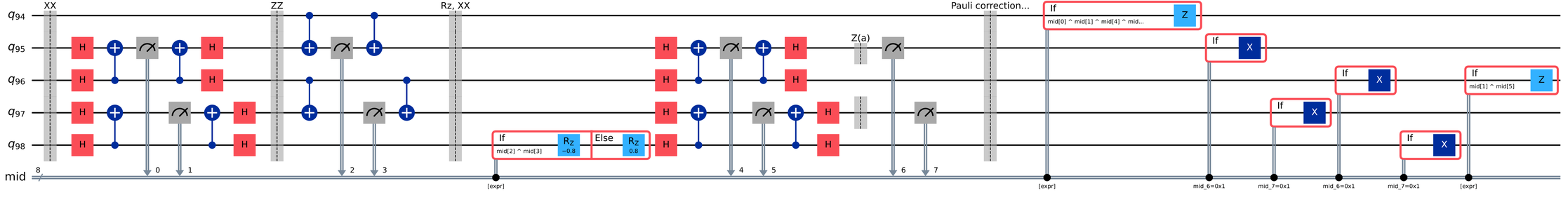

[Z2] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [16, 18]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1


[Z2] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1

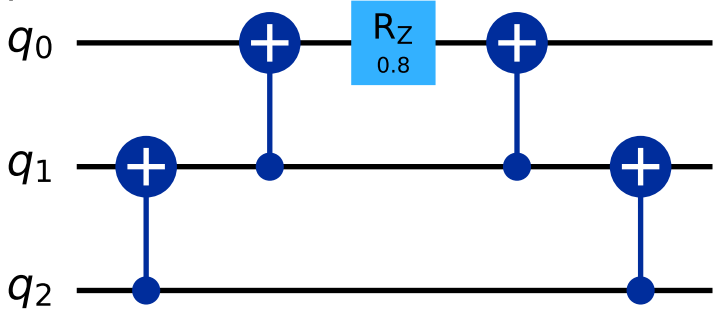
[Z2] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [16, 18]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 6



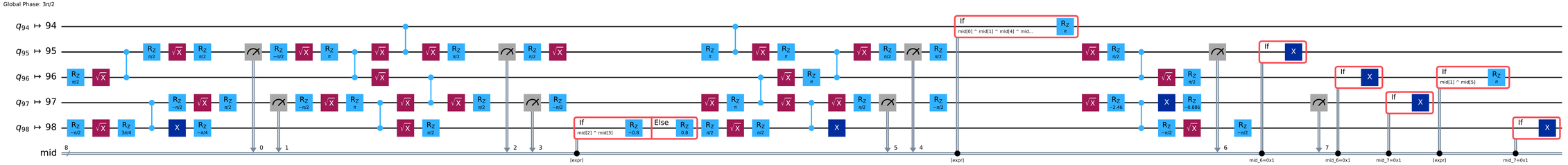
[Z3] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, 8 mid-circuit measurements, logical→physical [94, 96, 98]



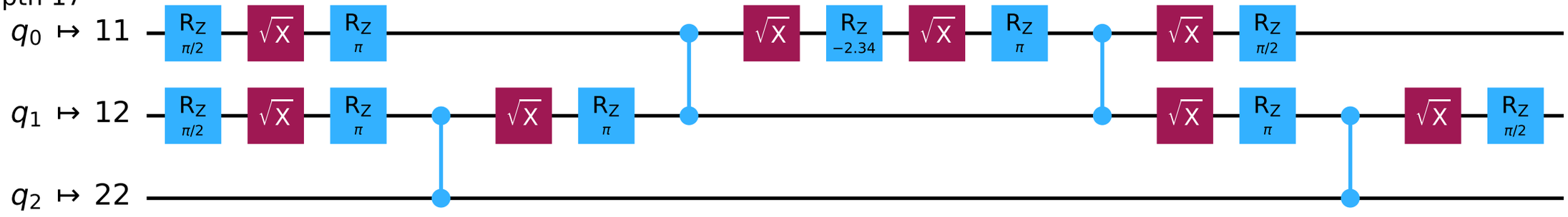
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



[Z3] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:33 sx:24 cz:12 measure:8 if_else:7 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, total depth 35

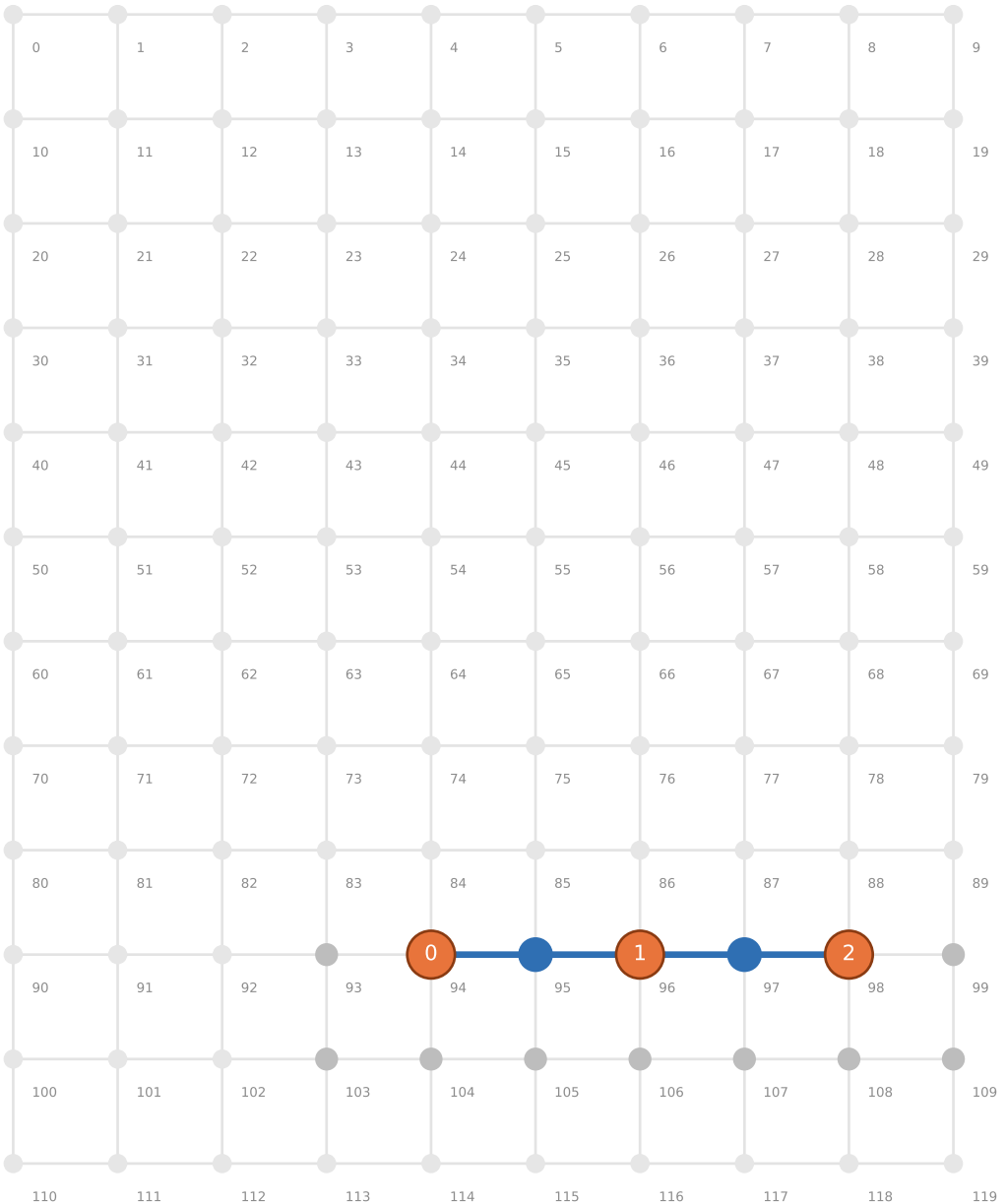


[Z3] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 17

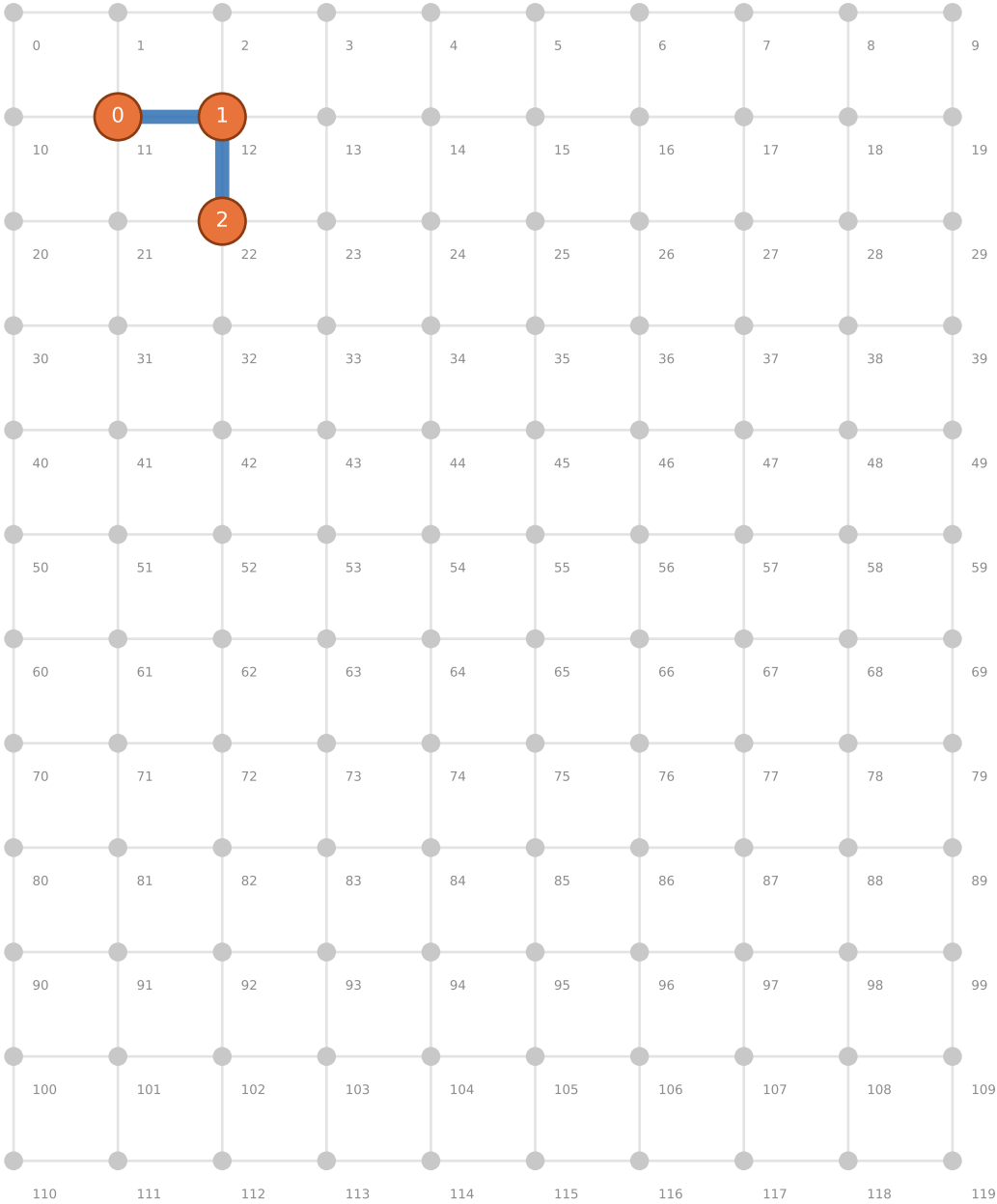


[Z3] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [94, 96, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

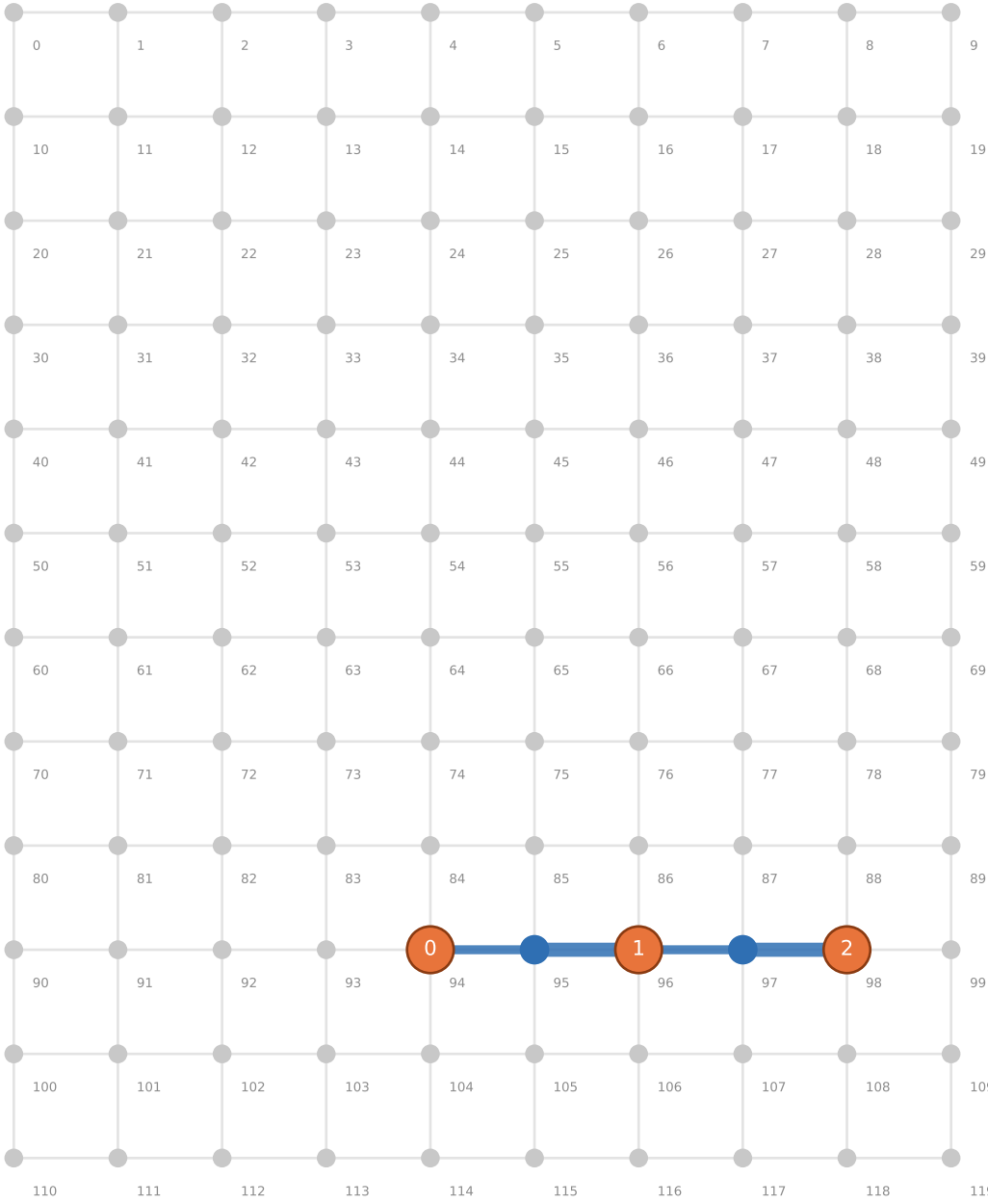
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers



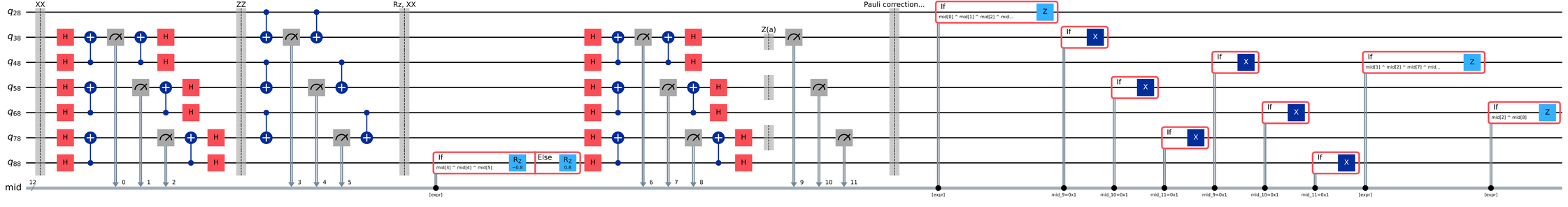
[Z3] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



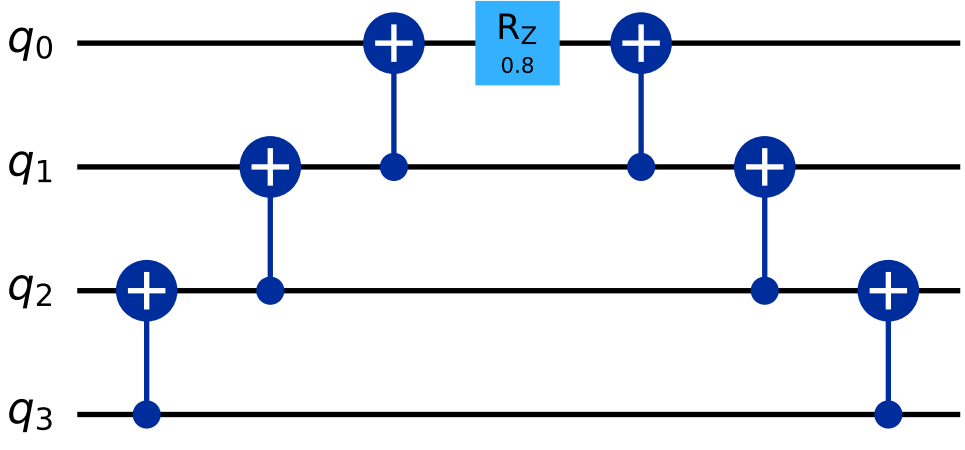
[Z3] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [94, 96, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 12



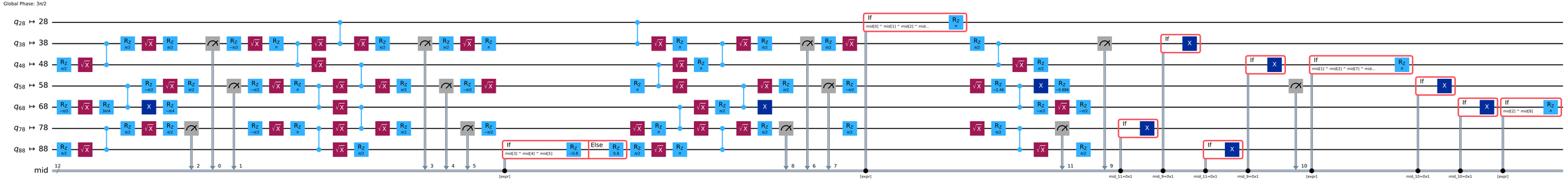
[Z4] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, 12 mid-circuit measurements, logical→physical [28, 48, 68, 88]



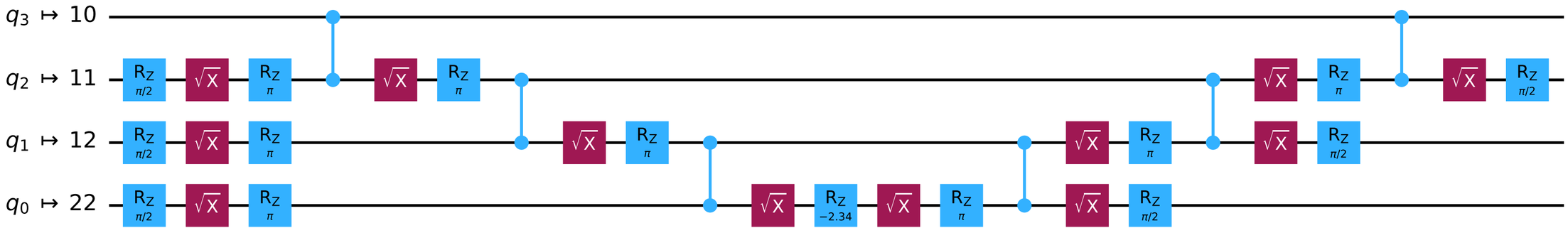
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6

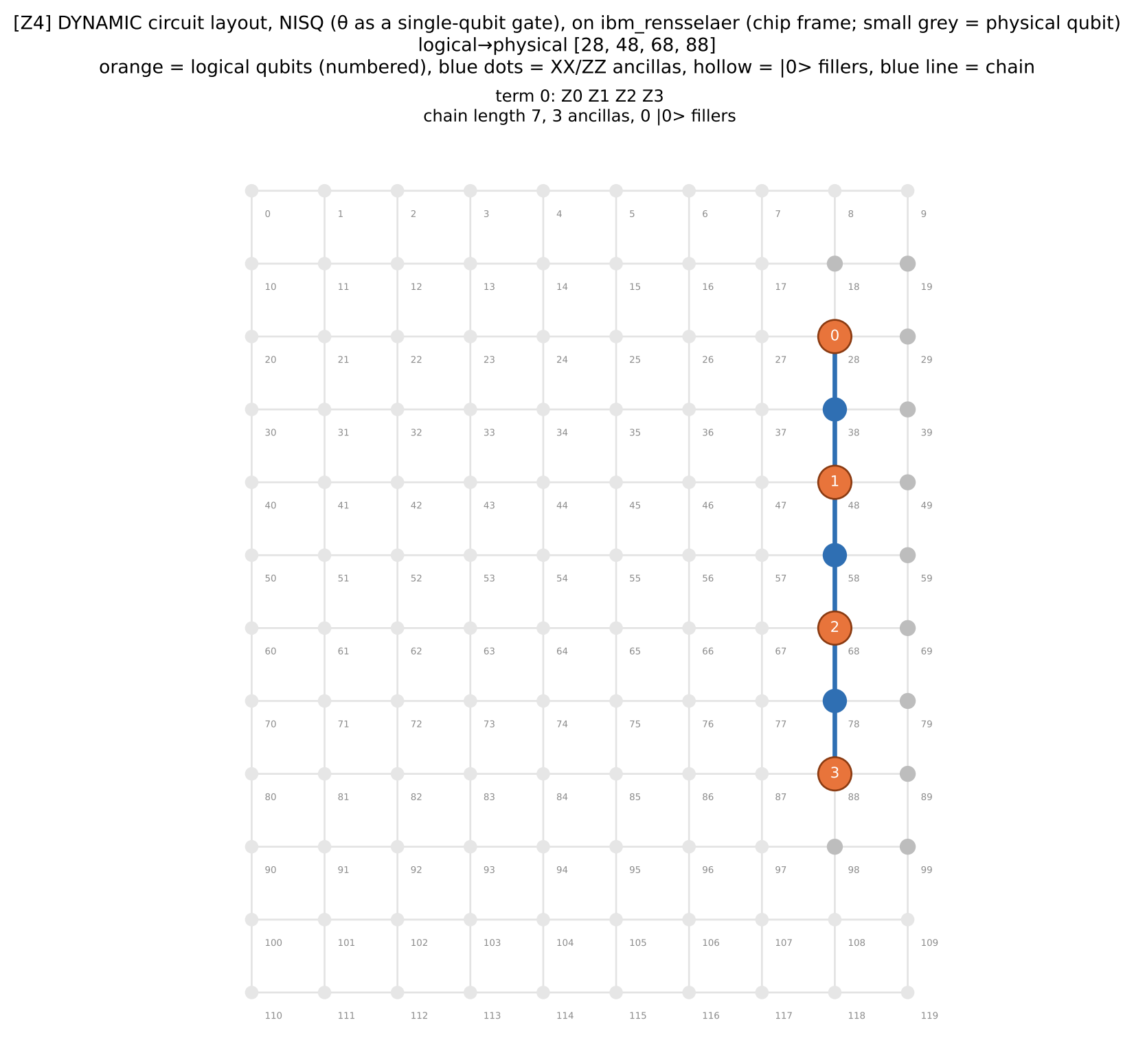


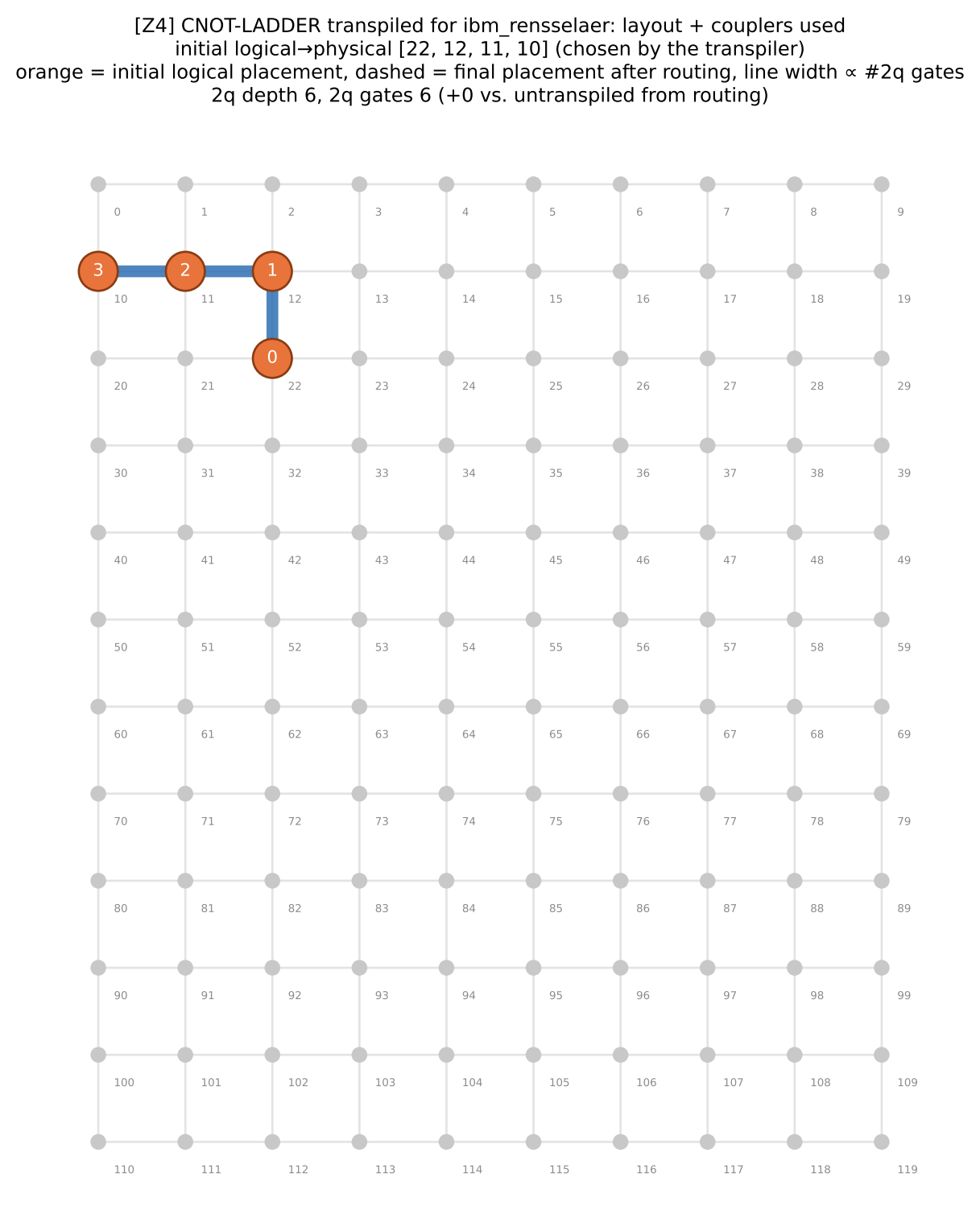
[Z4] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:46 sx:36 cz:18 measure:12 if_else:10 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, total depth 37



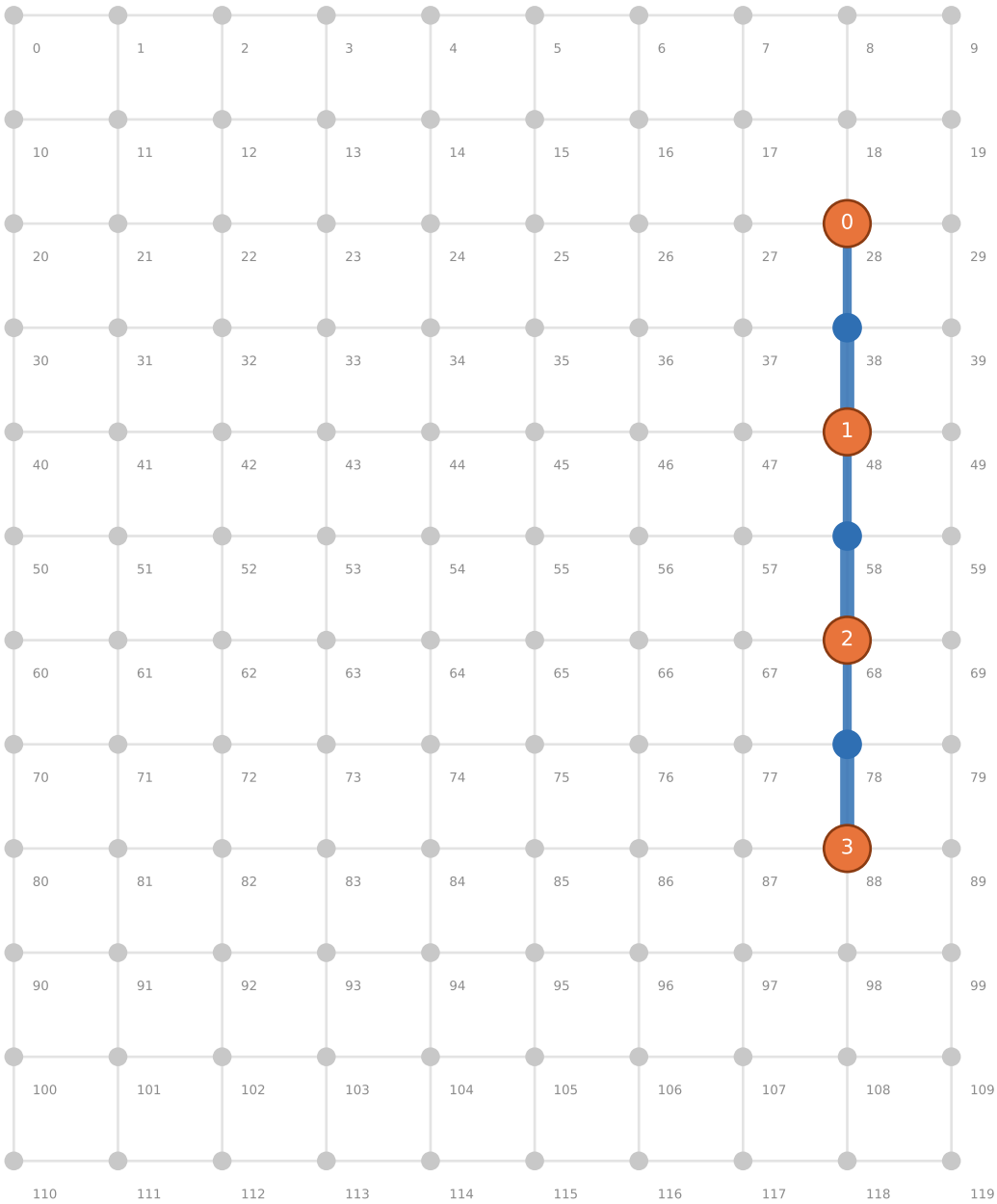
[Z4] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23



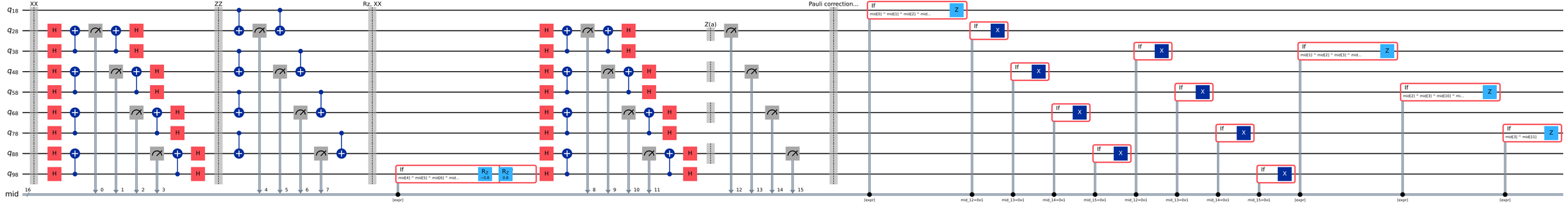

[Z4] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [28, 48, 68, 88]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3


[Z4] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
2
1
0
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

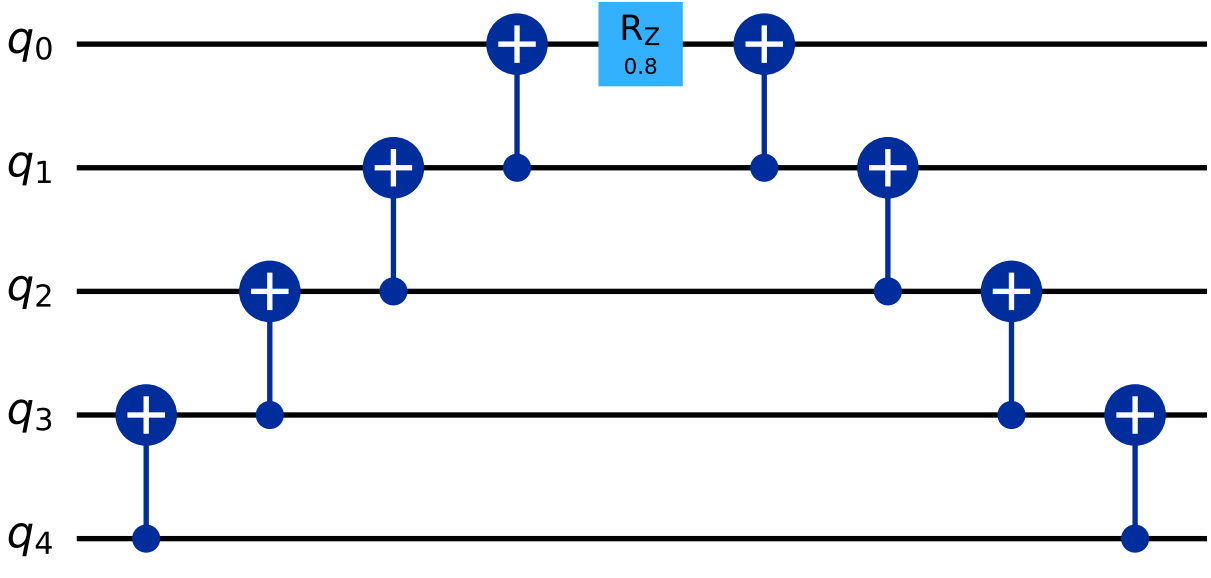
[Z4] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [28, 48, 68, 88]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 18



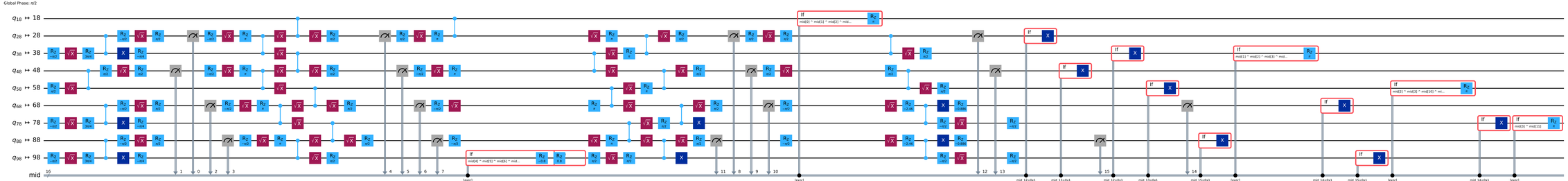
[Z5] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, 16 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



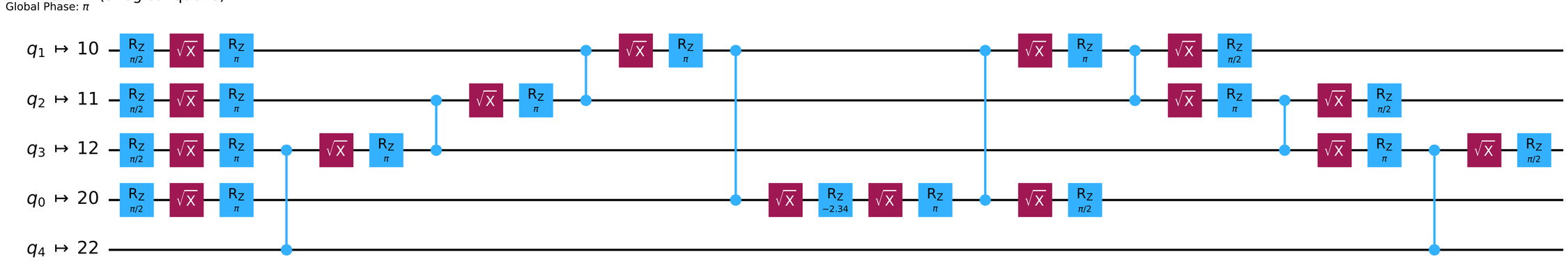
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8



[Z5] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:65 sx:48 cz:24 measure:16 if_else:13 x:7)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, total depth 37

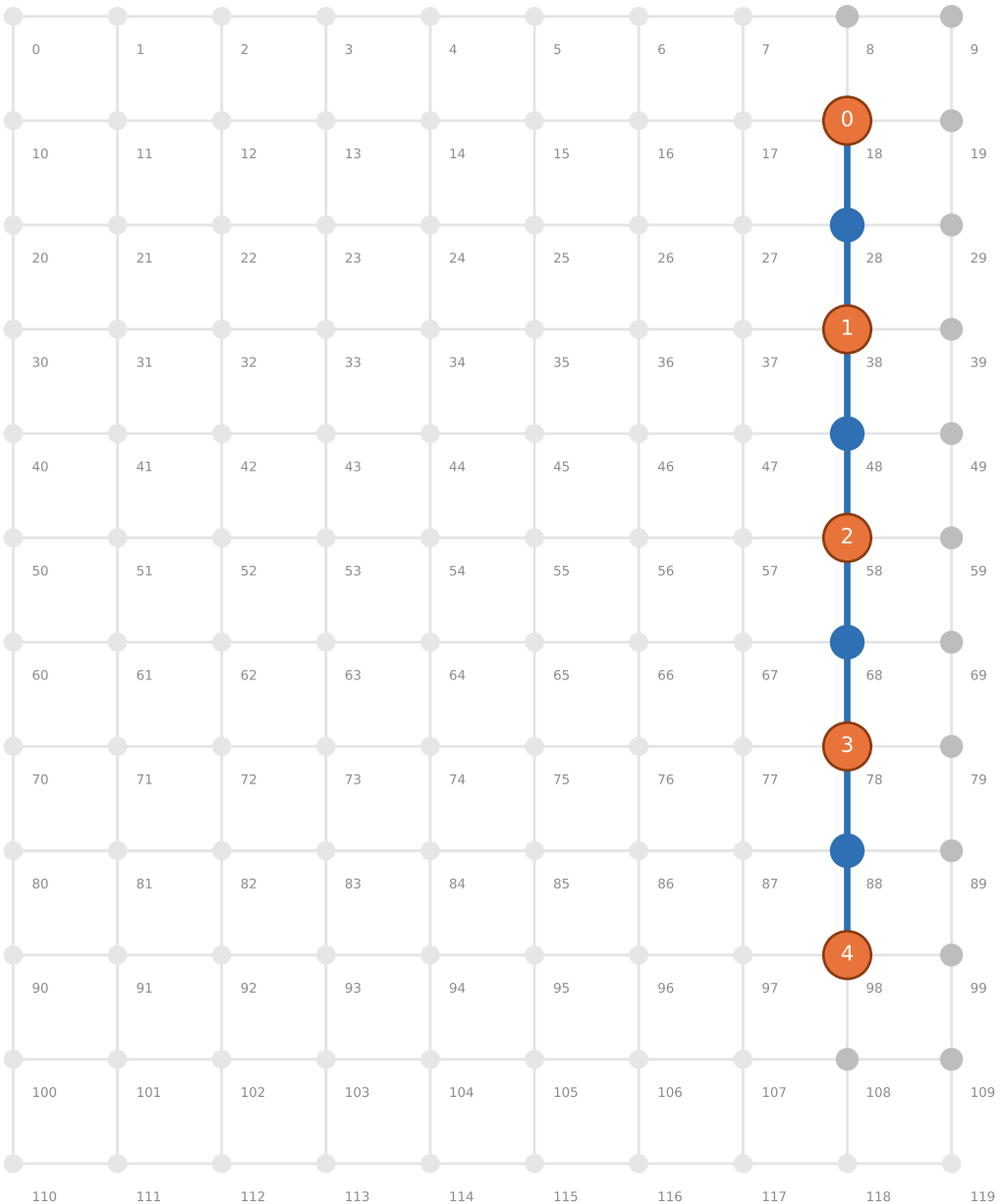


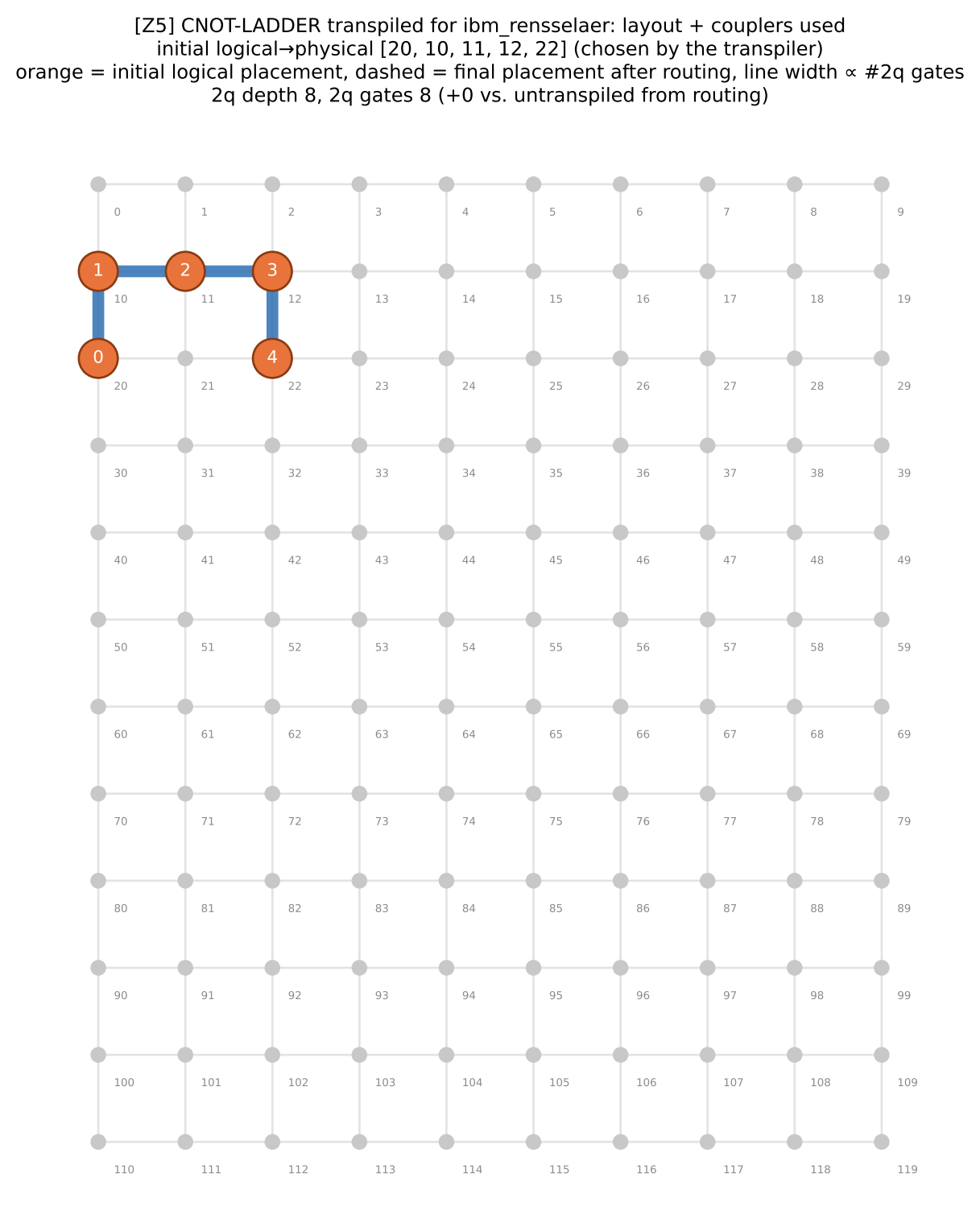
[Z5] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29



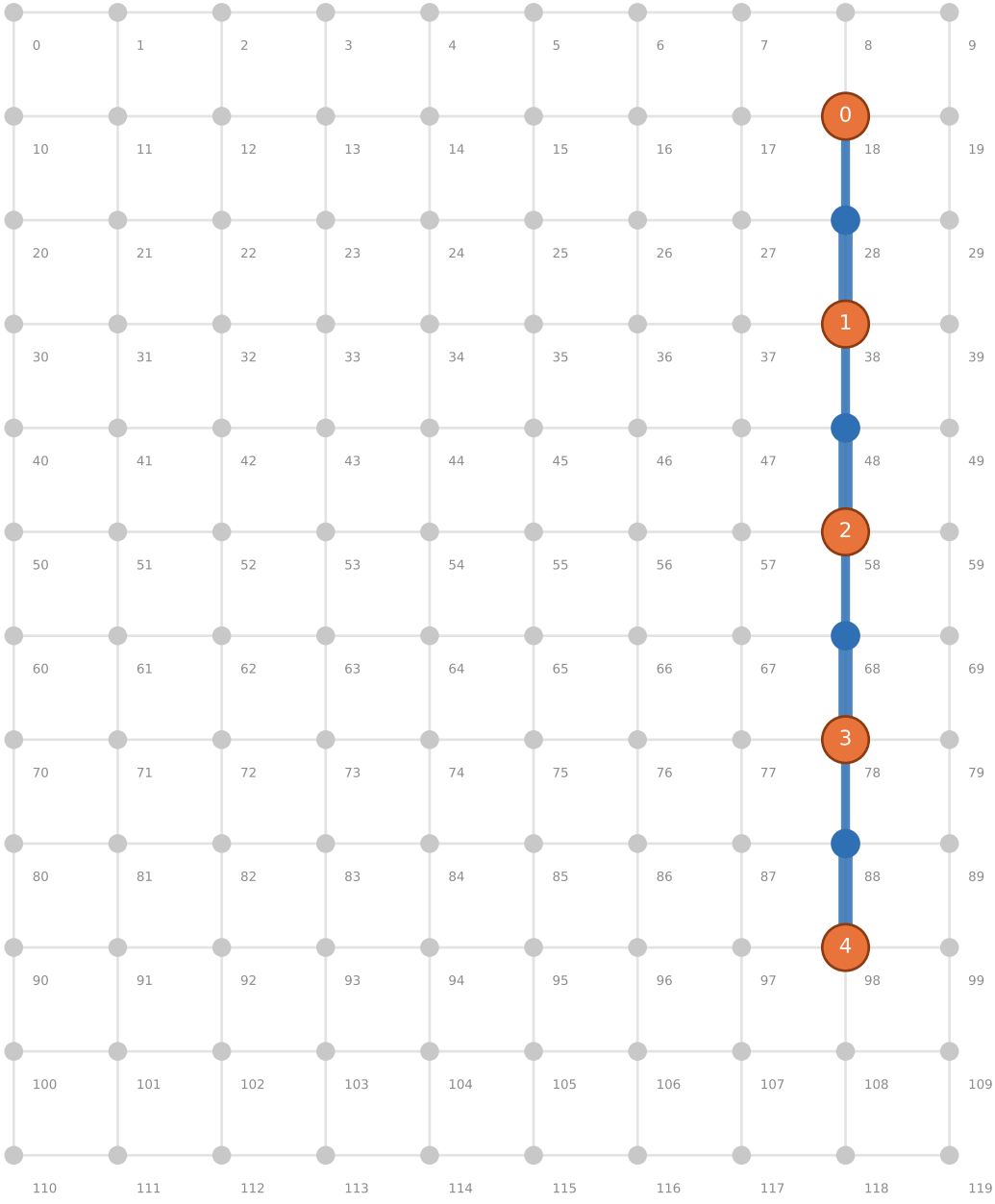
[Z5] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers




[Z5] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
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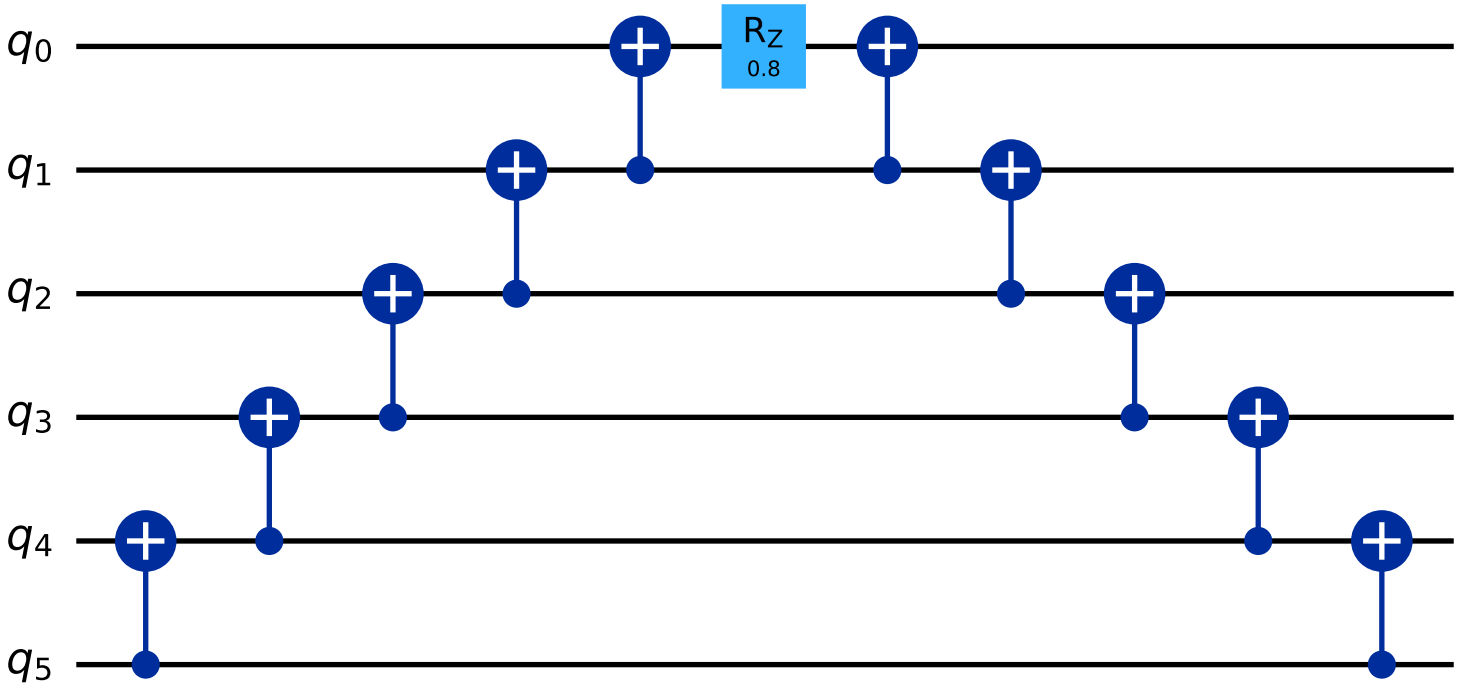
[Z5] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 24



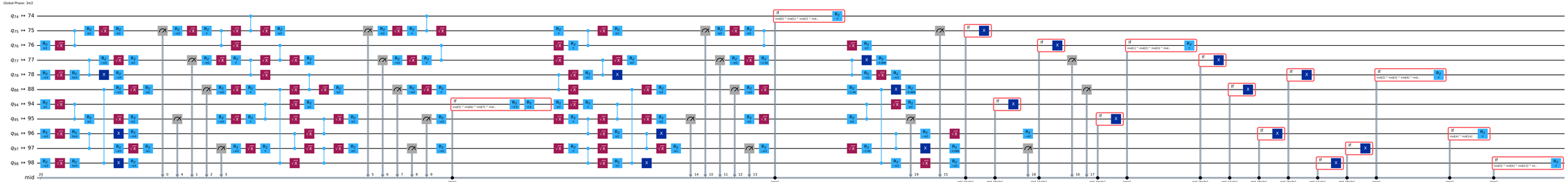
[Z6] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 6, 20 mid-circuit measurements, logical→physical [74, 76, 78, 98, 96, 94]



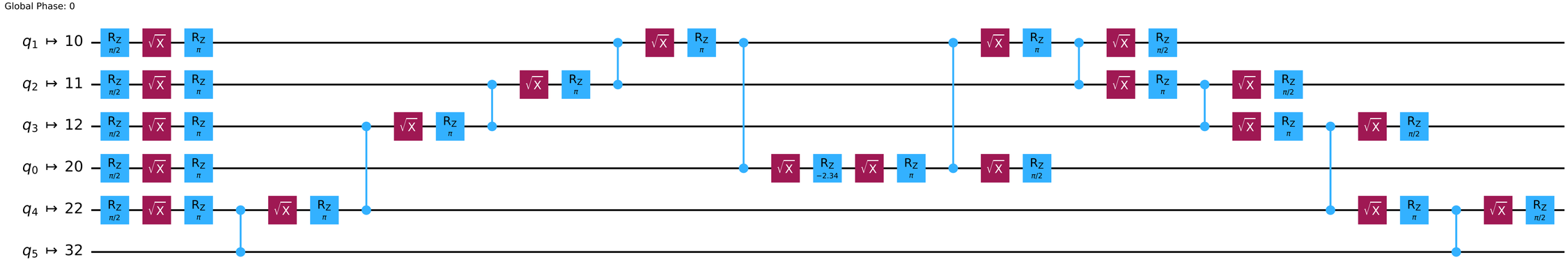
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10



[Z6] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:80 sx:60 cz:30 measure:20 if_else:16 x:9)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 6, total depth 39

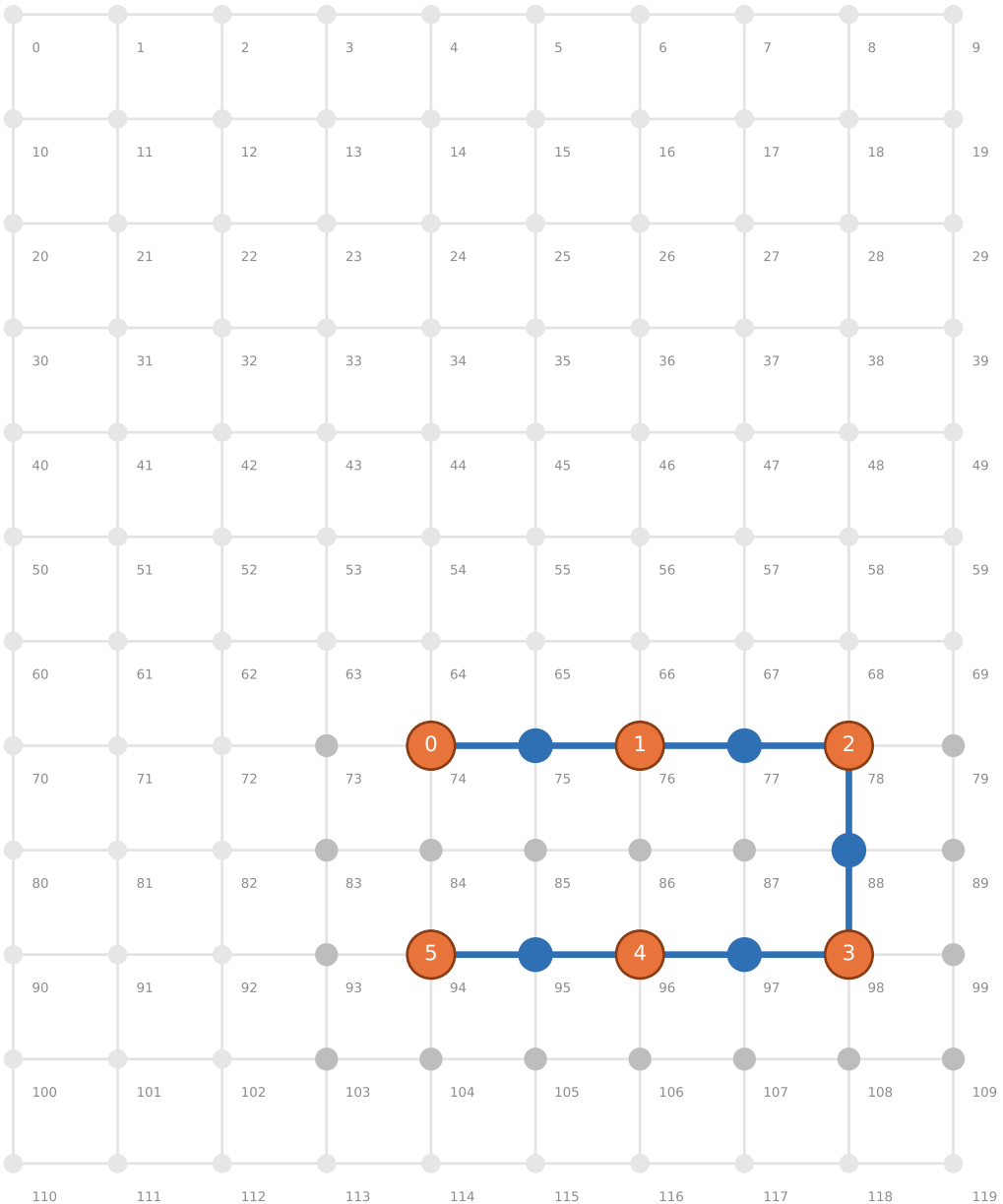


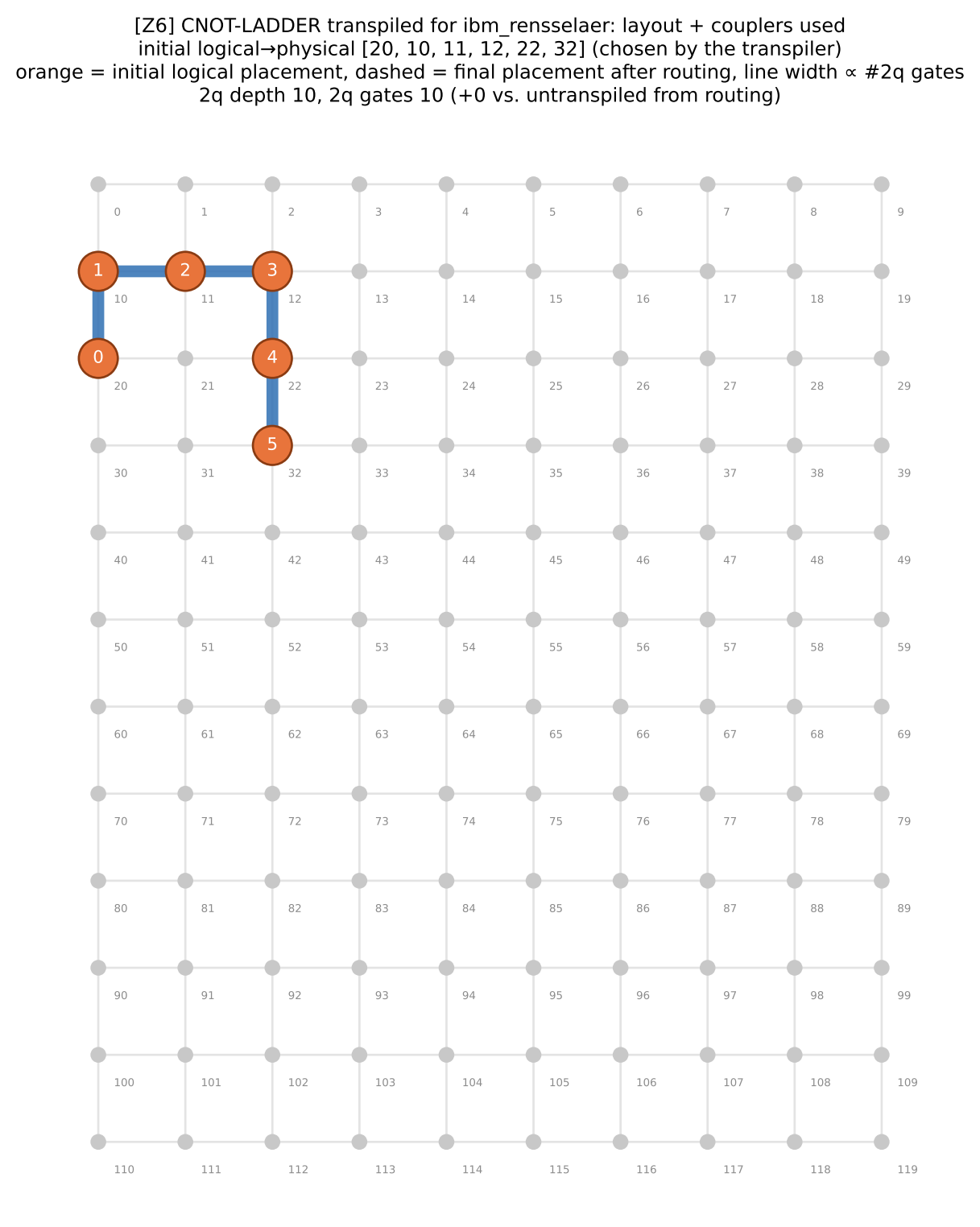
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 10, total depth 35



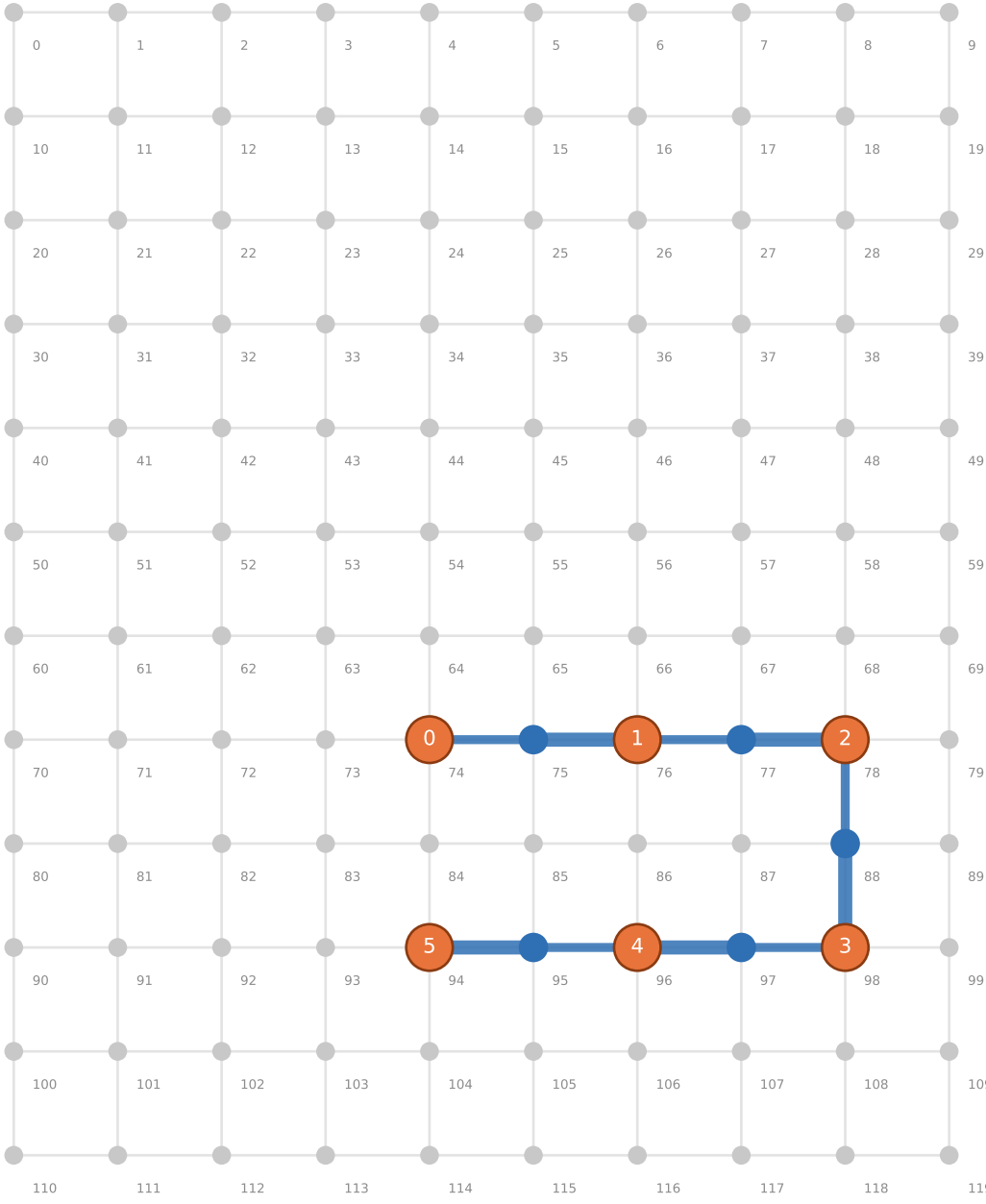
[Z6] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers

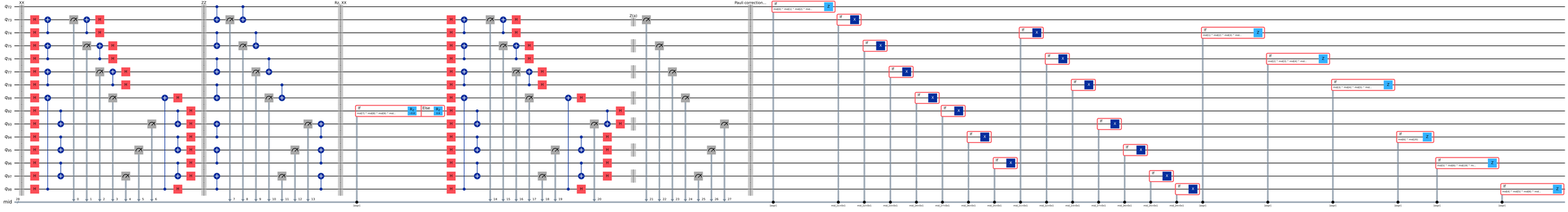



[Z6] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22, 32] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
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87
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91
92
93
94
95
96
97
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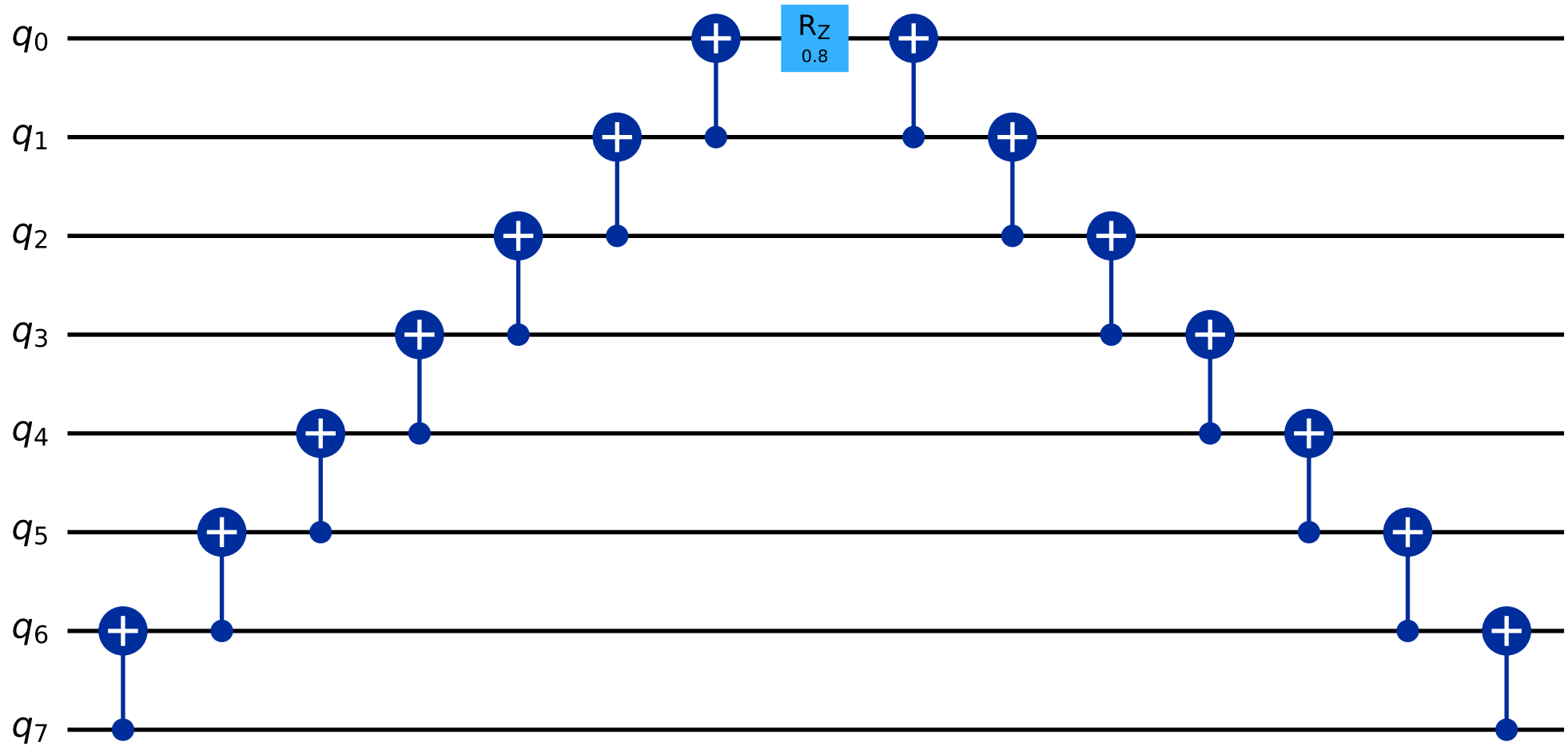
[Z6] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 30



[28] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = ZQ Z1 Z2 Z3 Z4 Z5 Z6 Z7 (8 logical qubits)
2q depth 6, 28 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]



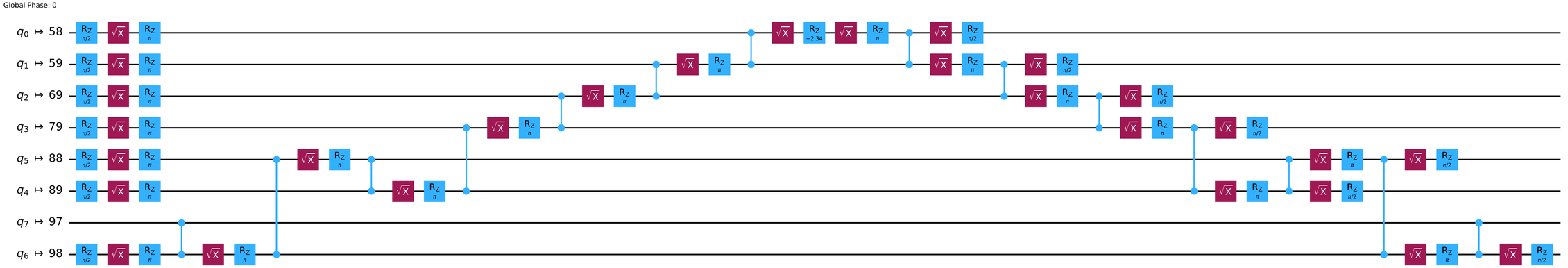
[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14

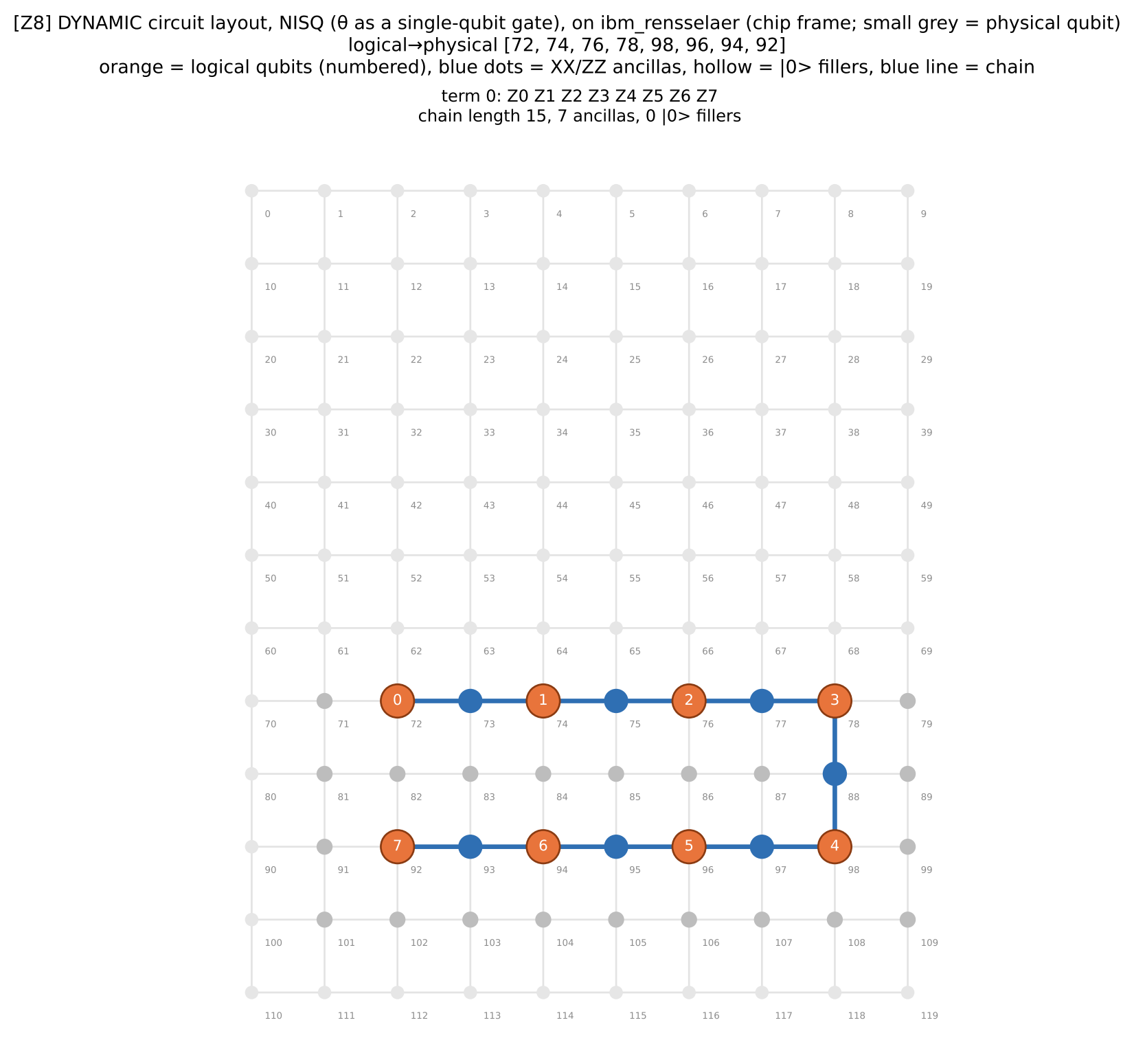


[28] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:110 sx:84 cz:42 measure:28 if_else:22 x:12)
exp(-i θ P_0, θ=0.4: P_0 = ZY Z1 Z2 Z3 Z4 Z5 Z6 Z7 (8 logical-qubits)
2q depth 6, total depth 40

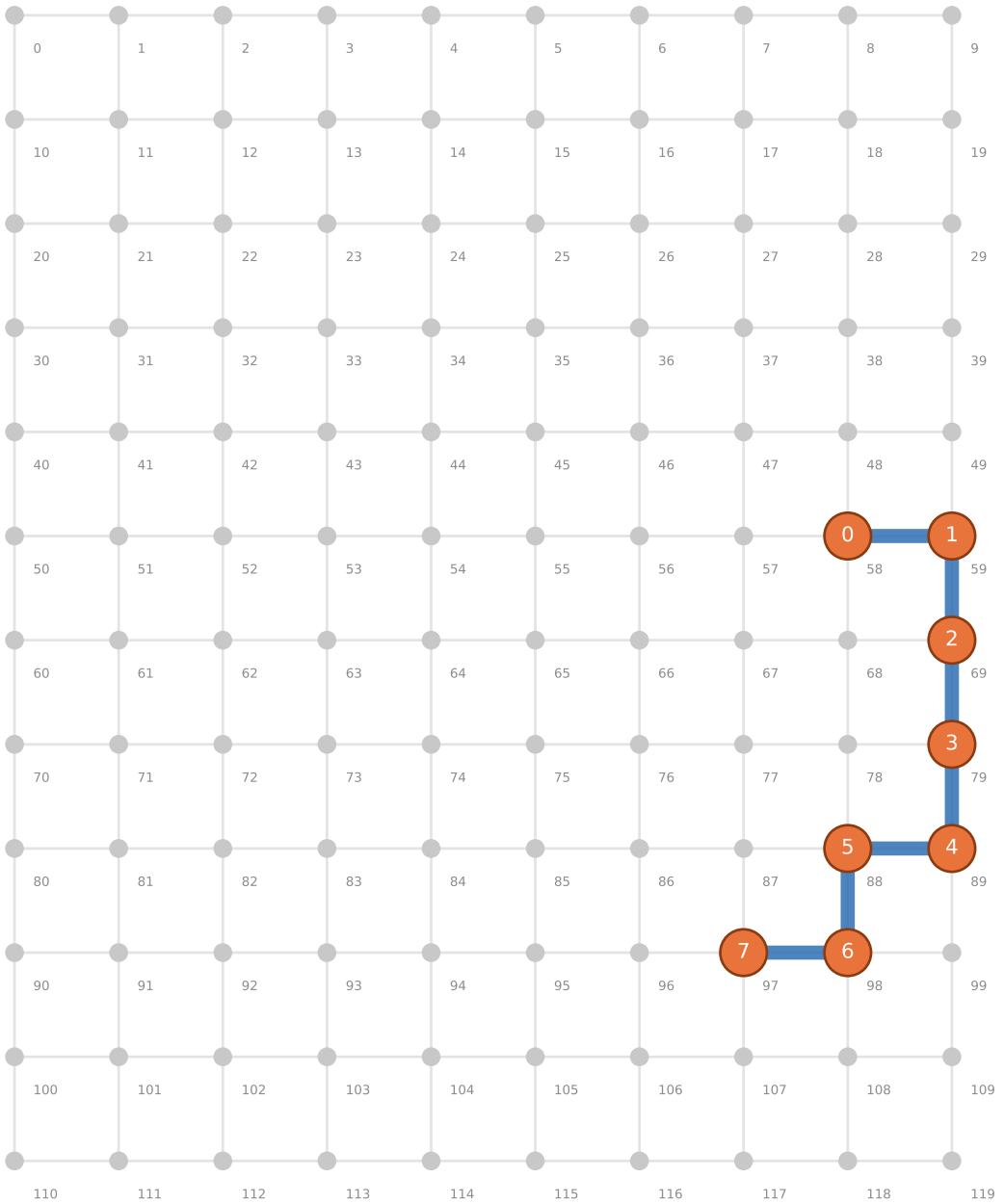


[Z8] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

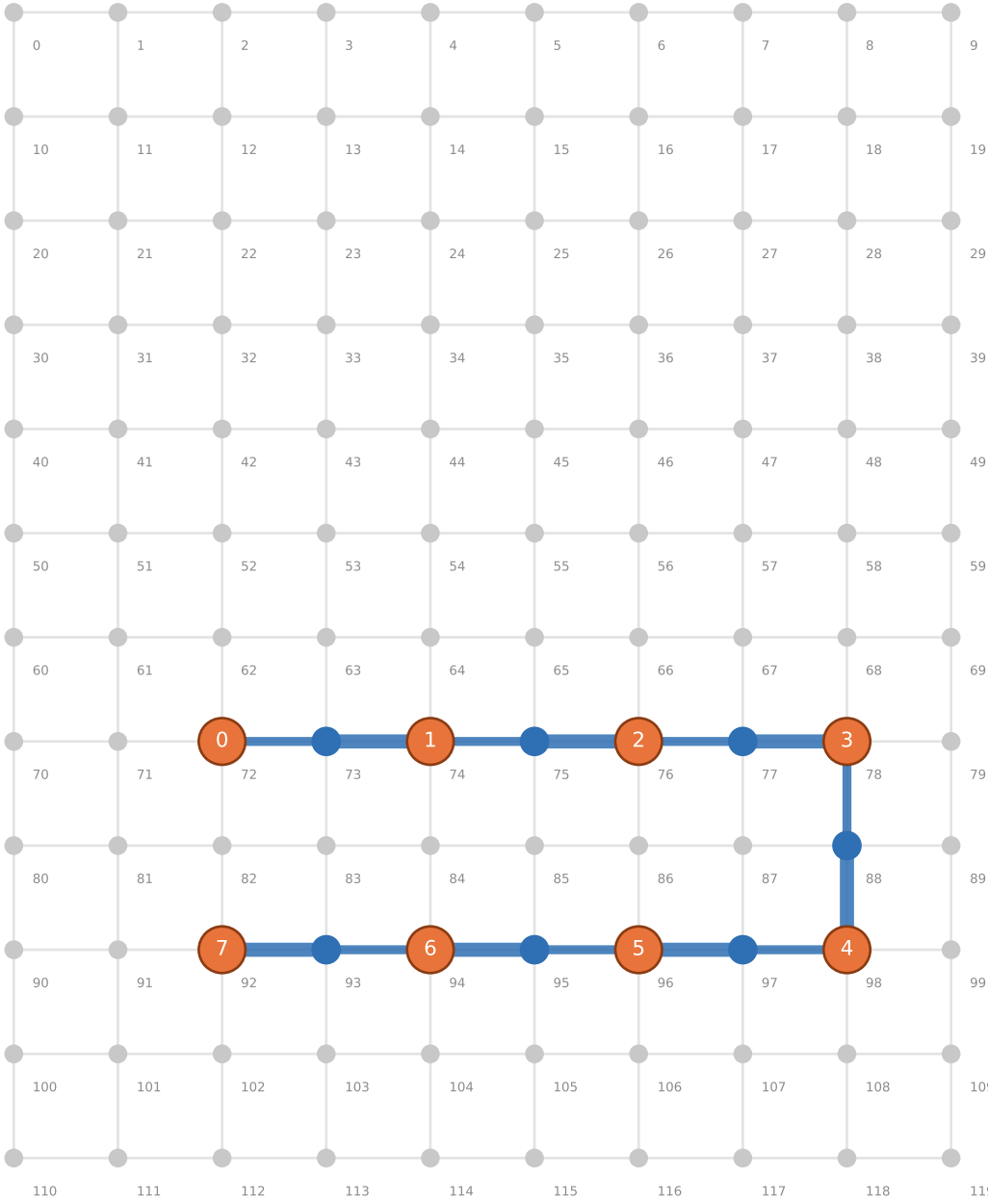



[Z8] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
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4
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6
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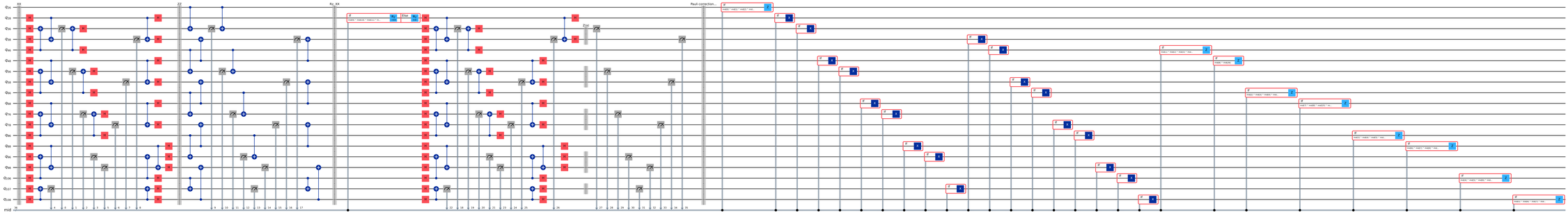
[Z8] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



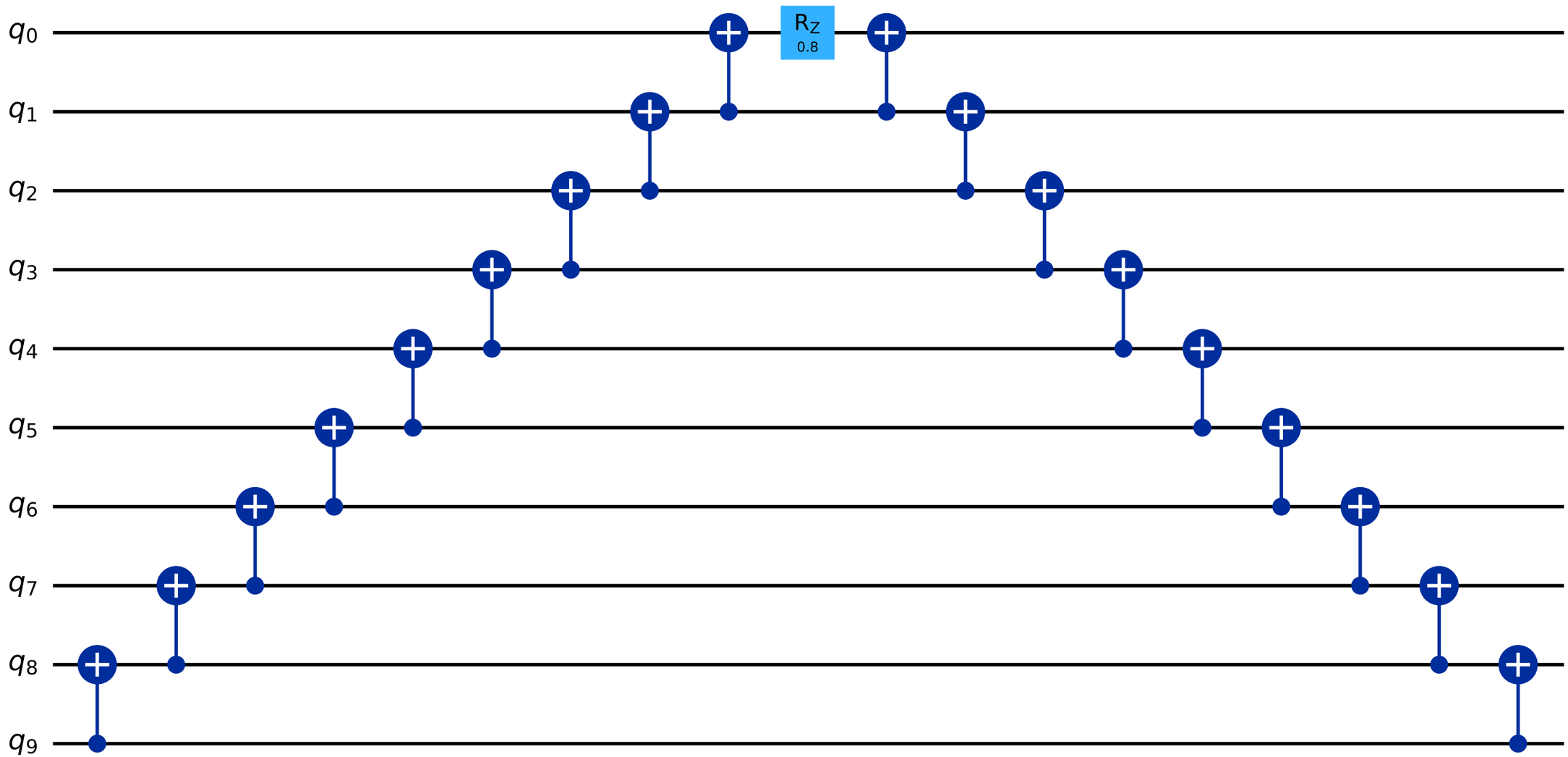
[Z8] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 42



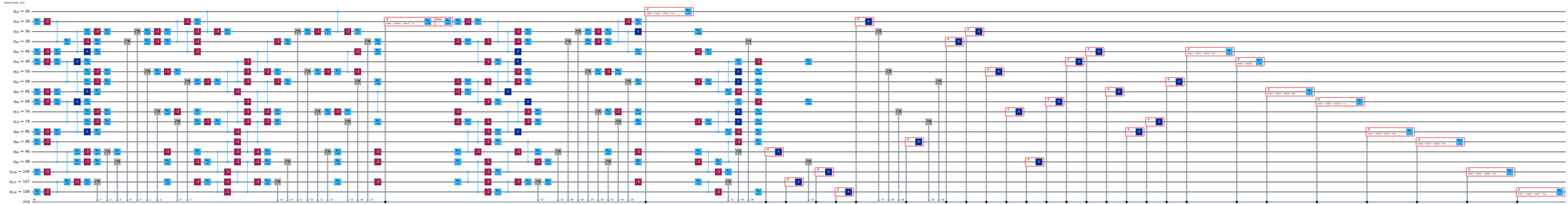
[2210] DYNAMIC constant-depth circuit, N(SQ:9) as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = 2D Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 6, 36 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]



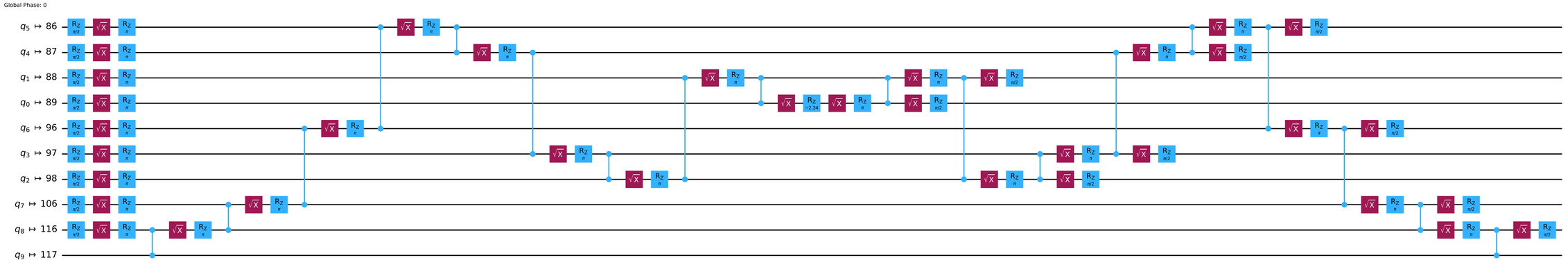
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18



[210] DYNAMIC circuit, N050 (9 as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:140 sx:108 cz:54 measure:36 if_else:28 x:15)
expct-9 Pi, th=0.4, P => 20 21 22 23 24 25 26 27 28 29 · (10 logical qubits)
2q depth 6, total depth 44

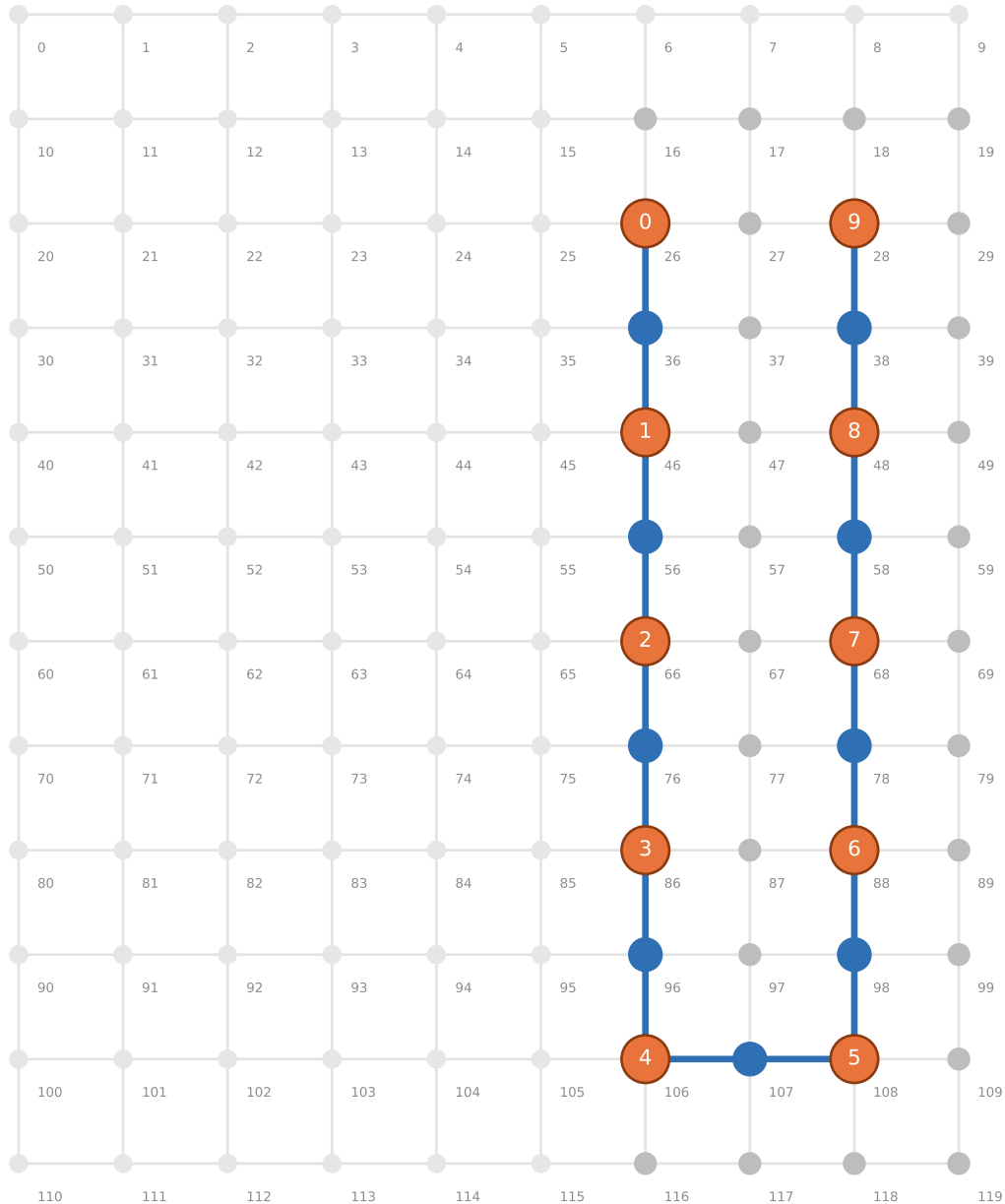


[Z10] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18, total depth 59

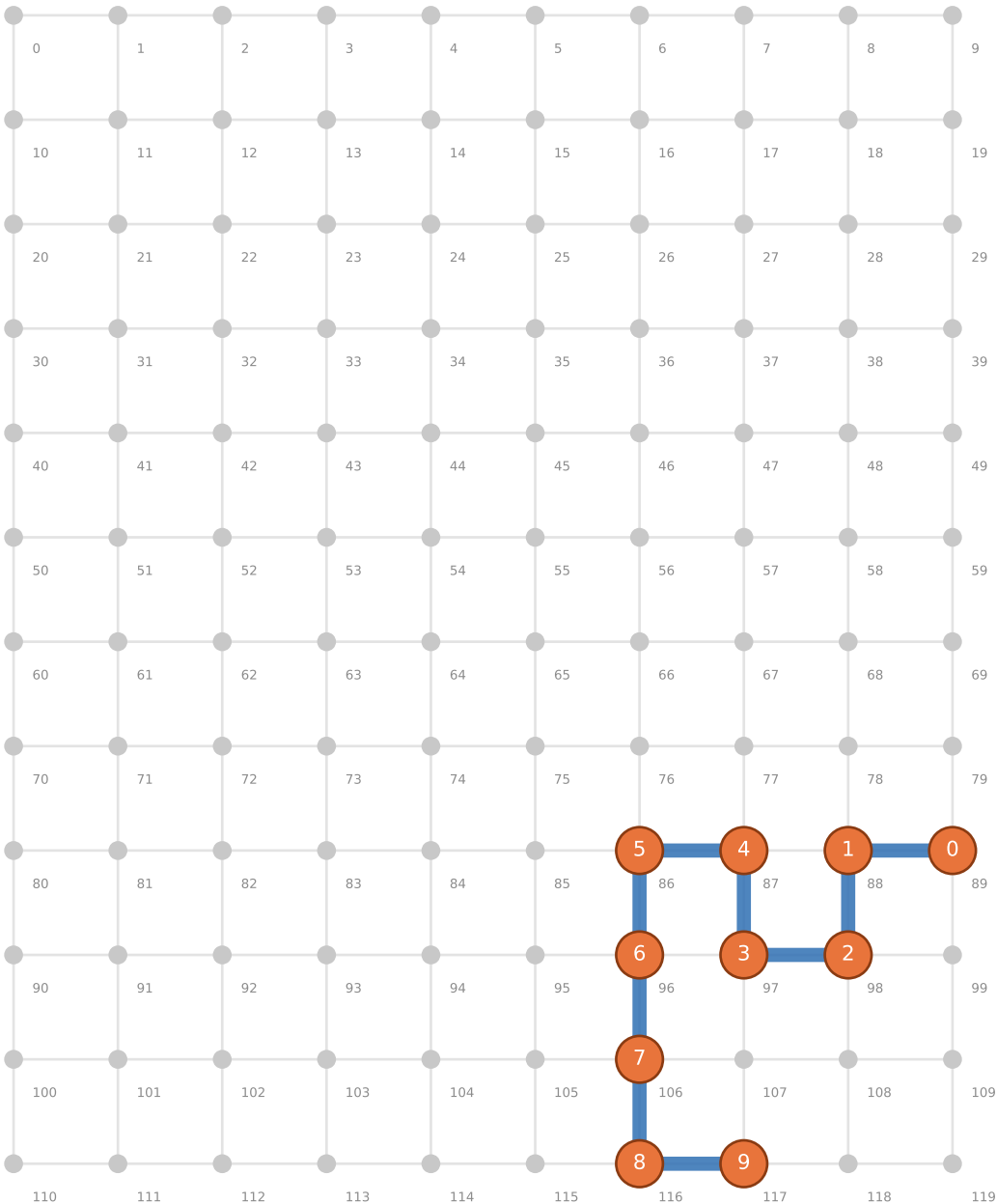


[Z10] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

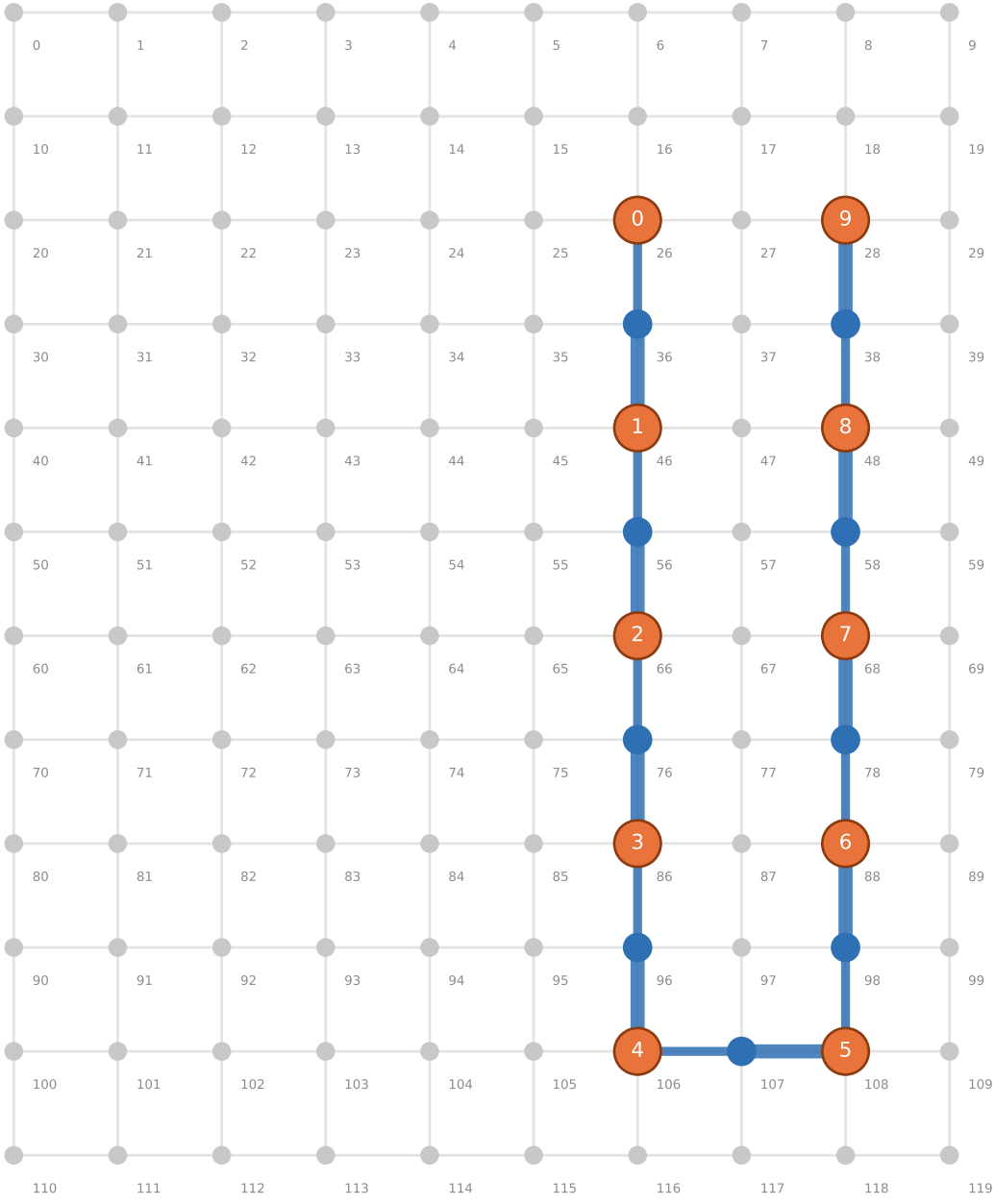
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers



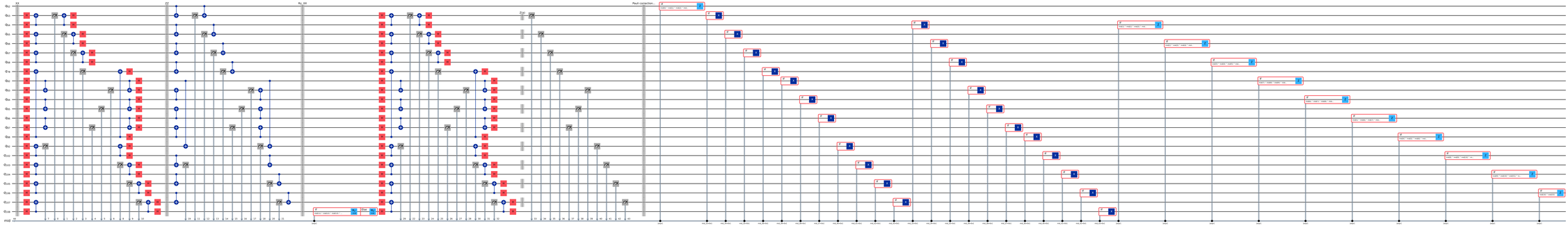
[Z10] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)



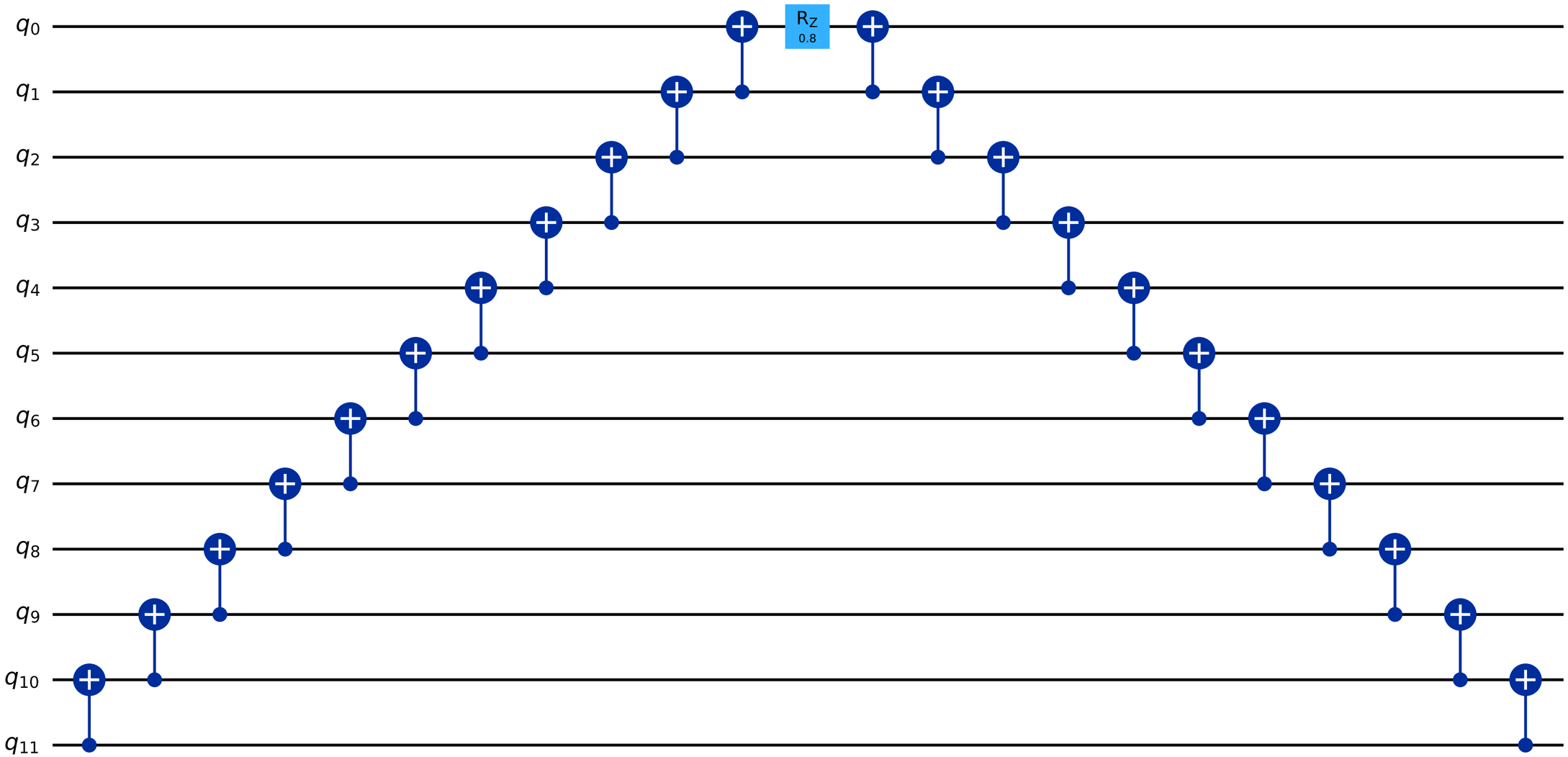
[Z10] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 54



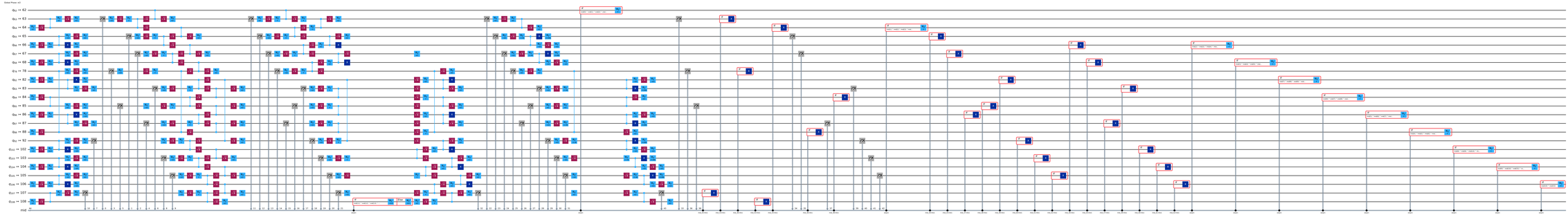
[213] DYNAMIC constant-depth circuit, NISQ IR as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
expr(-9.9), 6×0.4: 9 → 29 21 22 23 24 25 26 27 28 29 210 213 . (11 logical qubits)
2q depth 8, 44 mid-circuit measurements, logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]



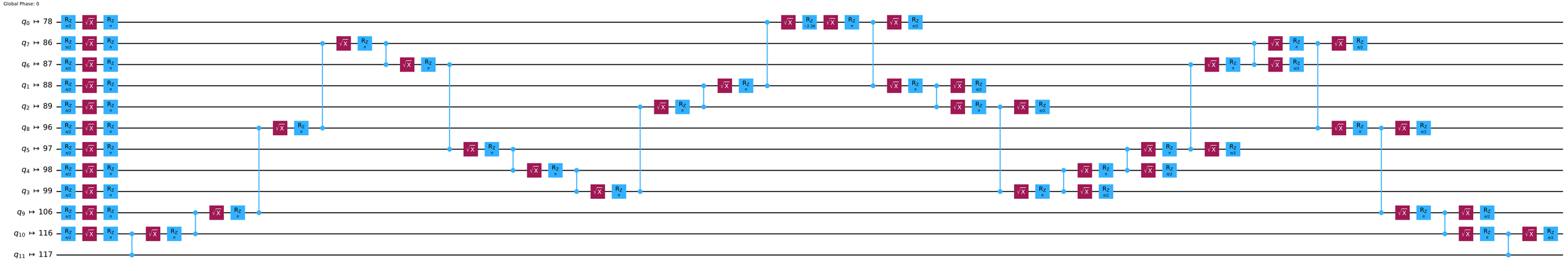
[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22



[Z1] DYNAMIC circuit, MSQ (8 as a single qubit gate), TRANSPILED for ibm_nmsoteler (ISA: rz:174 sx:132 cz:66 measure:44 if_else:34 x:21) expr: 8·F(8·cX·F · P·× 21 22 23 24 25 26 27 28 29 210 211 · I)2 logical qubits: 22 depth: 6, total depth: 45

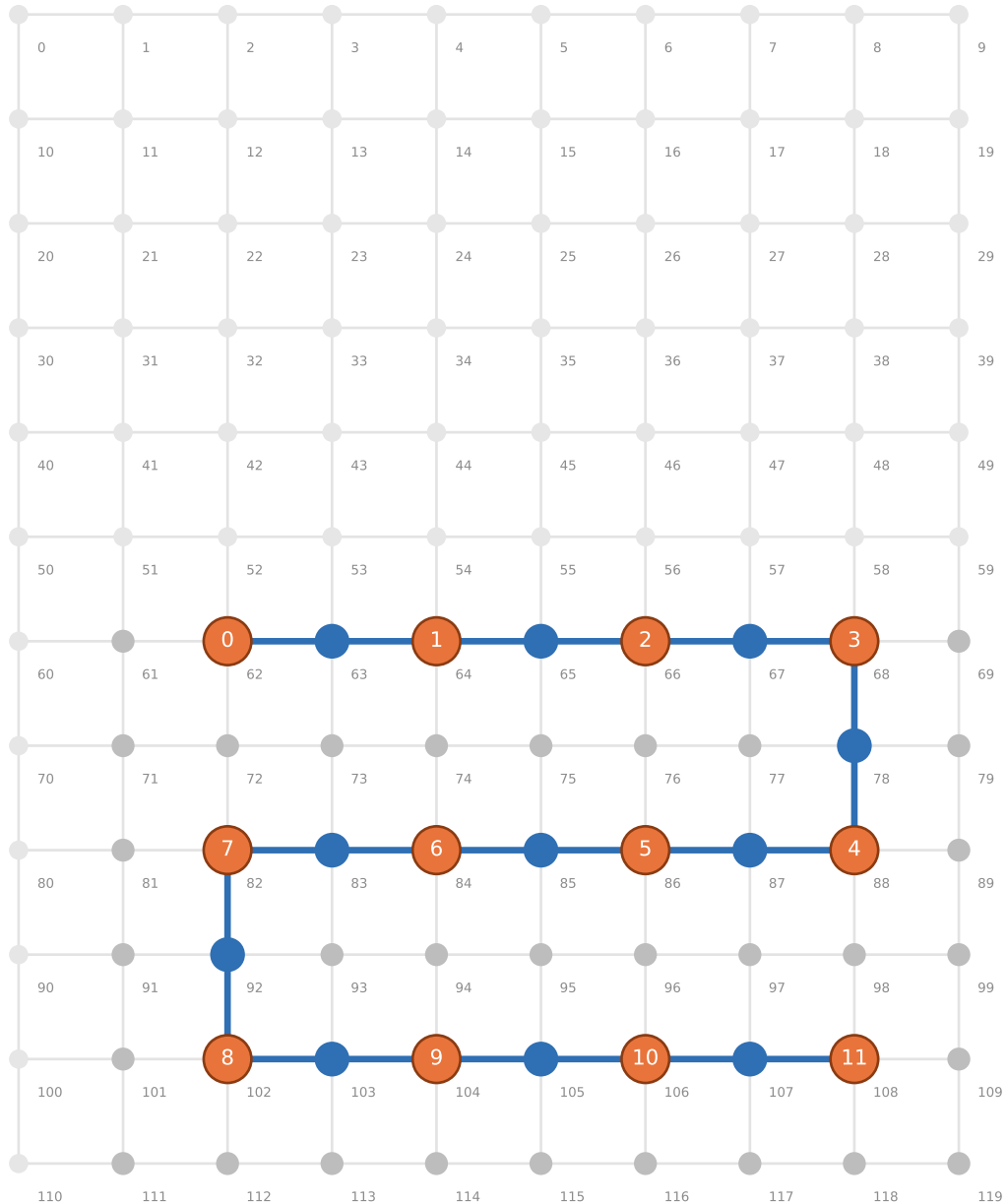


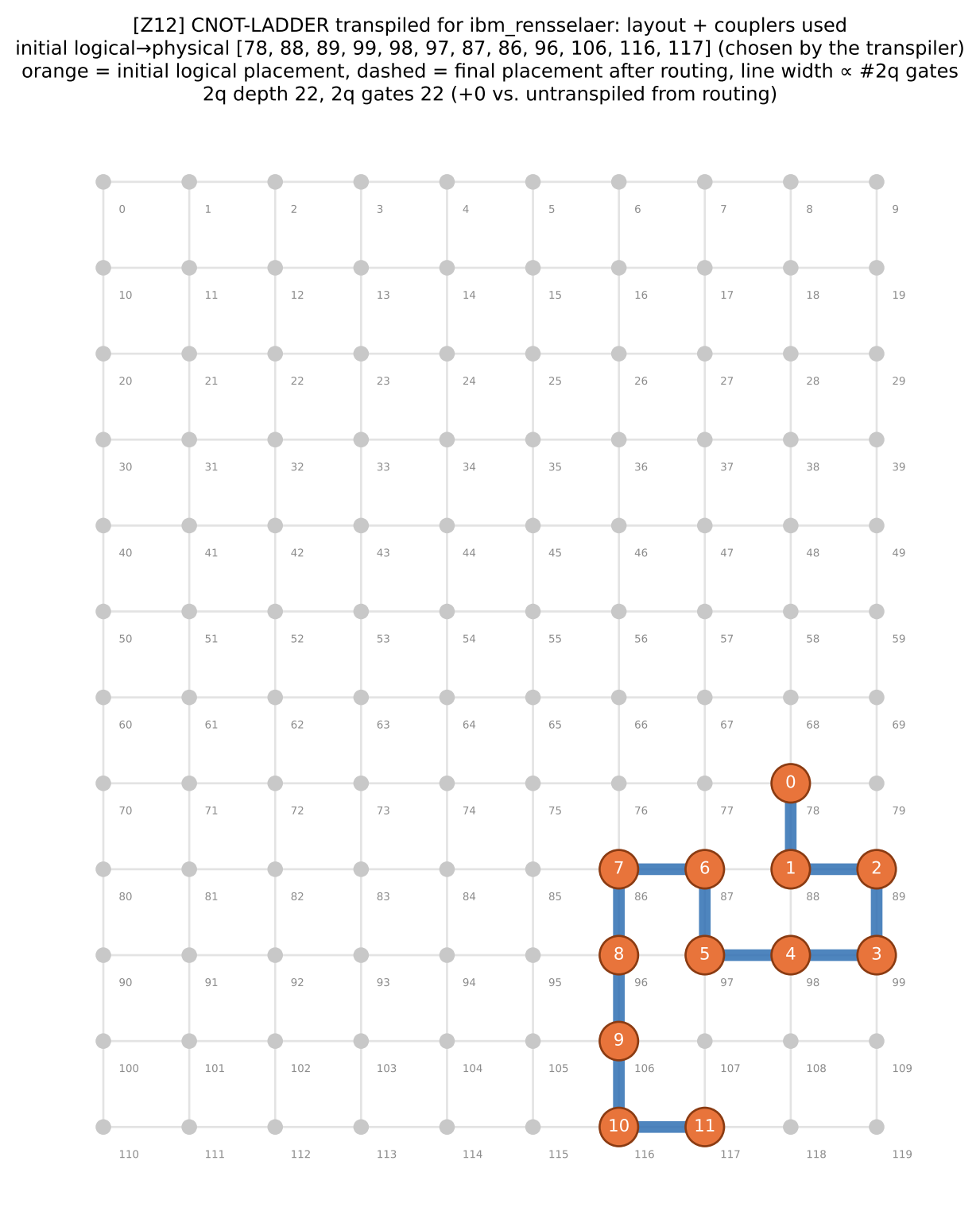
[Z12] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 22, total depth 71



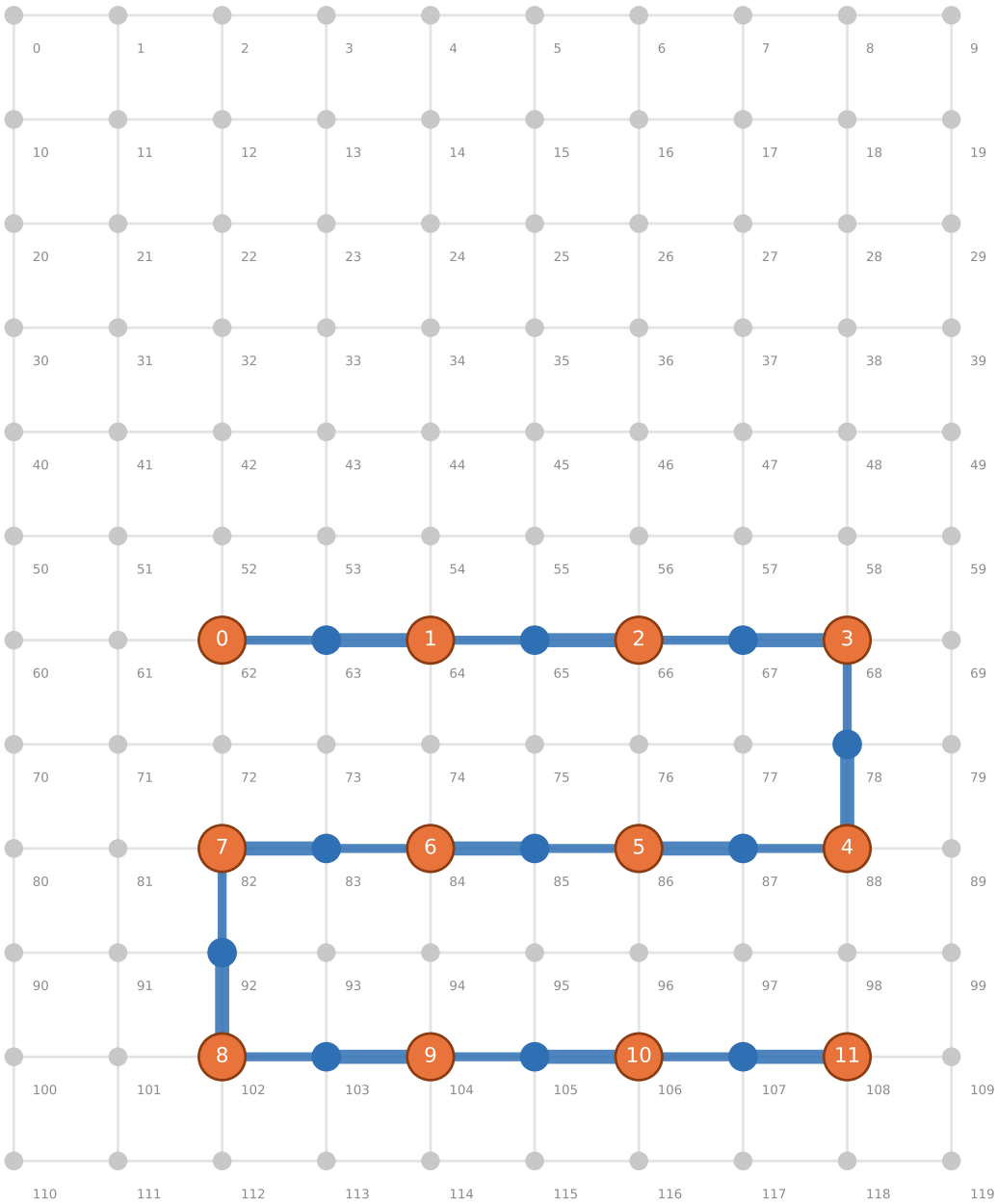
[Z12] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers

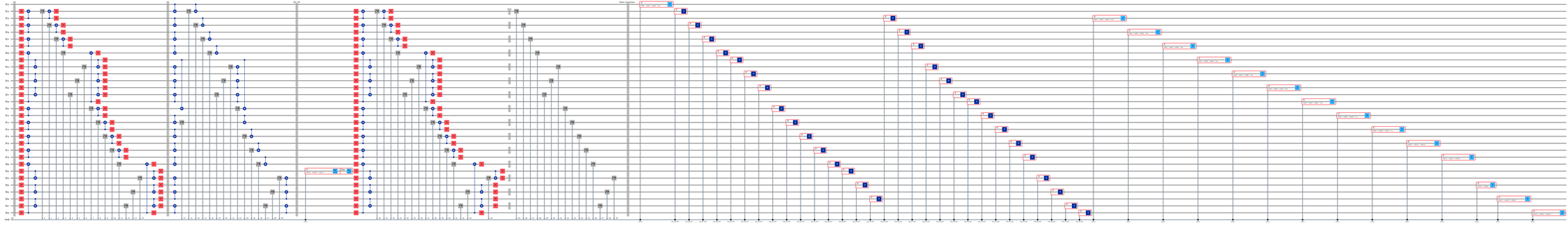



[Z12] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [78, 88, 89, 99, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
7 6 1 2
8 5 4 3
9
10 11

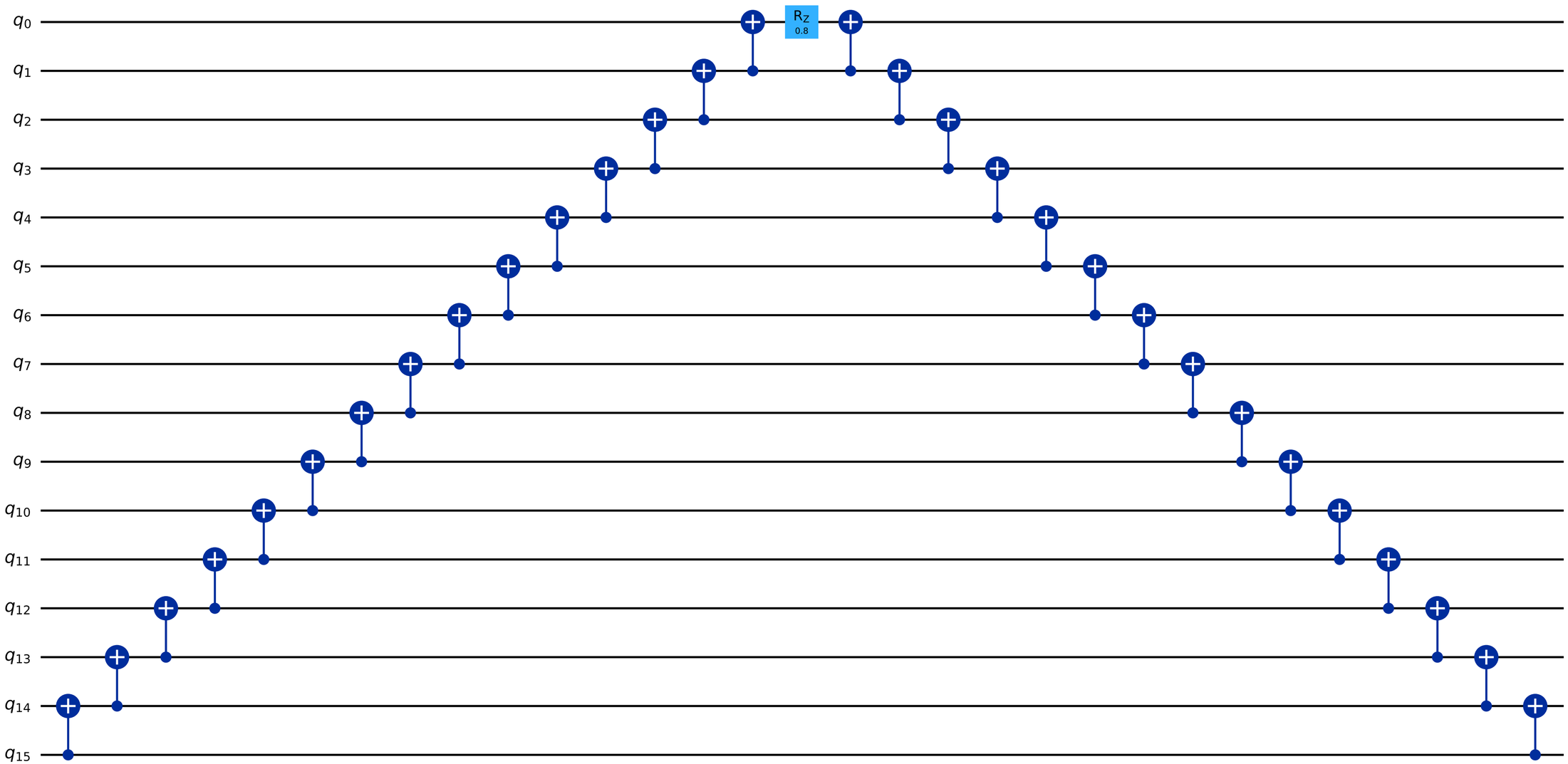
[Z12] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 66



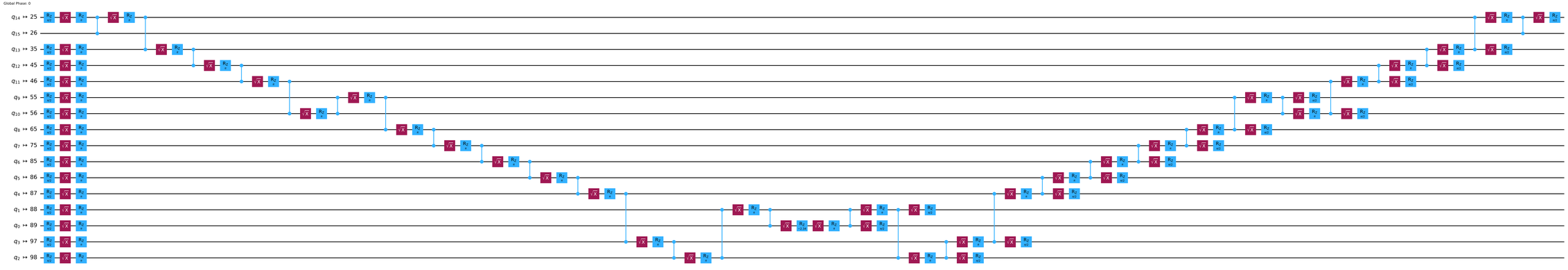
[2/16] DRAWING physical-depth-circuit, NISQ-III on a single-qubit-gate (unconnected, physical qubits of 64+, simulator)
[3/16] DRAWING 64-bit-full-circuit-measurements, NISQ-III on a single-qubit-gate (unconnected, physical qubits of 64+, simulator)
[4/16] 64-bit-full-circuit-measurements, logical-physical-[0, 54, 56, 58, 59, 60, 61, 62, 63, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82]



[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30

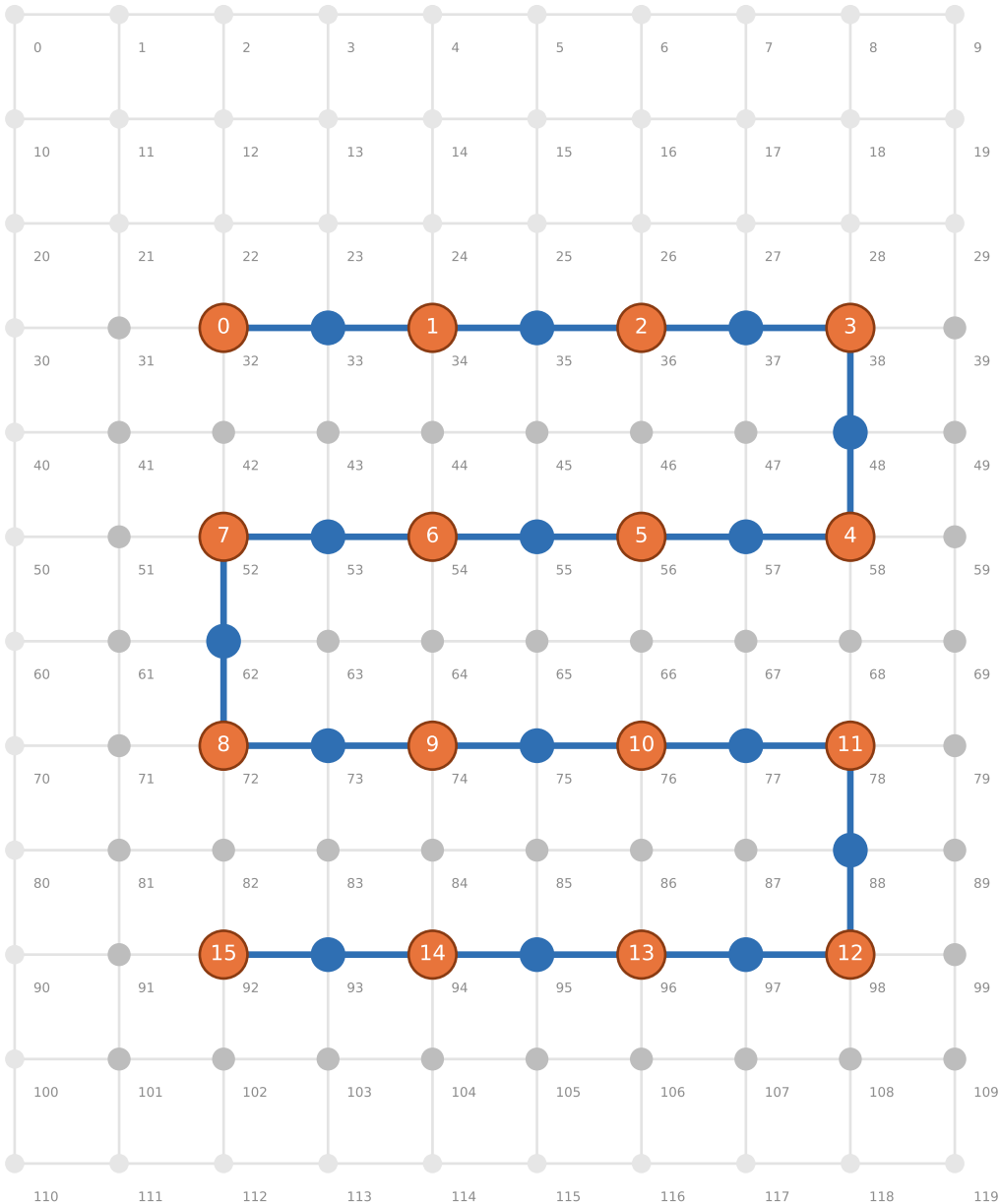


[Z16] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

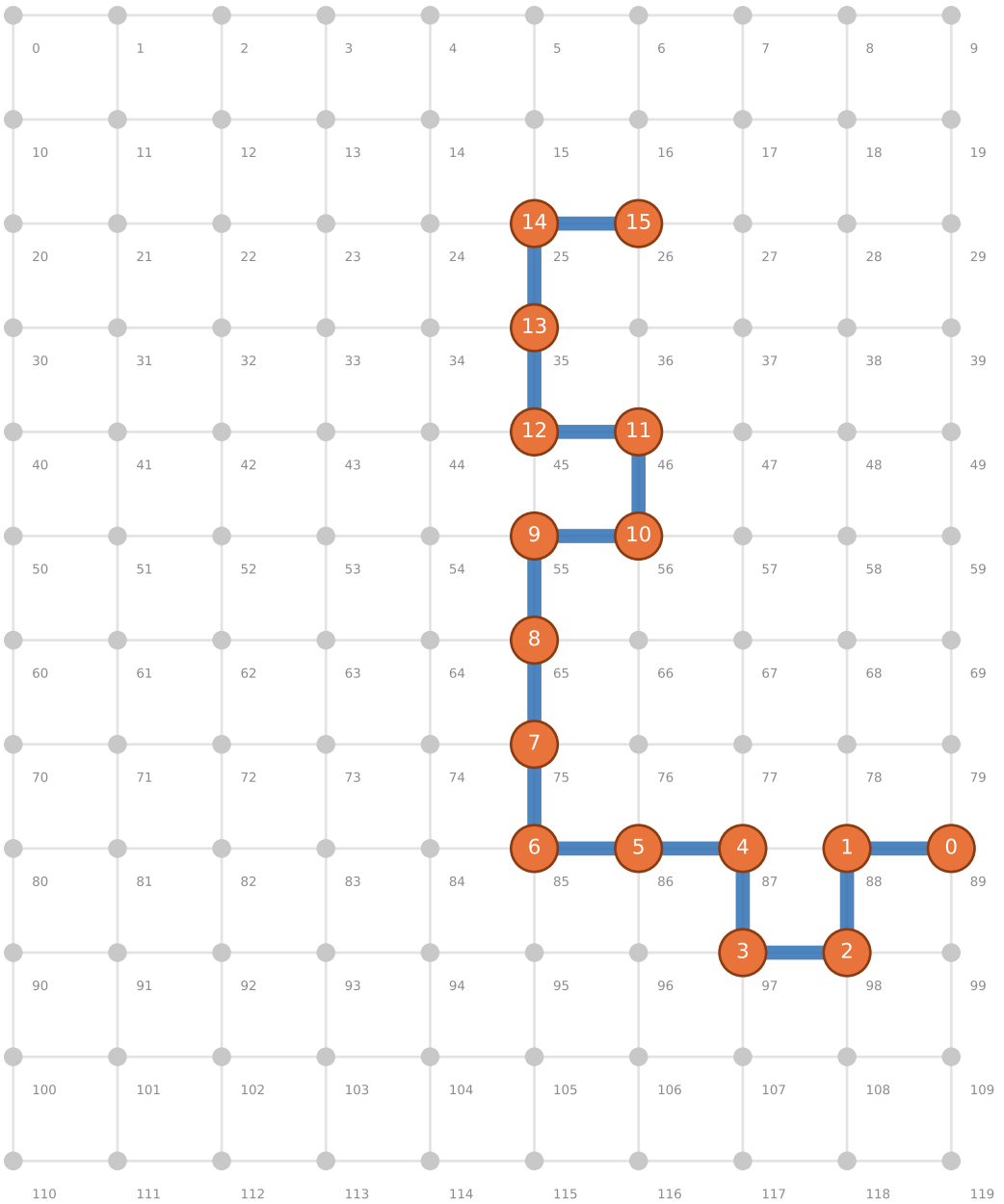


[Z16] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

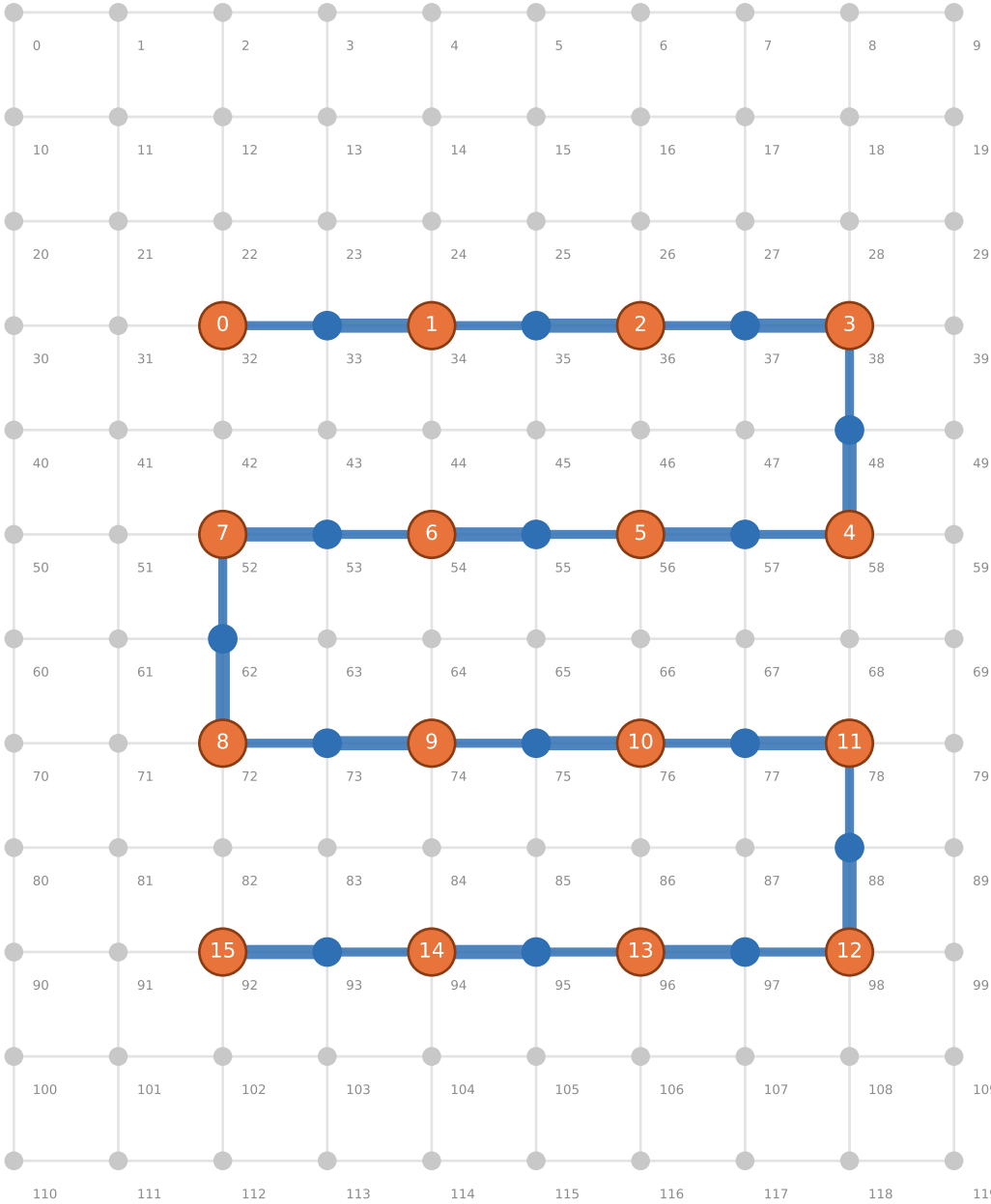
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



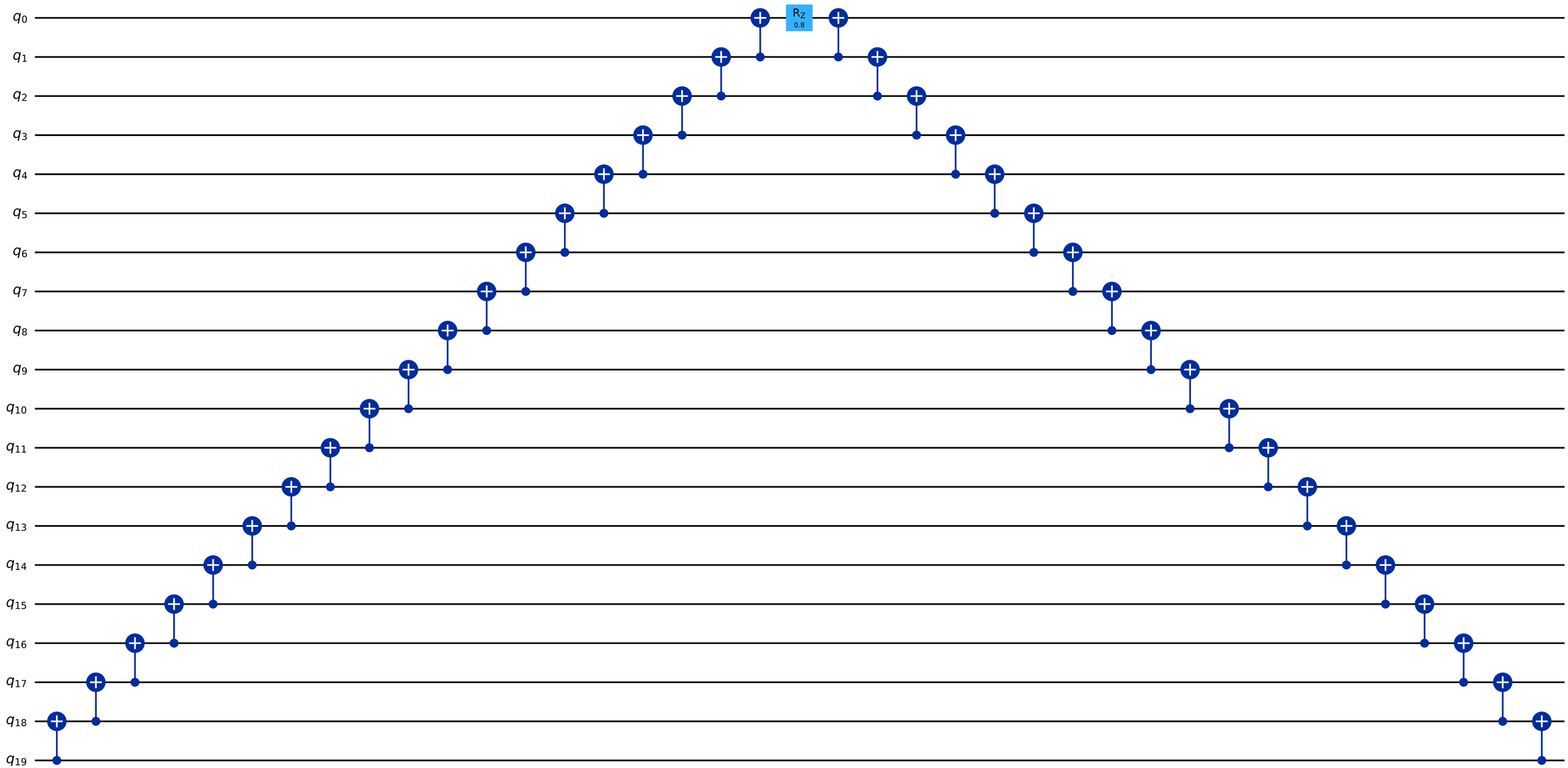
[Z16] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 85, 75, 65, 55, 56, 46, 45, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)



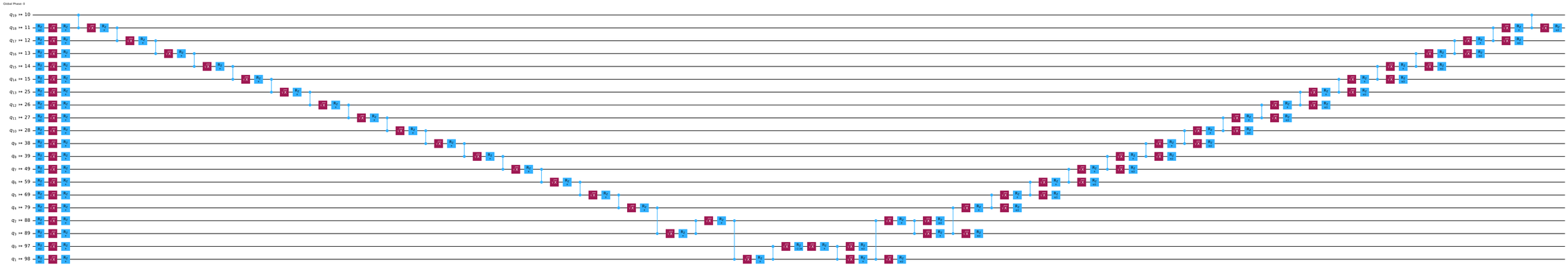
[Z16] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 90



[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

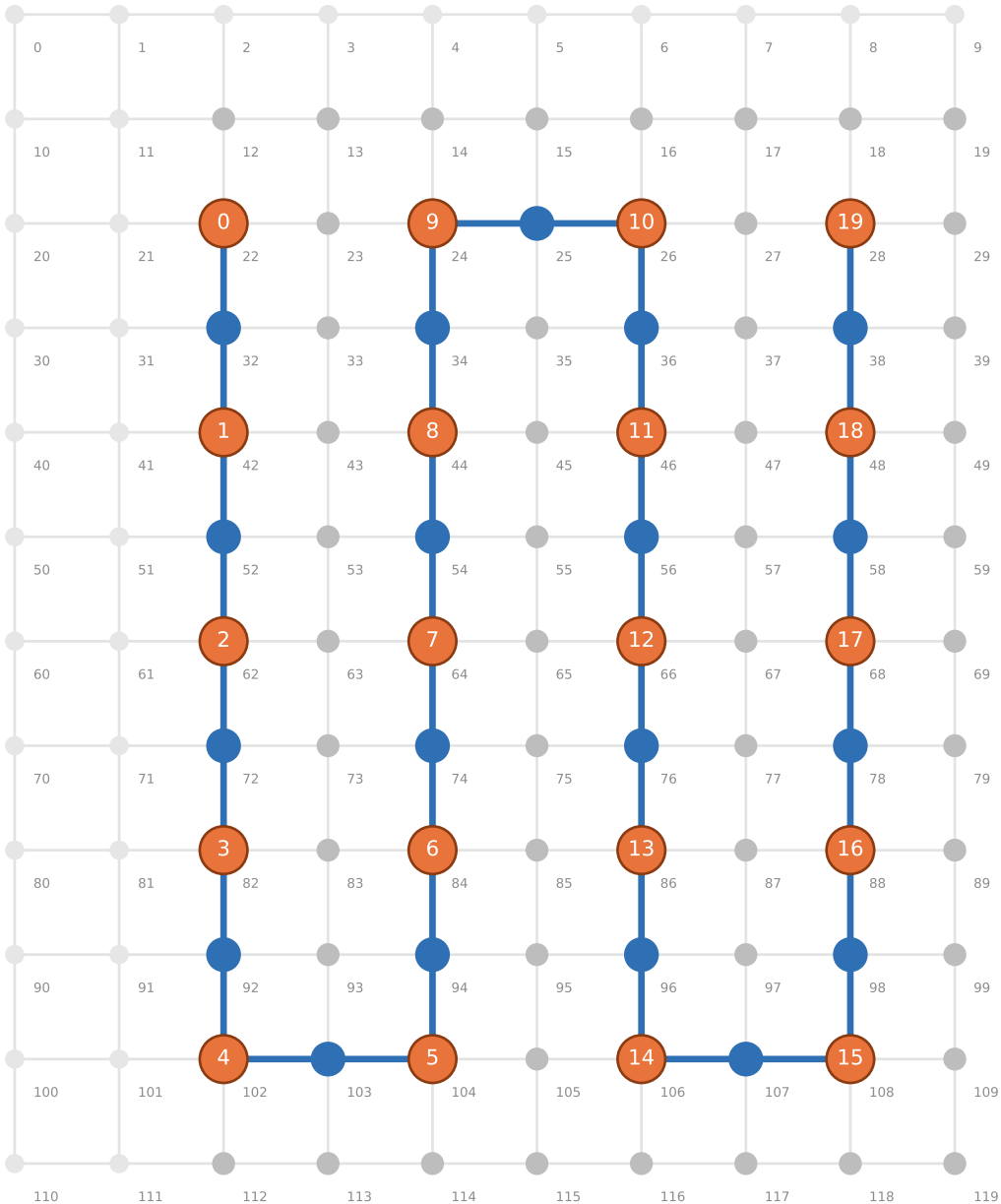


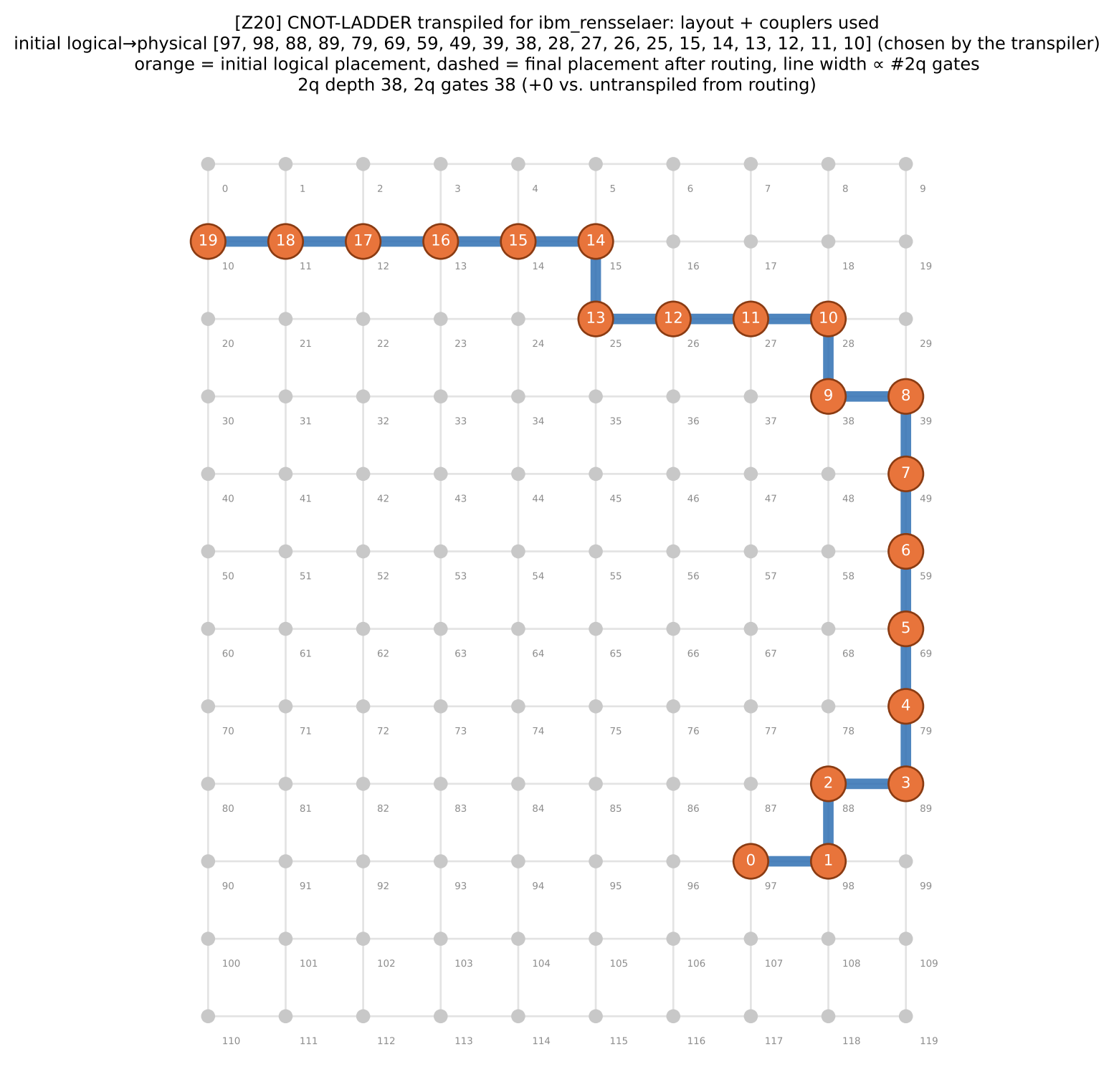
[Z20] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119



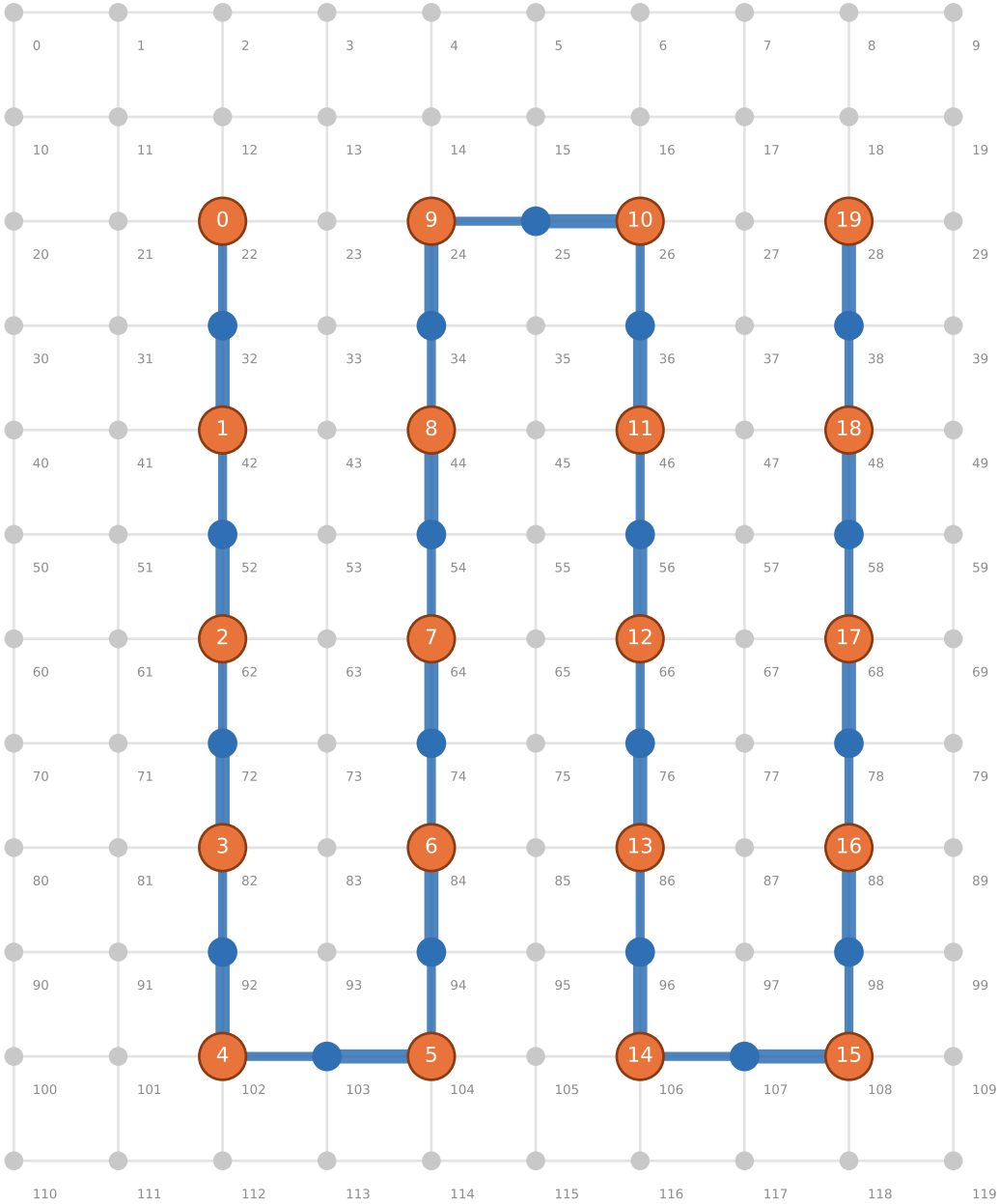
[Z20] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

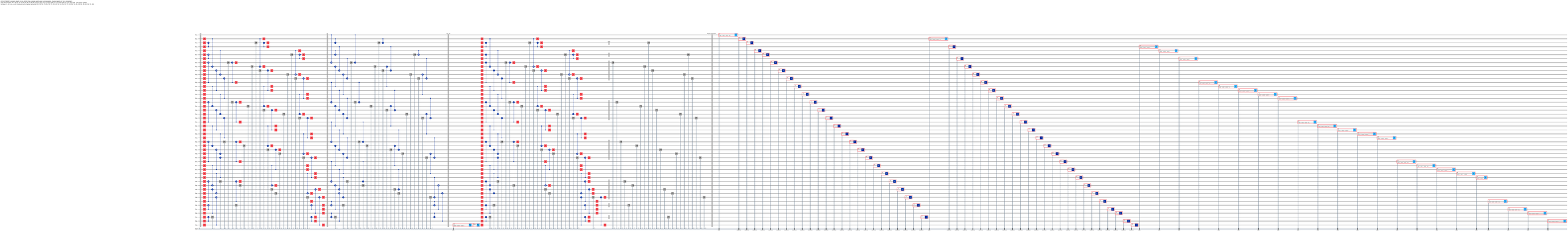
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers



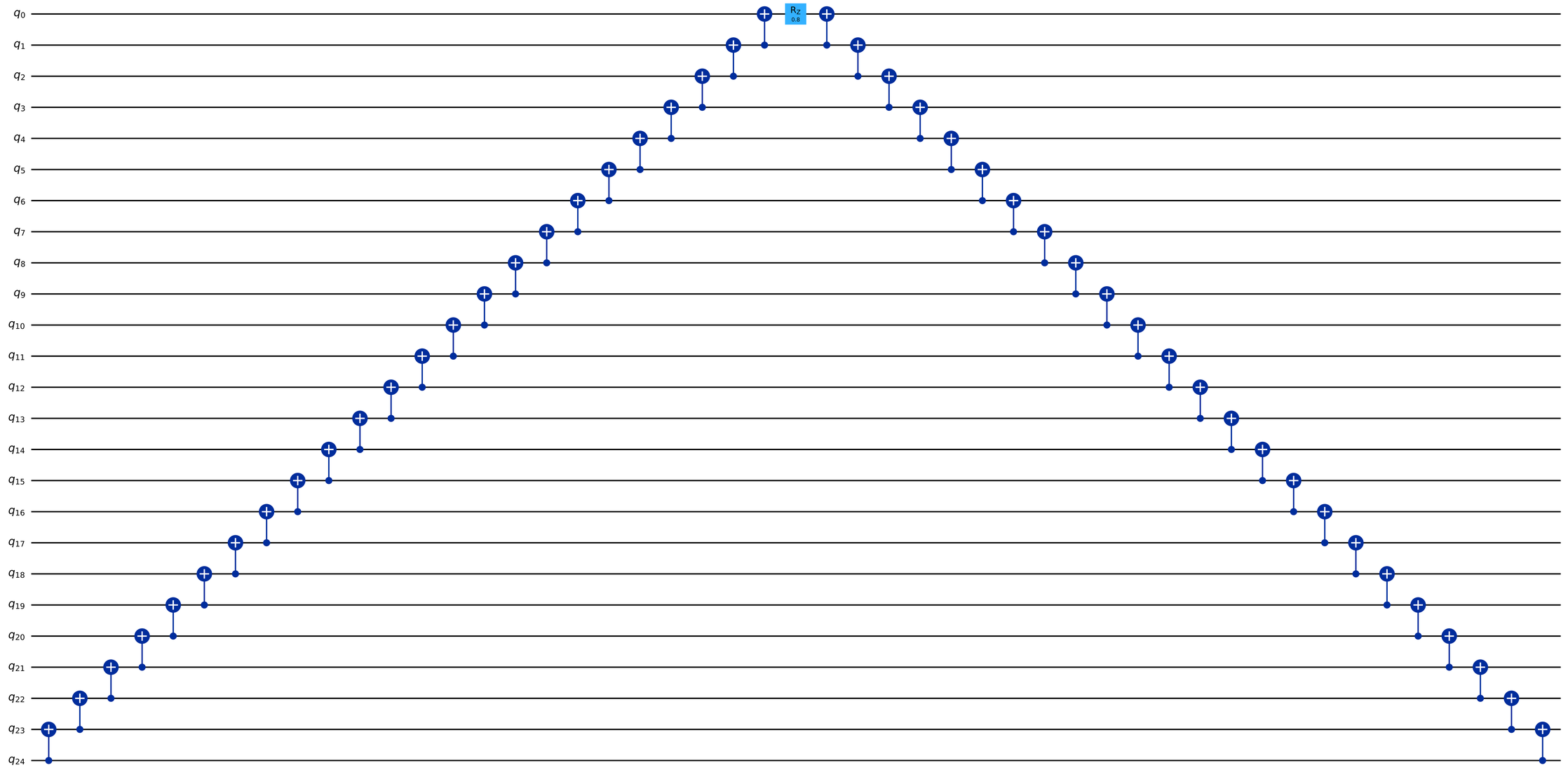

[Z20] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
19 18 17 16 15 14
10 11 12 13 14 15 16 17 18 19
13 12 11 10
20 21 22 23 24 25 26 27 28 29
9 8
30 31 32 33 34 35 36 37 38 39
7
40 41 42 43 44 45 46 47 48 49
6
50 51 52 53 54 55 56 57 58 59
5
60 61 62 63 64 65 66 67 68 69
4
70 71 72 73 74 75 76 77 78 79
2 3
80 81 82 83 84 85 86 87 88 89
0 1
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[Z20] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 114

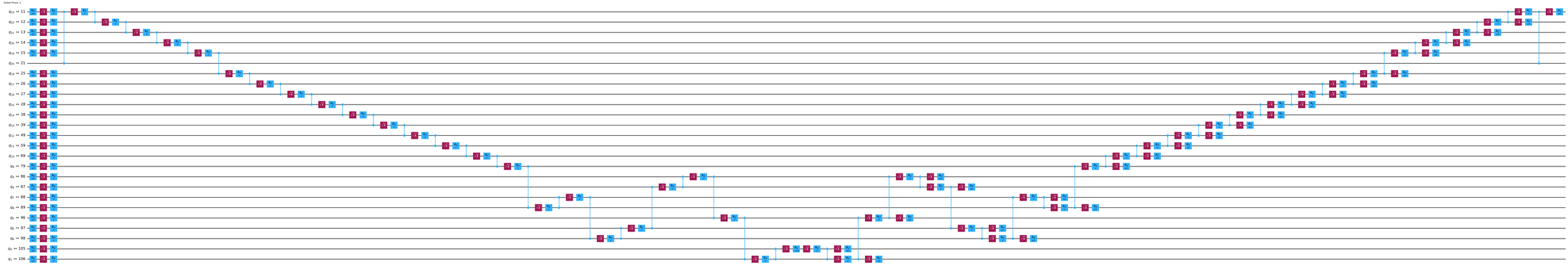




[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48

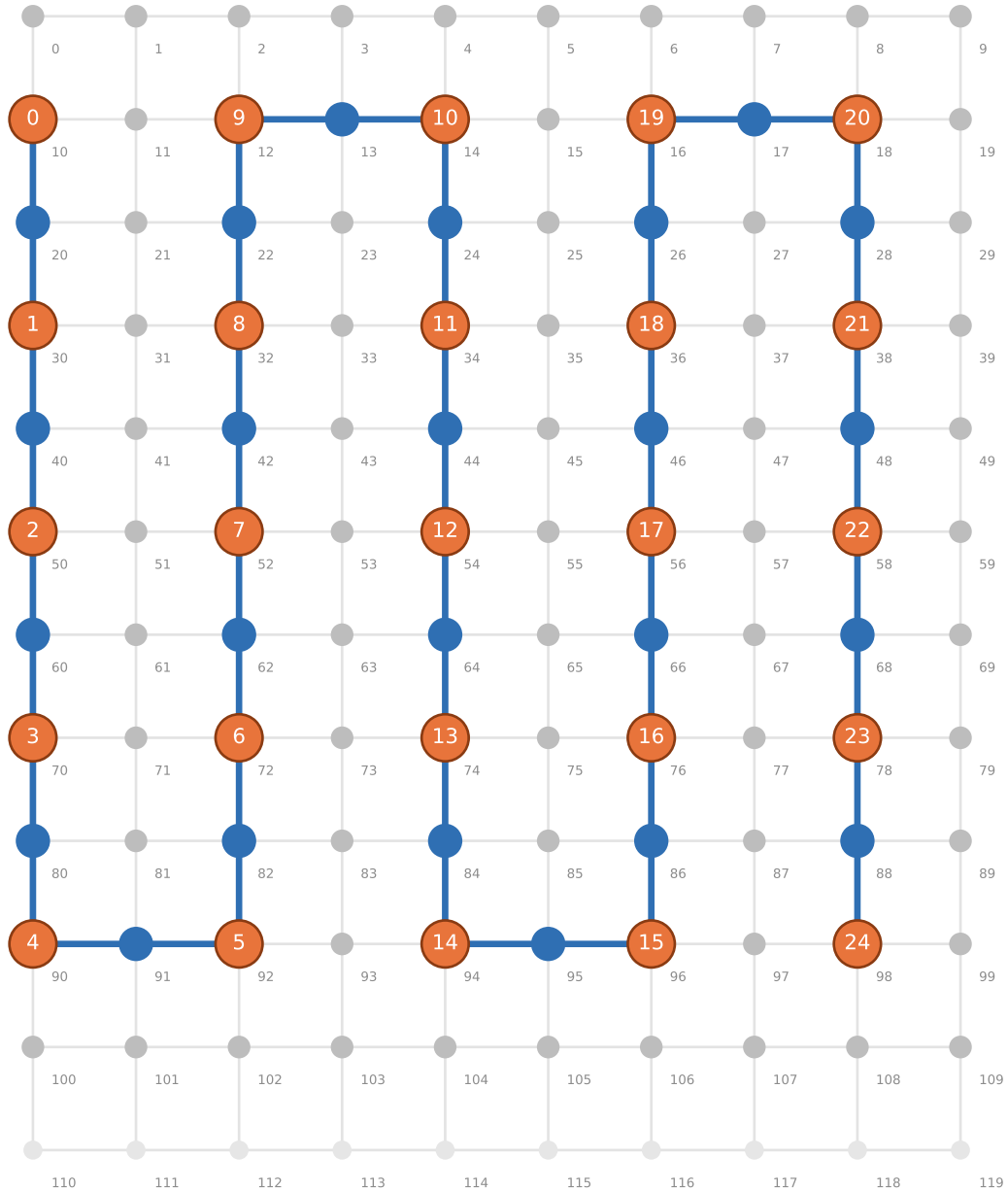


[225] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz/120 sx/96 cz/481)
expl-i 9 P1, th=0.4: P = 20 21 22 23 24 25 26 27 28 29 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 (25 logical qubits)
2q depth 48, total depth 149

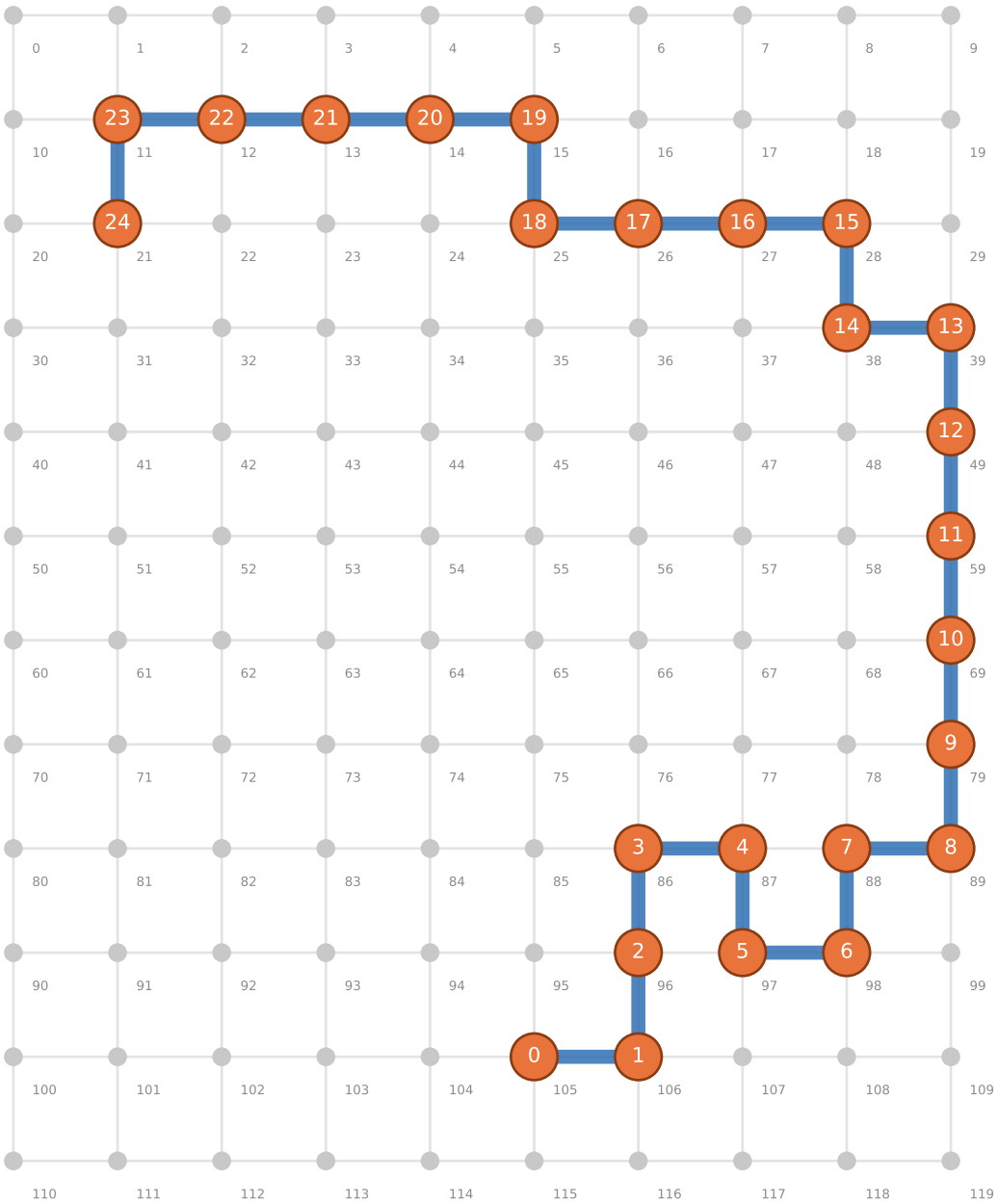


[Z25] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

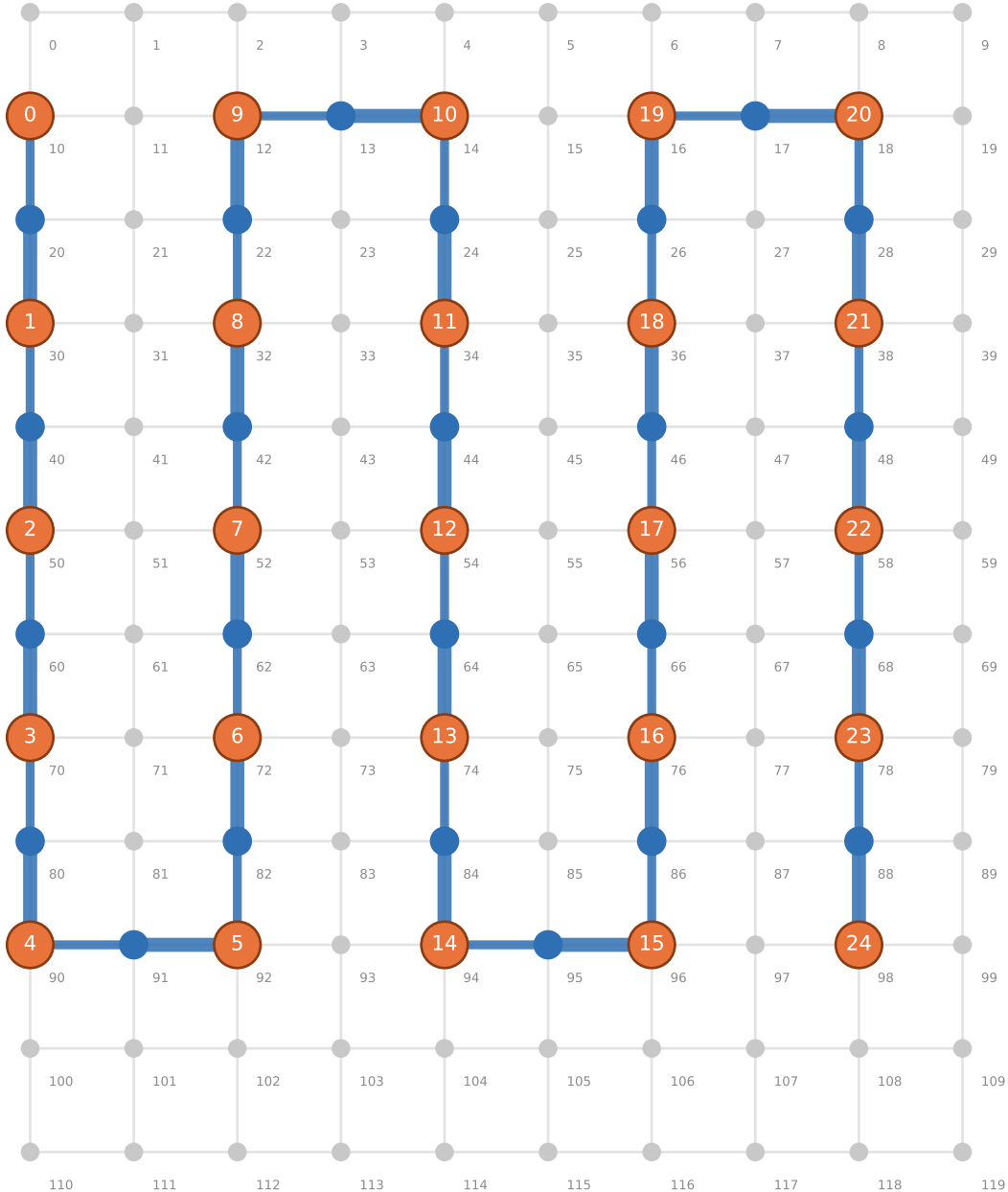
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers



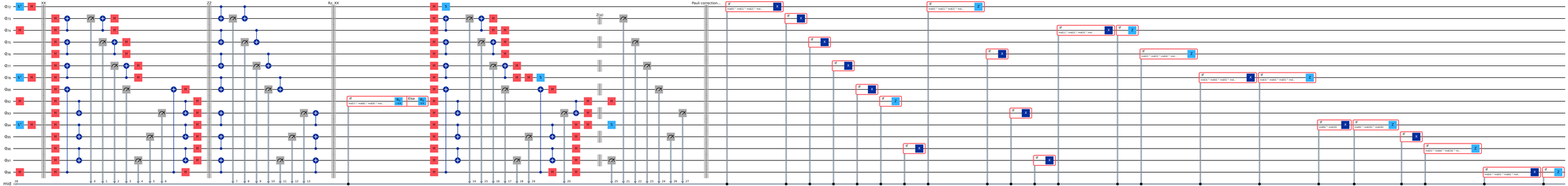
[Z25] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [105, 106, 96, 86, 87, 97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)



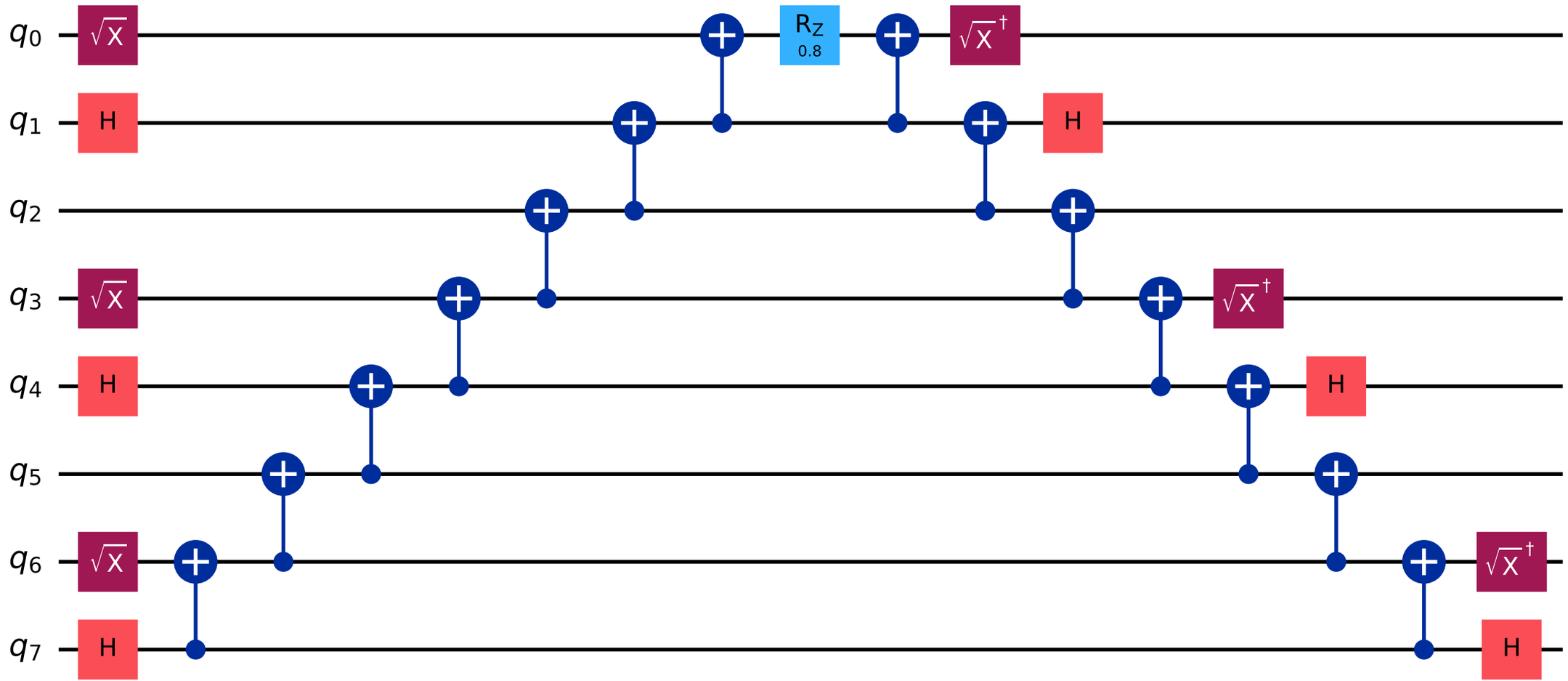
[Z25] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 144



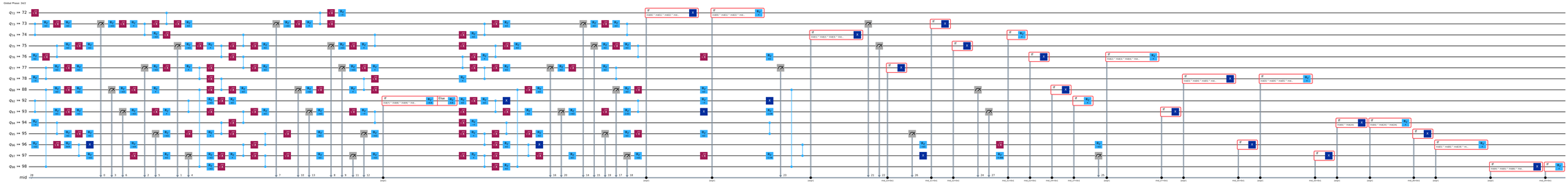
[XYZXYZXY] DYNAMIC constant-depth circuit, NISQ (8 as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P_0 P_1 ... P_{N-1}/4 · P_i = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 · (8 logical qubits)
2q depth 6, 28 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]



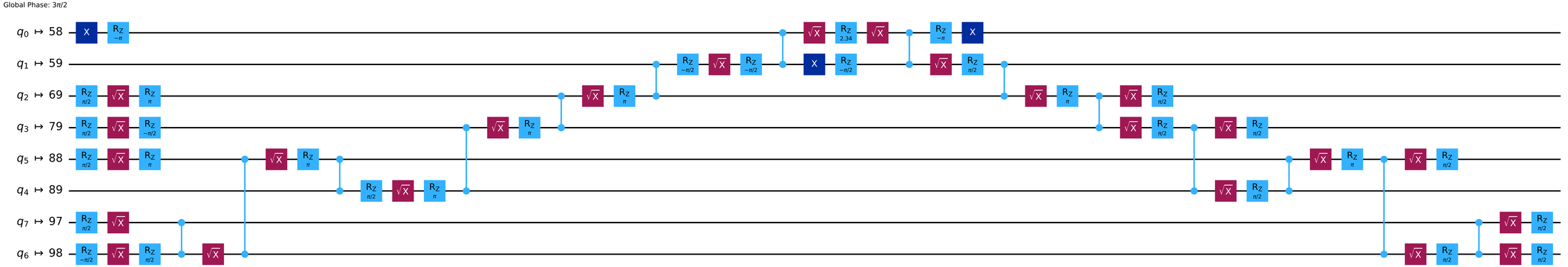
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14



[XYZXYZXY] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:95 sx:76 cz:42 measure:28 if_else:23 x:6)
exp(i·θ·P), θ=π/4 · P → YO X1 Z2 Y3 X4 Z5 Y6 X7 · (8 logical qubits)
2q depth 6, total depth 41

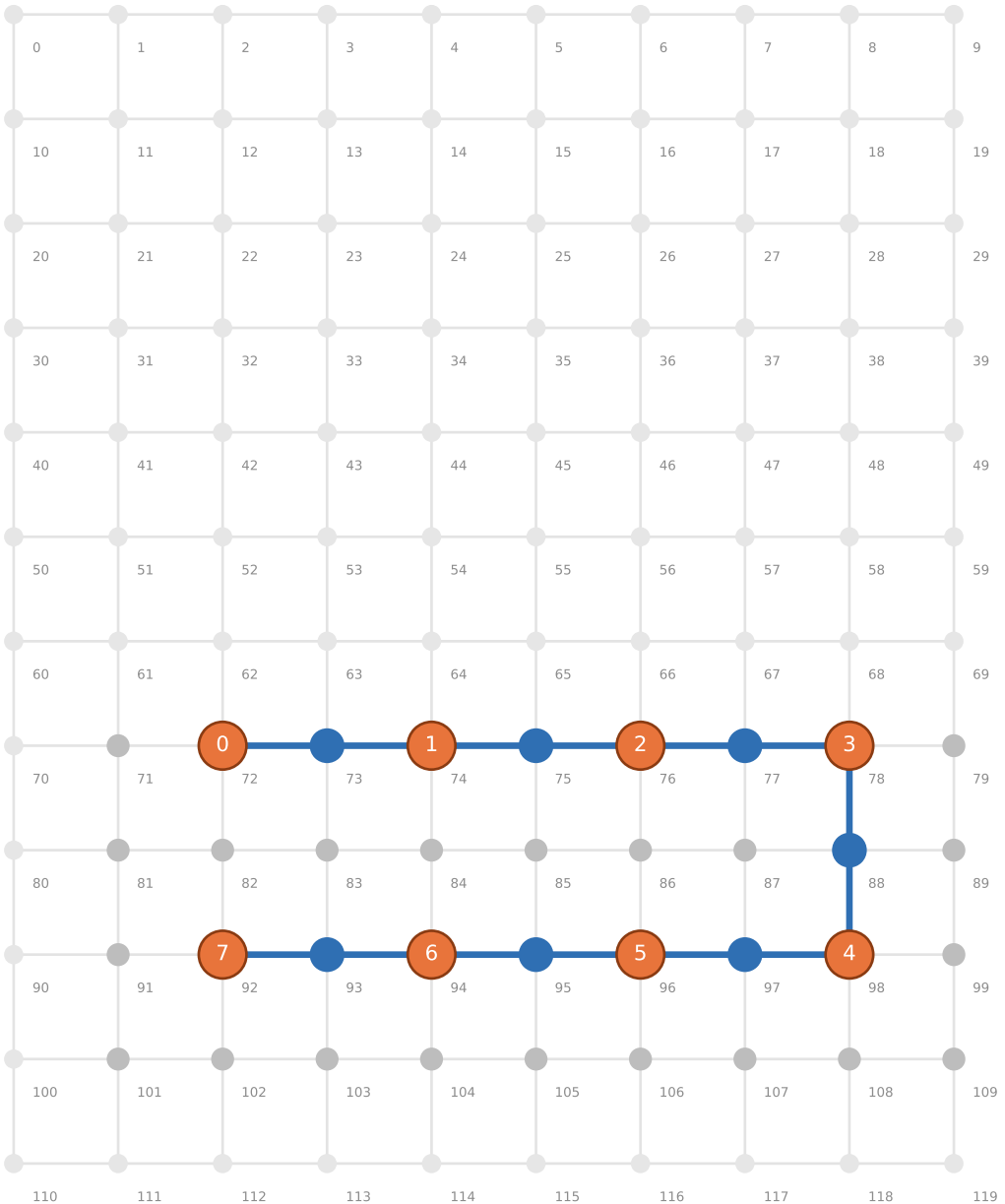


[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:31 sx:24 cz:14 x:3)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 47

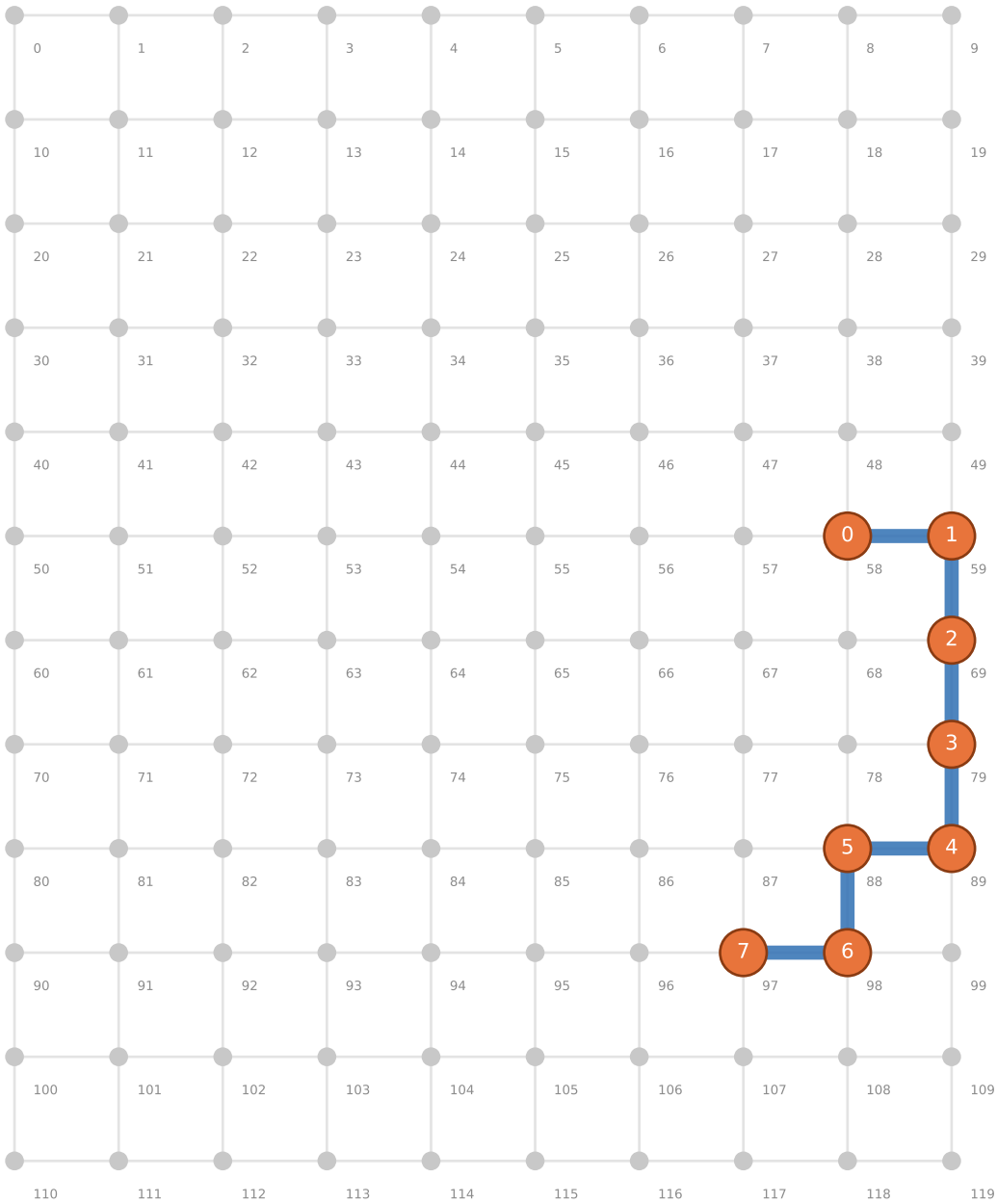


[XYZXYZXY] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

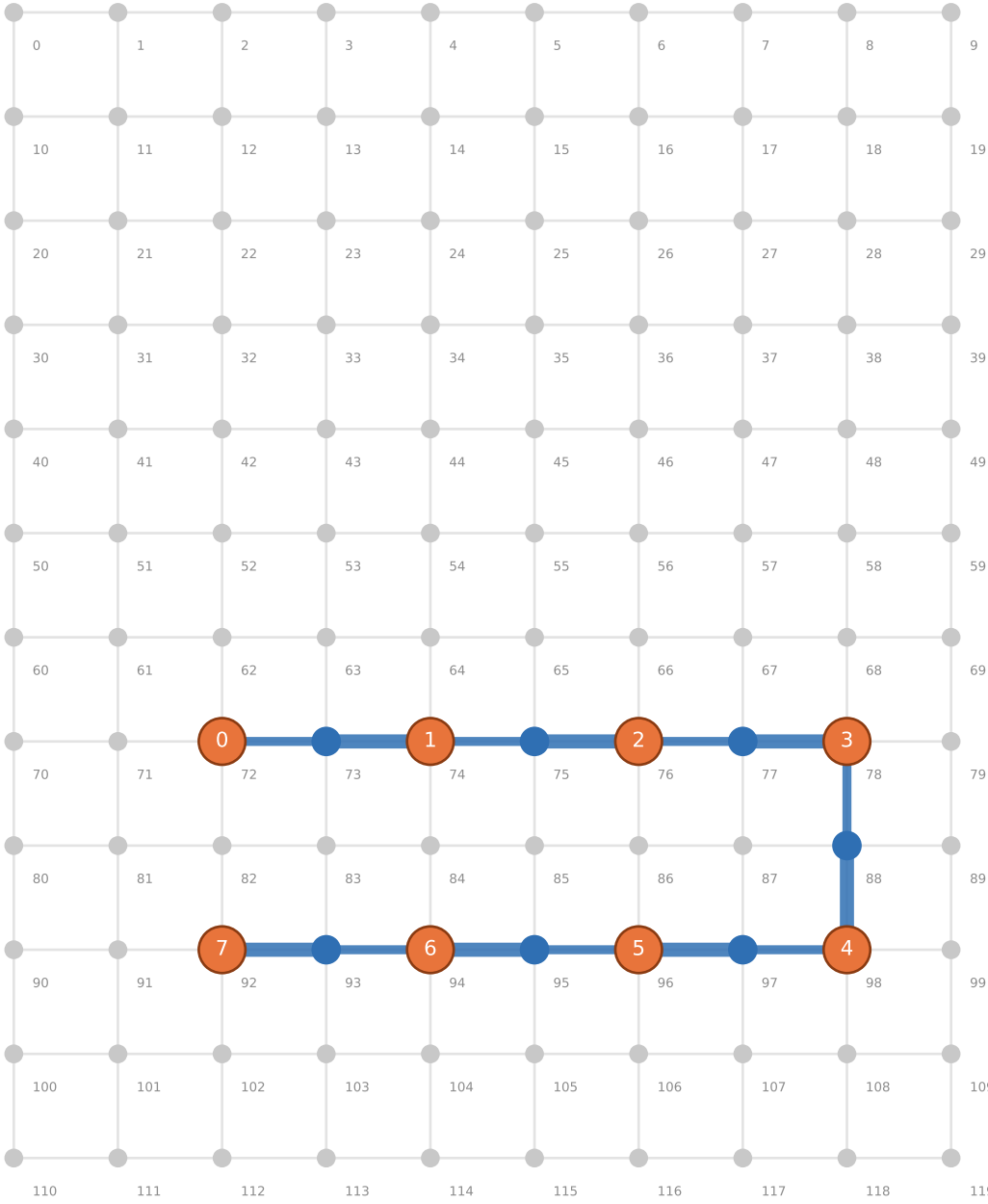
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers



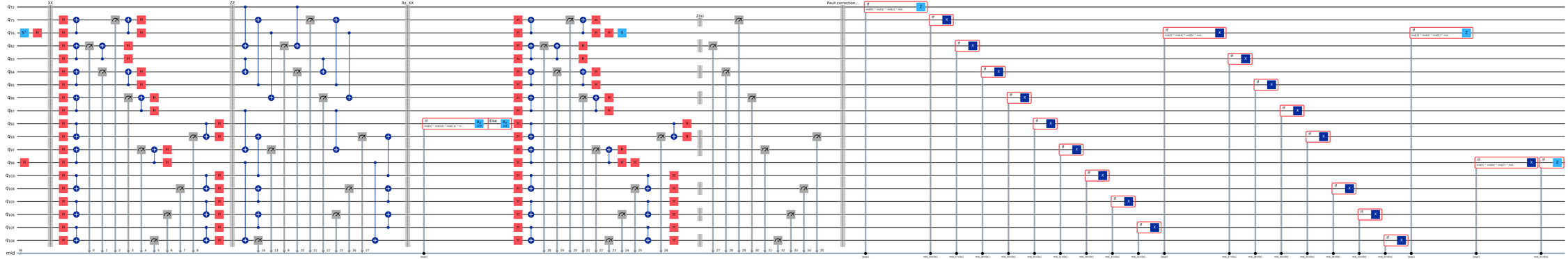
[XYZXYZXY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



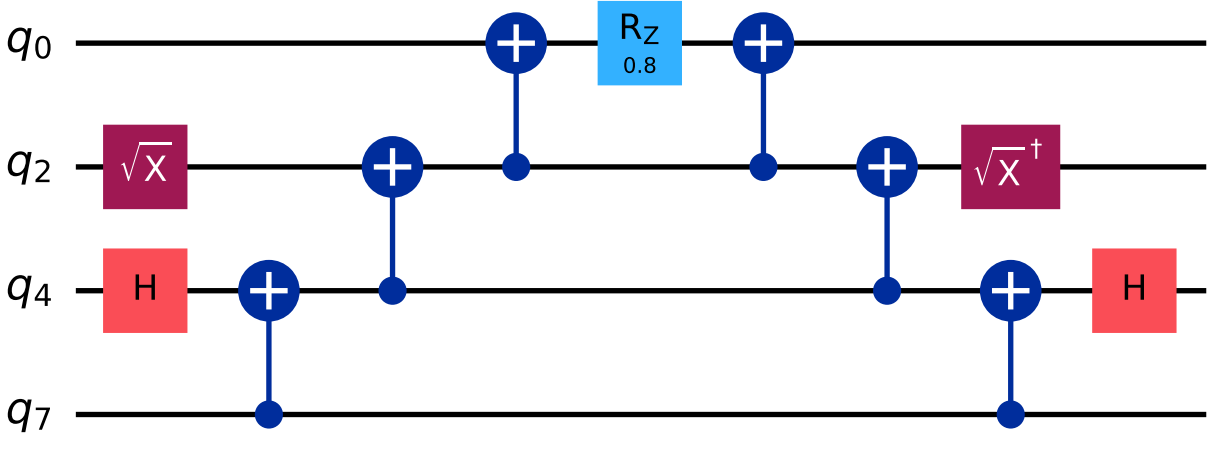
[XYZXYZXY] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 42



[ZIXYIIZ_gapped] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = ZIIYIZZ X4 Z7 (8 logical qubits)
2q depth 6, 36 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]



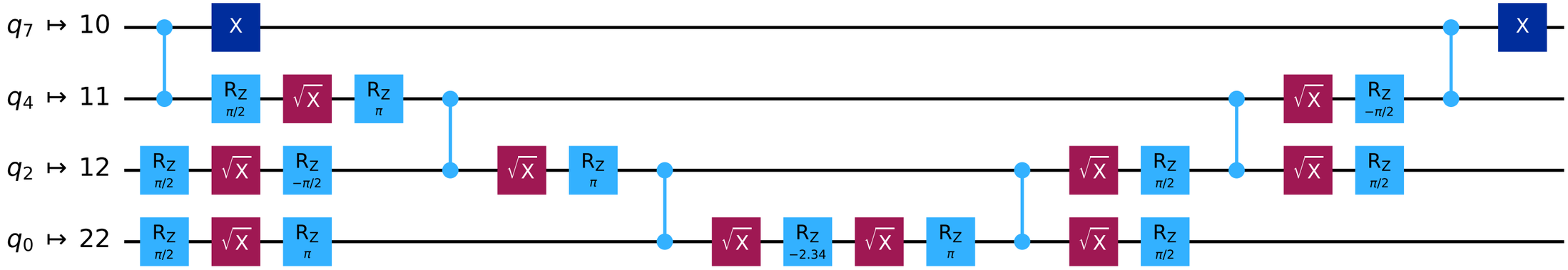
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6



[ZIXIYIZ_gapped] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:130 sx:104 cz:54 measure:36 if_else:22 x:9)
exp(-i 8 P1, 8=0.4) P = ZI Y2 YI XI ZT (8 logical qubits)
2q depth 6, total depth 36

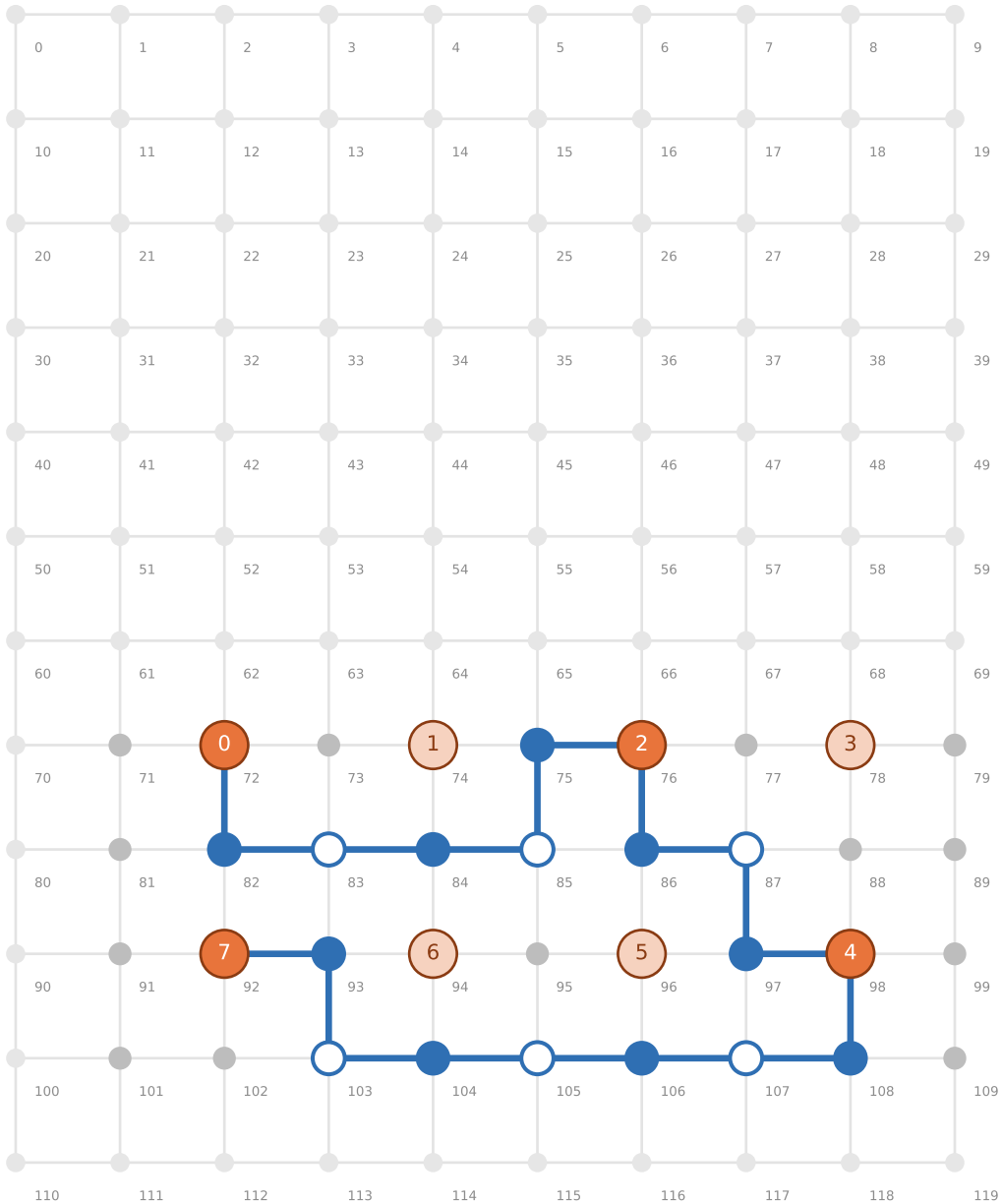


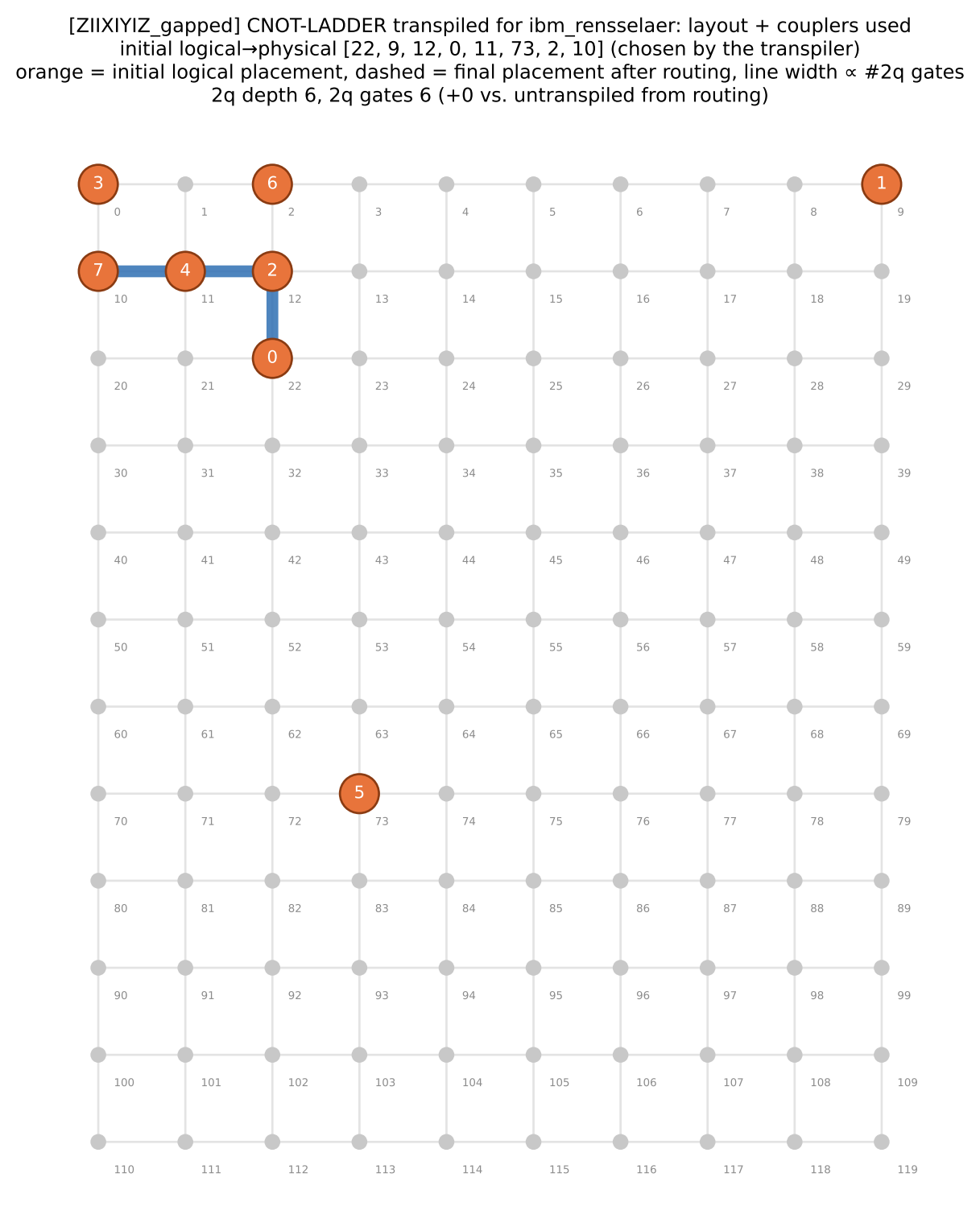
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



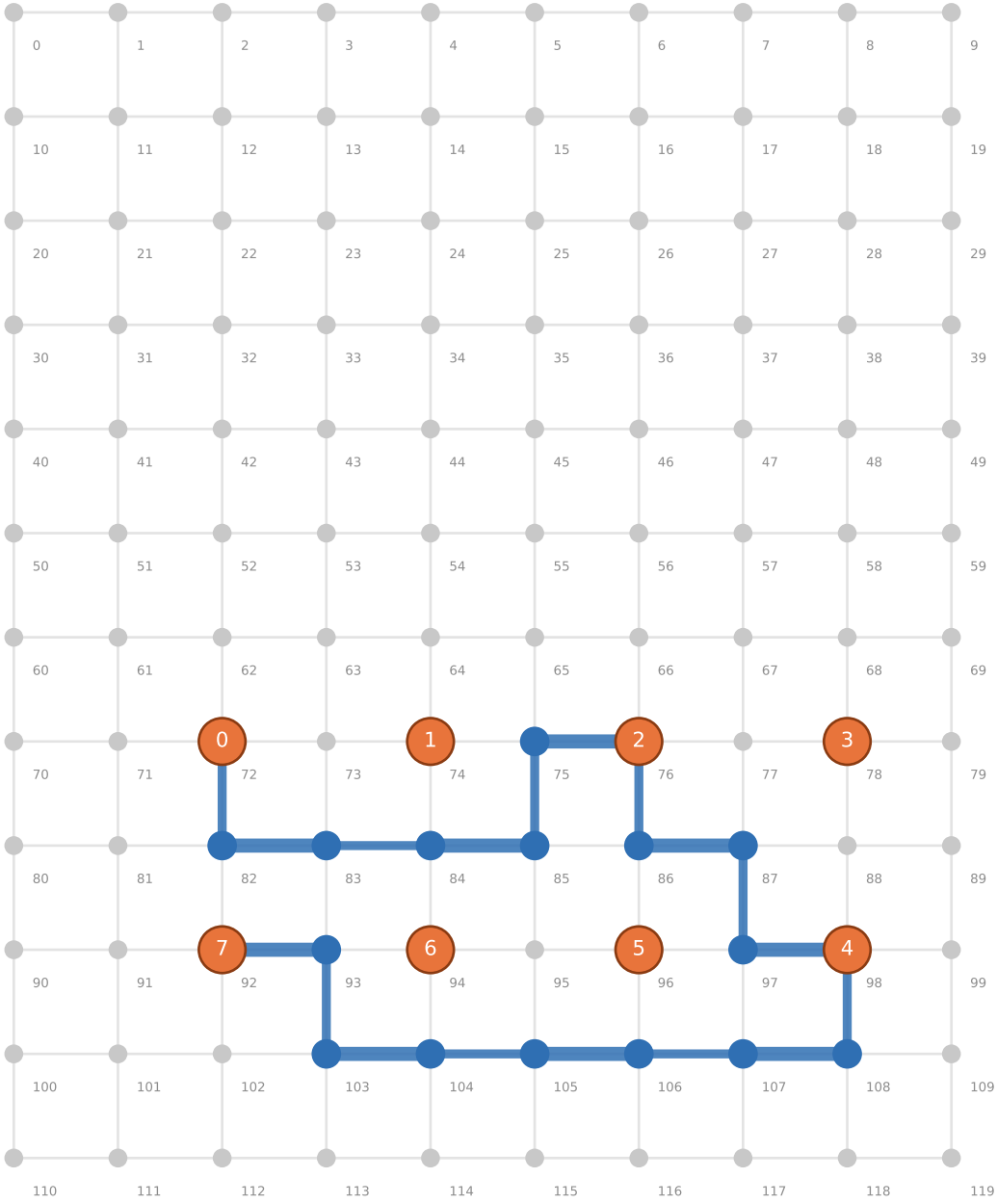
[ZIIXIYIZ_gapped] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers

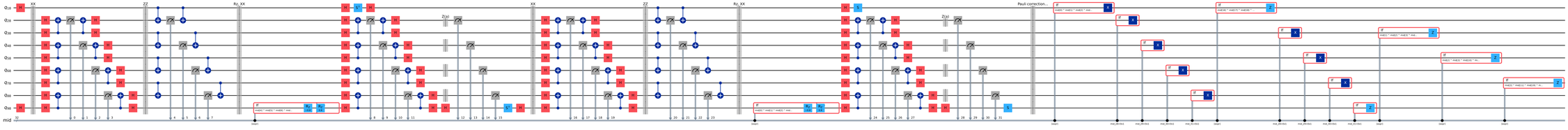



[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 9, 12, 0, 11, 73, 2, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
6
1
7
4
2
0
5
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

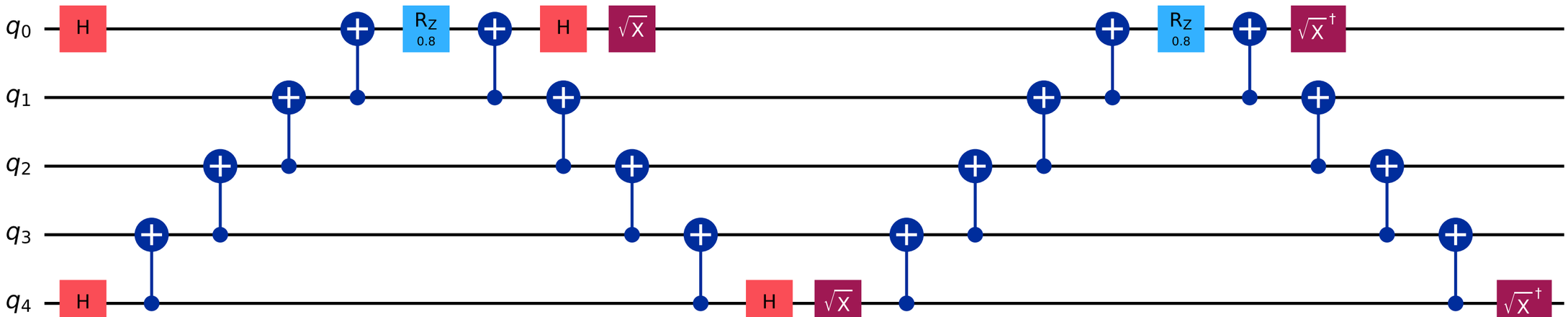
[ZIIXIYIZ_gapped] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 54



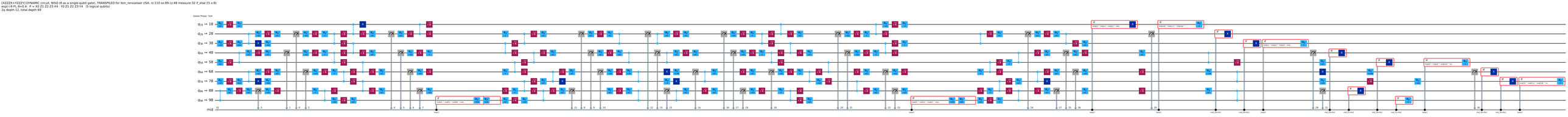
[XZZZX+YZZZY] DYNAMIC constant-depth-circuit, NISQ (8 as a single-qubit gate) (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical-qubits)
2q depth 12, 32 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



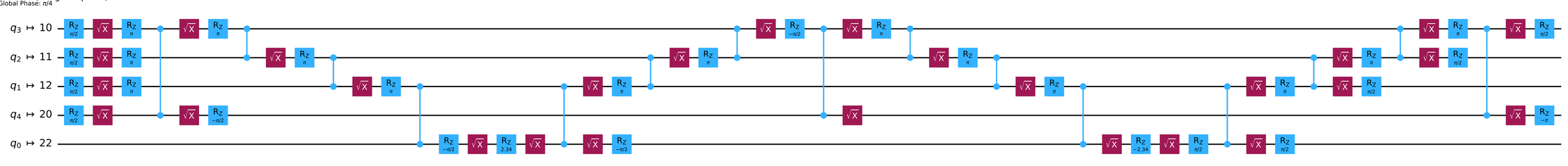
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



[XZZZX+YZZZY].DYNAMIC circuit, NISQ (It as a single-qubit gate), TRANSPILED for ibm_rensselaer (ISA: rz:110 sx:89 cz:48 measure:32 if_else:15 x:8)
exp(i·π·P), P=I⊗…⊗X⊗Z⊗Z⊗Z⊗X⊗…⊗I, P → X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical qubits)
2q depth 12, total depth 69

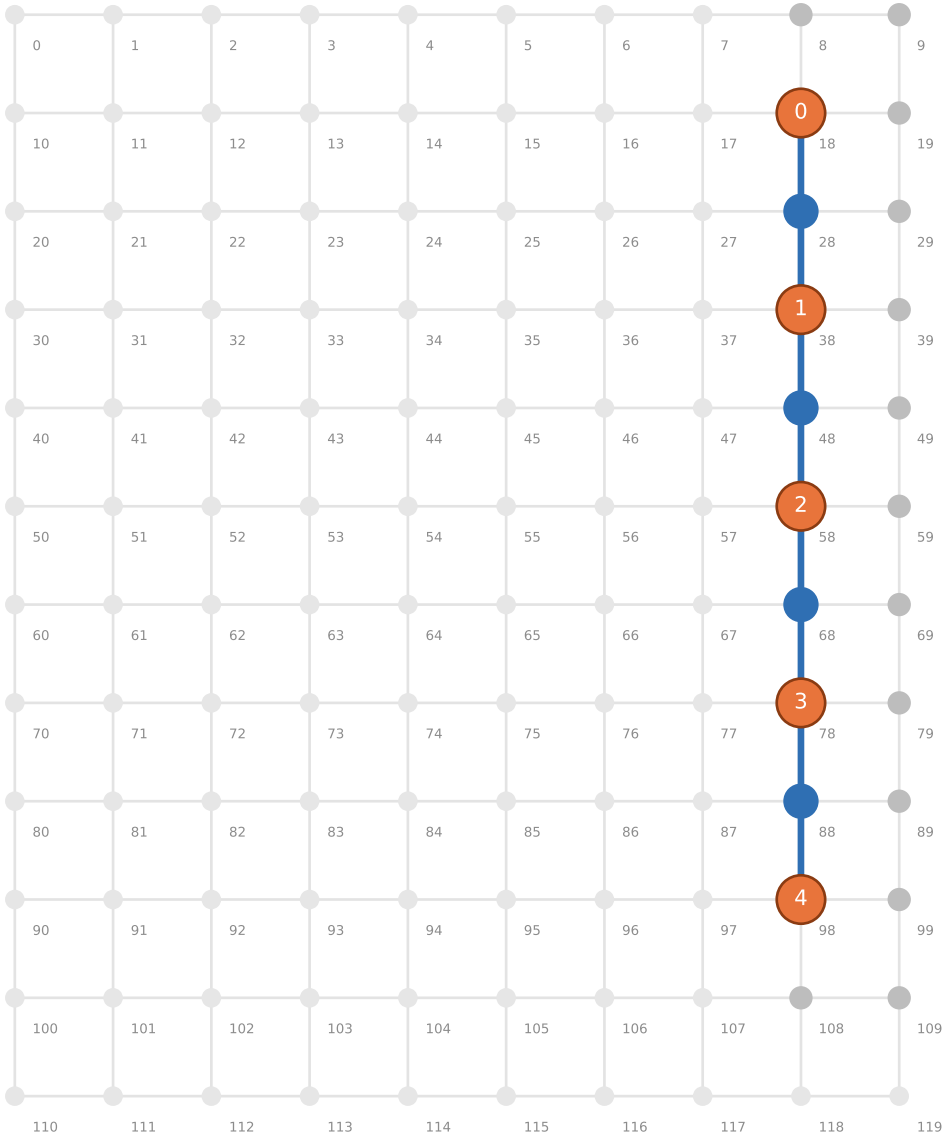

Global Phase: 7π/4
q$_{18}$ ↦ 18
q$_{28}$ ↦ 28
q$_{38}$ ↦ 38
q$_{48}$ ↦ 48
q$_{58}$ ↦ 58
q$_{68}$ ↦ 68
q$_{78}$ ↦ 78
q$_{88}$ ↦ 88
q$_{98}$ ↦ 98
mid
32
if
c4==0x1 ^ c4==0x3 ^ c4==0x5 ^ c4==...
(expr)

[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:30 sx:28 cz:15)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 15, total depth 52

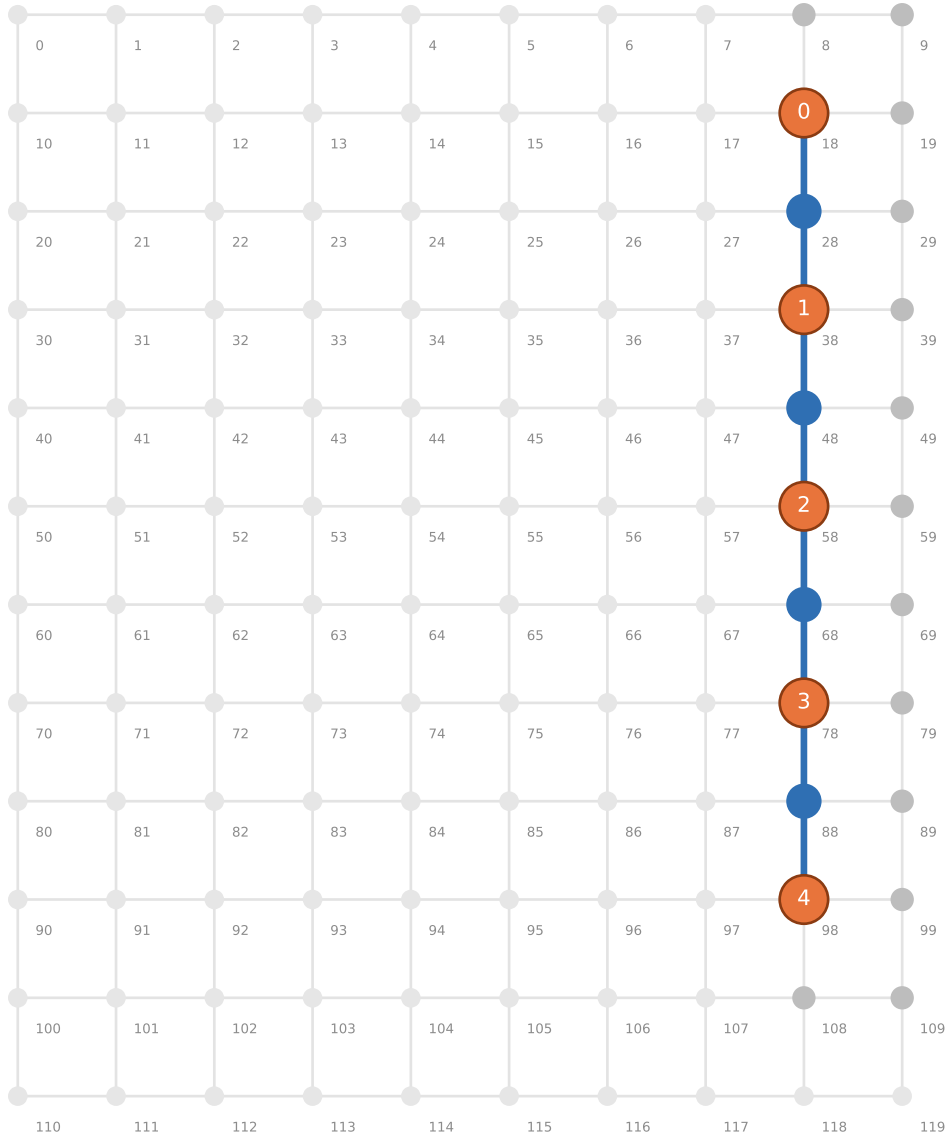


[XZZZX+YZZZY] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

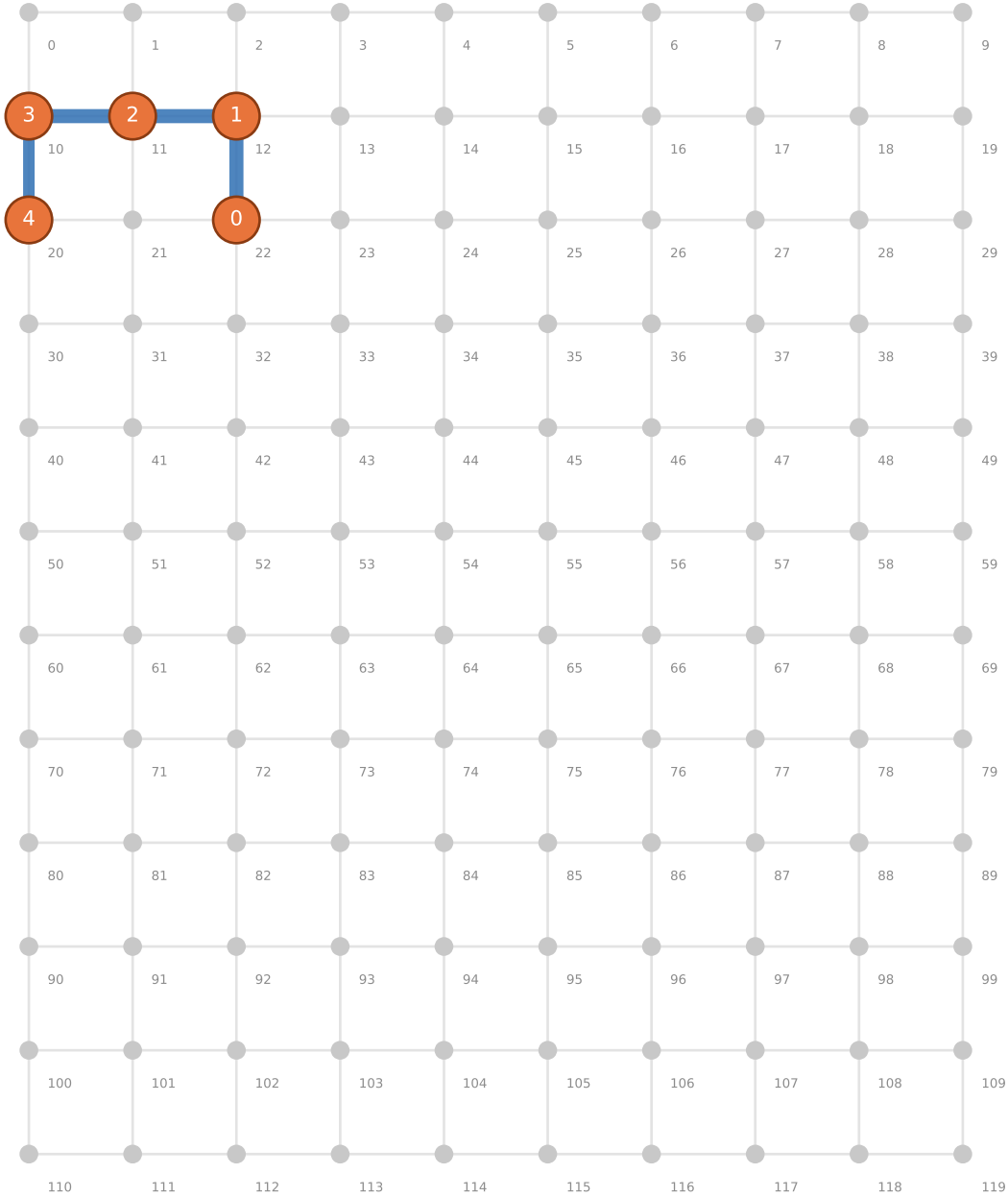
term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers



[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10, 20] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)



[XZZZX+YZZZY] DYNAMIC circuit (nisq) transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 12, 2q gates 48

